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School of Computing and Information Systems

Lesson 1:
Becoming Customer Focused

Self Introduction

Learning Outcomes for today’s lesson

- **Course Curriculum** and **Assessment Metrics**
- Understand the significance of **customer centricity**
- Ability to appreciate the practical challenges in **implementing a customer analytics** framework
- Form Groups

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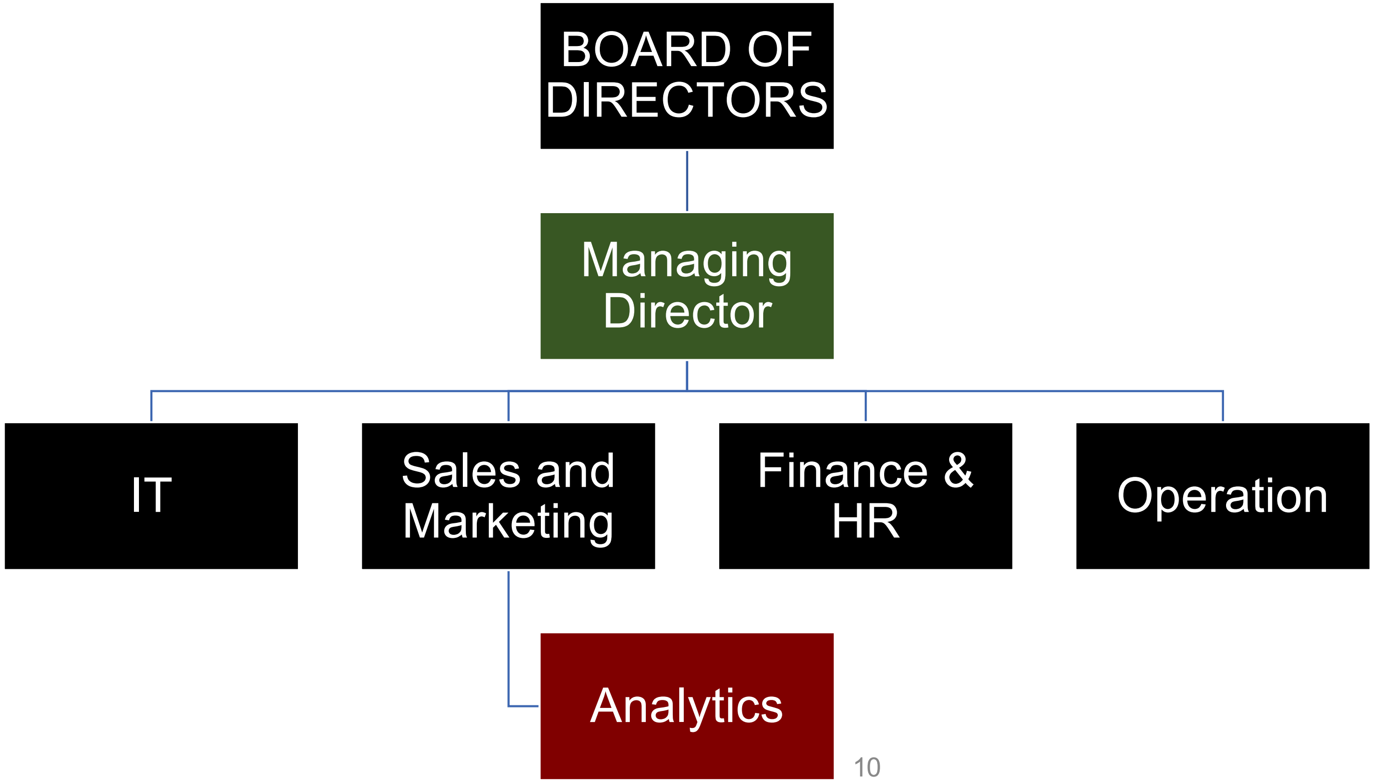
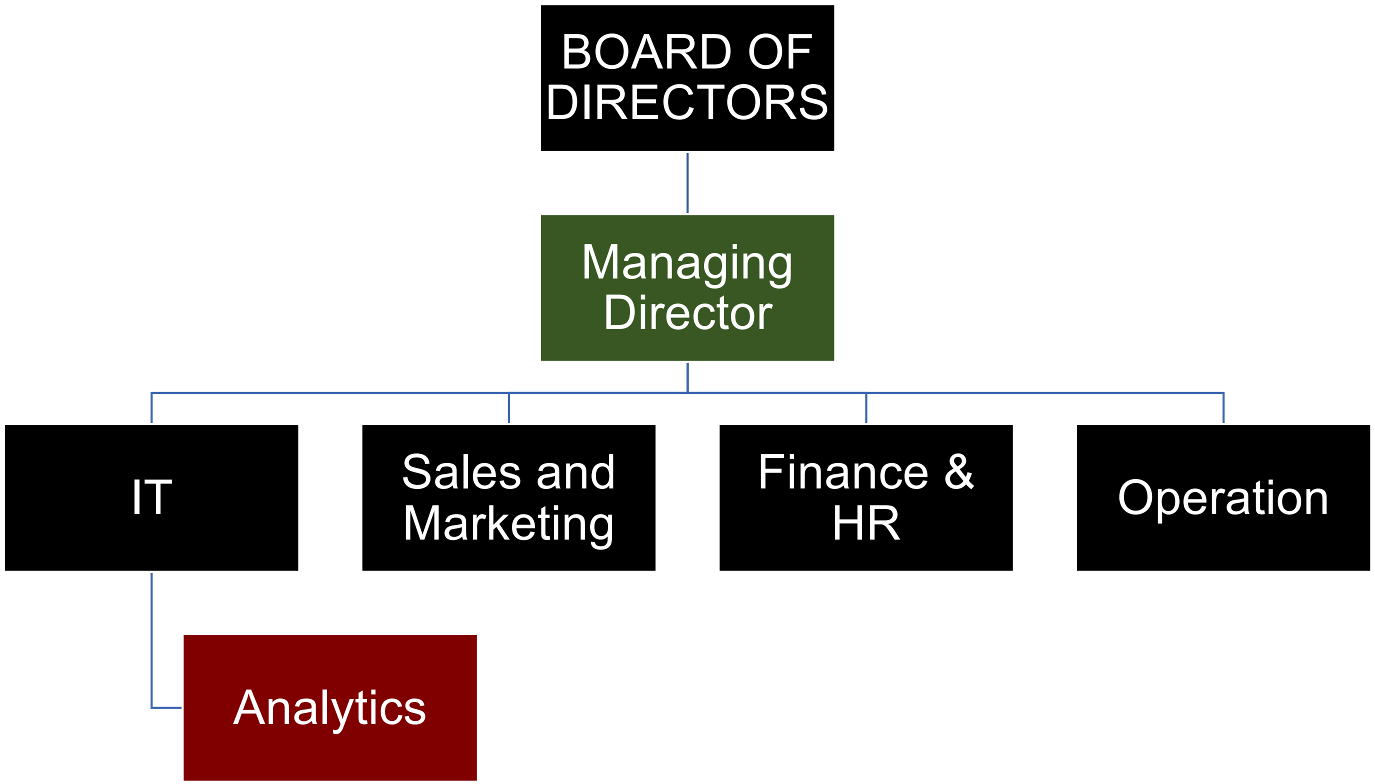
Typical Organisation Structure

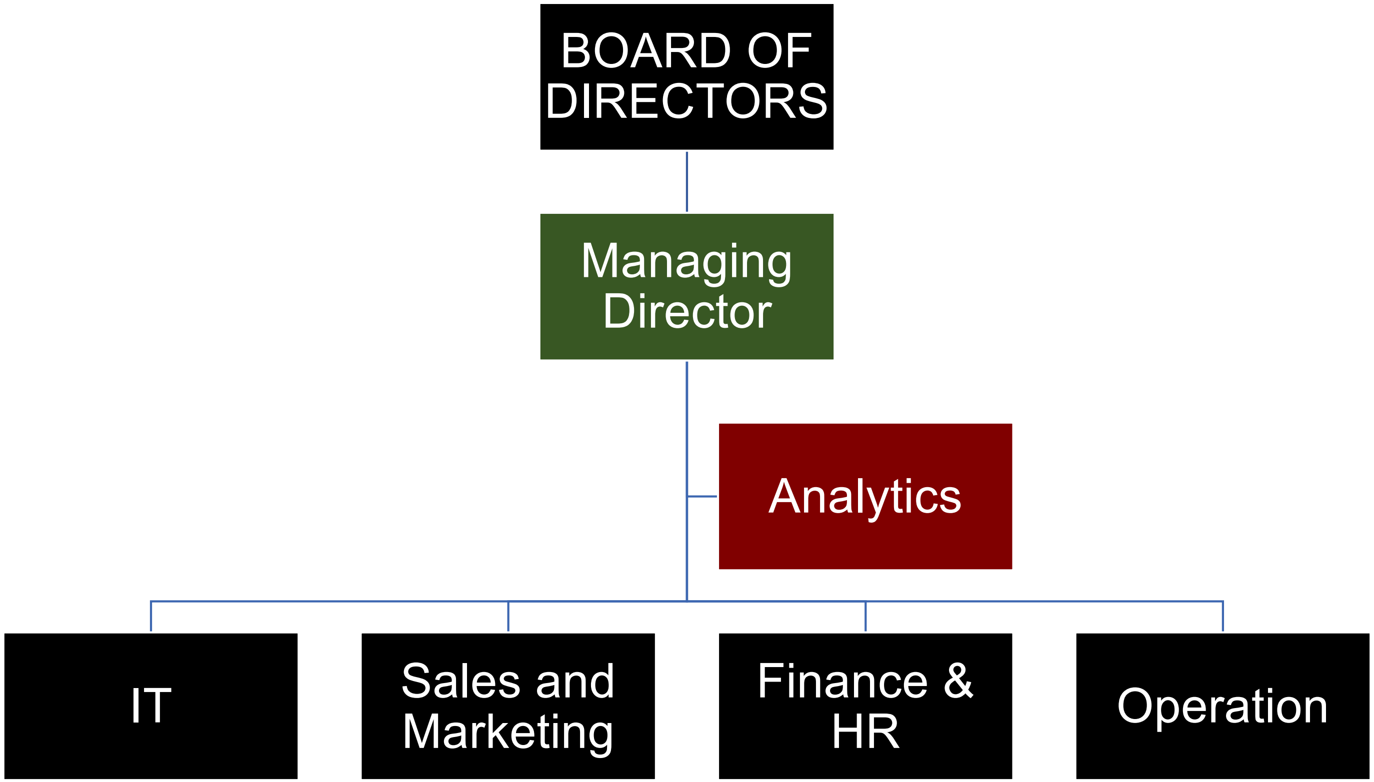
Customer Analytics within an Organisation



Customer Analytics within an Organisation



Customer Analytics within an Organization



The Analytics Maturity Ladder — "Sophisticated Equals Better"

- Four types of analytics capability:
  - **Descriptive** — *What is happening / what has happened?*
  - **Diagnostic** — *Why did it happen?*
  - **Predictive** — *What will happen next?*
  - **Prescriptive** — *What should we do about it?*
- The conventional belief: firms must *climb the ladder* from descriptive to prescriptive
- Value is assumed to increase with sophistication — the fancier the model, the bigger the payoff
- Reinforced by the current hype around machine learning and AI: *"sophisticated equals better"*
- Implication for practice: jump straight to predictive models, churn scoring, CLV forecasting, recommender systems

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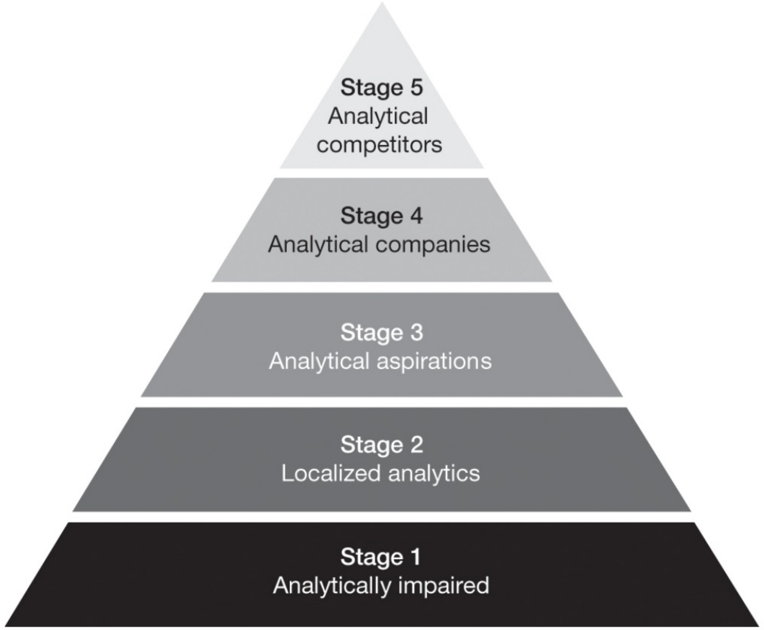
The five stages of analytical capability



	Distinctive capability/level of insights	Questions asked	Objective	Perceived value
Stage 1 Analytically impaired	Negligible; "flying blind"	What happened in our business?	Get accurate data to improve operations	None
Stage 2 Localized analytics	Local and opportunistic; may not be supporting company's distinctive capabilities	What can we do to improve this activity? How can we understand our business better?	Use analytics to improve one or more functional activities	ROI of individual applications
Stage 3 Analytical aspirations	Begin efforts for more integrated data and analytics	What's happening now? Can we extrapolate existing trends?	Use analytics to improve a distinctive capability	Future performance and market value
Stage 4 Analytical companies	Enterprise-wide perspective; able to use analytics for point advantage; know what to do to get to next level, but not quite there	How can we use analytics to innovate and differentiate?	Build broad analytic capability—analytics for differentiation	Analytics are an important driver of performance and value
Stage 5 Analytical competitors	Enterprise-wide, big results, sustainable advantage	What's next? What's possible? How do we stay ahead?	Analytical master—fully competing on analytics	Analytics are a primary driver of performance and value

Davenport, T. H., & Harris, J. G. (2017). *Competing on Analytics: The New Science of Winning* (Updated ed., Ch. 2). Harvard Business Review Press.

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Class Discussion

Exercise: Mapping the Generative AI Roadmap to the Five Stages of Analytical Capability

You have learned about Davenport & Harris's five stages of analytical competition, which describe how organizations progress from no real analytical capability to using analytics as the basis of competitive advantage. Most organizations today are now facing a similar journey with **Generative AI (GenAI)** — large language models, copilots, image and code generation, and agentic systems.

The question is: does GenAI follow the same maturity arc as traditional analytics, or is it fundamentally different? Can we map it onto the five-stage framework you have already learned.

Mapping the GenAI Roadmap to the Five Stages of Analytical Capability

From shadow use to competitive differentiation — and the barriers between each stage

Stage	Typical Use of GenAI	Leadership & Governance	Data & Infrastructure	Business Value	Key Barriers / Risks
1 — GenAI Impaired	None formally; "shadow" use of public ChatGPT on personal accounts, often against policy	Leadership unaware or actively blocking; no governance or policy	No enterprise data strategy; no tooling	None — only compliance exposure	Data leakage and IP risk; lack of awareness; bans push usage underground; reputational and regulatory exposure
2 — Localized GenAI	Team-level experiments — marketing copy, GitHub Copilot, HR CV summarisation	Aware but not driving adoption; no central governance; team subscriptions	Free or team-tier tools; no integration with internal data	Small, siloed productivity gains	Fragmentation and duplicated spend; inconsistent quality; uneven security; pilot purgatory; no measurement of ROI
3 — GenAI Aspirations	Pilots tied to a stated enterprise strategy; early enterprise tools (Copilot, Gemini, private LLM)	CEO / CIO / CAIO publicly committed; vision articulated; risk framework being built	Enterprise licensing in place; data foundations still weak; pilots not yet scaled	Promise of value; limited proof points	Will outpacing skill; talent and change-management gap; weak data foundations; pilots that fail to scale; vendor lock-in
4 — GenAI Companies	Embedded across many functions; RAG over internal data; reusable prompt libraries; agent prototypes	Mature enterprise governance for safety, IP, privacy; strong executive sponsorship	Internal data platform connected to LLMs; LLMOps capability; monitoring and evaluation	Measurable productivity and quality gains across functions	GenAI is enabler, not differentiator; complacency; hallucination at scale; cost of inference; provider concentration; reskilling
5 — GenAI Competitors	Proprietary / fine-tuned models; customer-facing GenAI products; agentic workflows reshape unit economics	Passionate, hands-on senior leadership; operating model redesigned around AI-augmented work	Proprietary data moats; advanced model ops; in-house model development	GenAI is core to competitive advantage; visible to customers and competitors	Regulatory and ethical scrutiny; defending data and model moat; talent war; reputational risk; geopolitical exposure

Source: Adapted from Davenport & Harris, Competing on Analytics (HBR Press, 2017), Ch. 2.

Class Activity — What's Really Going on with GenAI Adoption?

Crowdsource the real-world barriers from people doing this work today

The Activity

Step 1 — Reach out

Identify 2–3 working professionals in your group who have used GenAI in their organisation in the last 6 months. Aim for variety: a developer, a marketer, an analyst, a manager.

Step 2 — Ask the questions

Ask each contact:

1. What GenAI tools is your organisation actually using?
2. What problems have you personally hit when trying to adopt GenAI at work?
3. Where does your firm sit on the 5-stage maturity curve, and what is blocking the next step?

Step 3 — Map and share

Map each story onto a stage of the GenAI maturity table. Bring 1 surprising barrier back to class for group discussion.

Themes to Listen For

Data quality and access

Models hallucinate confidently when underlying data is dirty, siloed, or missing context.

Governance & IP exposure

Who owns the output? Is sensitive data leaking into prompts? Most firms cannot answer.

Workflow integration

Tools sit alongside existing systems instead of inside them — friction kills adoption.

Trust & verification

Users either over-trust outputs or refuse to use them at all. Few firms have a verification habit.

Skills and change management

Prompting, evaluation, and redesigning work around AI are not yet core competencies.

Measuring real ROI

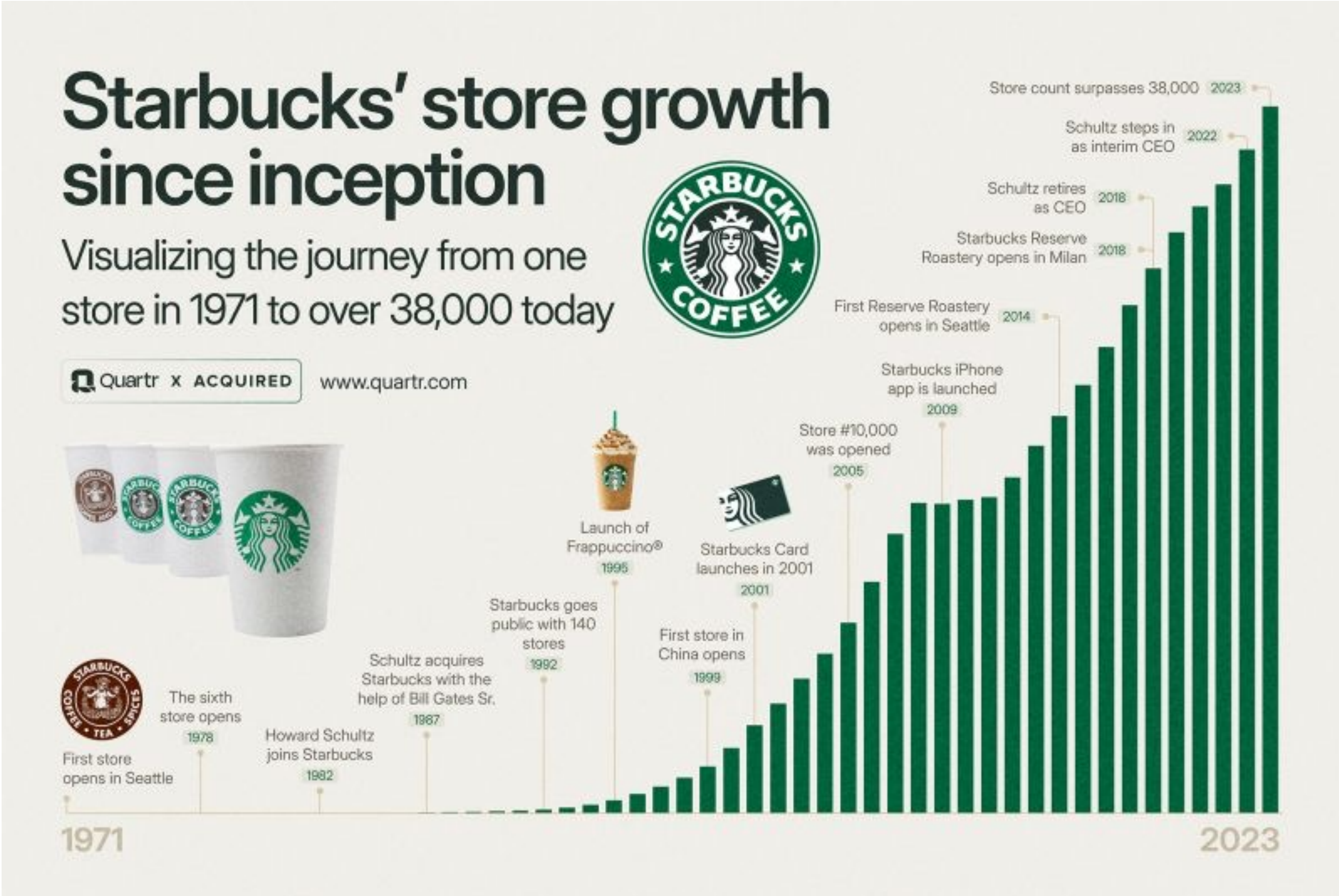
Pilots impress; sustained P&L impact is rare. Vanity metrics dominate.

CASE STUDY

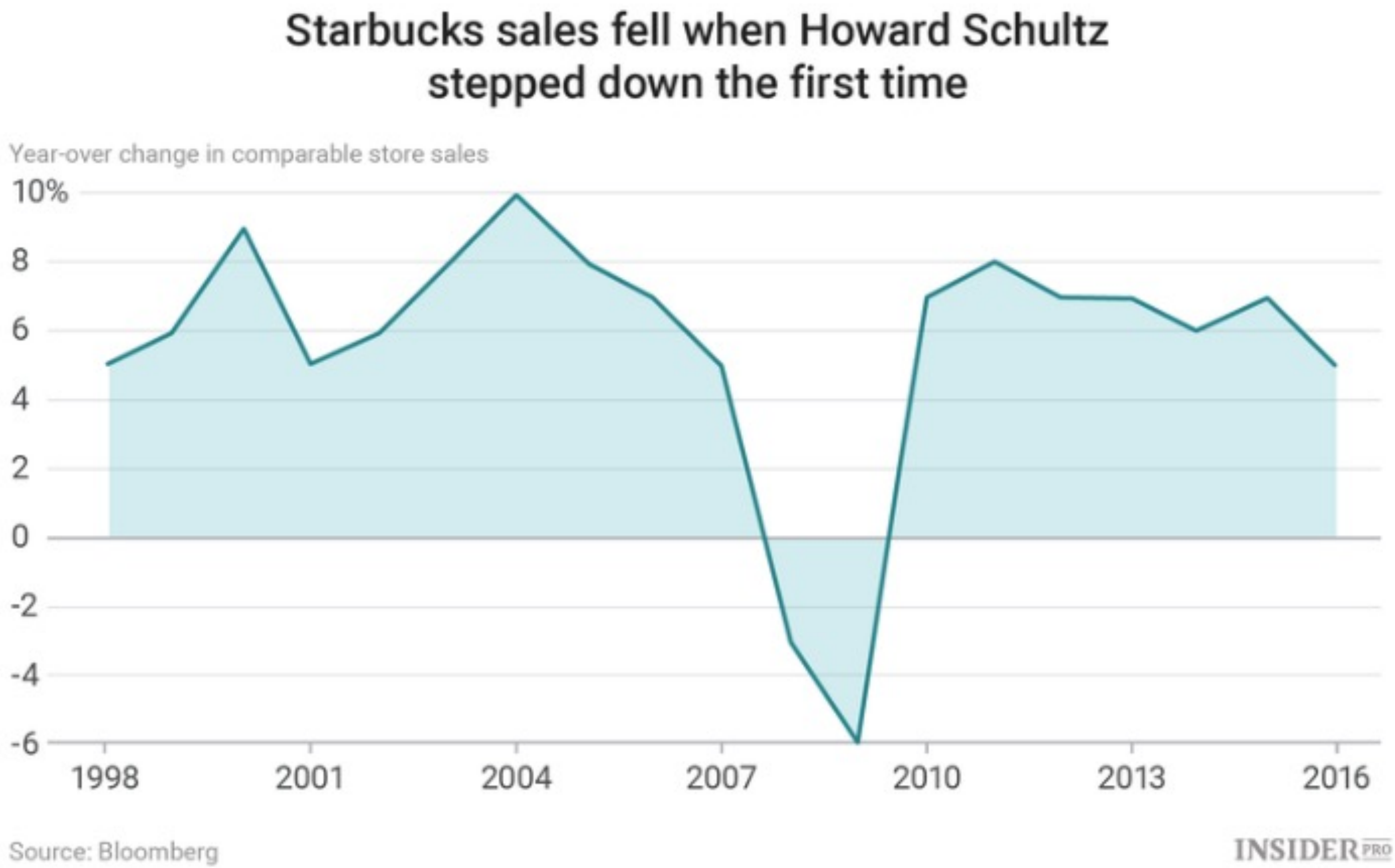
"We sell 4 billion cups a year. Why is growth slowing?"

By 2008, Starbucks had 16,000 stores worldwide but was posting its first-ever decline in same-store traffic. Howard Schultz returned as CEO and discovered the issue wasn't the coffee — it was that the company had become a store-opening machine. Growth was measured by new locations, not by whether existing customers were coming back more often or spending more. The company couldn't answer a basic question: who are our best customers, and are we losing them?

Schultz, H. (2011), Onward: How Starbucks Fought for Its Life Without Losing Its Soul, Rodale Press.



Starbucks' store growth since inception
Visualizing the journey from one store in 1971 to over 38,000 today
Quartr x ACQUIRED
www.quartr.com
STARBUCKS COFFEE
First store opens in Seattle
The sixth store opens
1978
Howard Schultz joins Starbucks
1982
Schultz acquires Starbucks with the help of Bill Gates Sr.
1987
Starbucks goes public with 140 stores
1992
Launch of Frappuccino®
1995
First store in China opens
1999
Starbucks Card launches in 2001
2001
Store #10,000 was opened
2005
Starbucks iPhone app is launched
2009
First Reserve Roastery opens in Seattle
2014
Starbucks Reserve Roastery opens in Milan
2018
Schultz retires as CEO
2018
Schultz steps in as interim CEO
2022
Store count surpasses 38,000
2023
1971
2023


Starbucks sales fell when Howard Schultz stepped down the first time
Year-over change in comparable store sales
10%
8
6
4
2
0
-2
-4
-6
1998
2001
2004
2007
2010
2013
2016
Source: Bloomberg
INSIDER PRO

STARBUCKS

What They Did

Starbucks launched Starbucks Rewards (2009), reorienting the entire business around the individual customer as the unit of analysis. Instead of tracking revenue per store, they began tracking visit frequency per member, spend per visit, and customer lifetime value by tier. By 2023, the programme had 30+ million active US members generating ~55% of US company-operated revenue. Every promotion, menu launch, and store design decision now runs through the lens of: does this increase frequency among our best customers?

LINK TO NEXT CONCEPT →

This is the shift from product-centric to customer-centric that the customer — not the SKU or the store — becomes the fundamental unit of analysis.

Two Approaches to Running a Firm

	Customer-Centric Firm	Product-Centric Firm
UNIT OF ANALYSIS	Customer is the fundamental unit of analysis	Product P&L is the fundamental unit of analysis
GROWTH ACCOUNTING	Acquisition, retention, and development at the core of growth	SKU expansion, product launches, and geographic penetration drive growth
DECISION LENS	Decisions made through the lens of long-term customer profitability (CLV)	Decisions driven by quarterly product sales and short-term margin
CUSTOMER TREATMENT	Recognises not all customers are equal; invests differentially	Treats all customers equally — same marketing, same service tiers

Adapted from Fader et al. (2022, p.6) and Kumar & Reinartz (2018, Ch 1 & 3)

How Organisation Reflects Strategy

Customer-Centric Organisation

- CRM vision that articulates the role of customer relationships
- Culture of customer orientation embedded in leadership
- Organisational processes aligned around customer needs
- Data & technology infrastructure supporting customer analytics
- Asks: "How many products can we develop for each customer?"

Product-Centric Organisation

- Strategy built around product lines and brand portfolios
- Success measured by units sold, market share, and product revenue
- Organised by product managers and category P&Ls
- No single owner of the customer relationship
- Asks: "How many customers can we get to buy our product?"

The pivot question: *Most firms say they are customer-centric but organise and measure like product-centric firms.*

Kumar & Reinartz (2018), Fig 3.2 CRM Strategy Elements, pp.36–38 | Fader et al. (2022), pp.5–8

CASE STUDY

Apple

"People don't know what they want until you show it to them. Our task is to read things that are not yet on the page."

— *Steve Jobs*

Product-Centric DNA

No focus groups for the iPhone, iPad, or Apple Watch. R&D and design intuition drove every product decision. The company was organised around product lines, not customer segments. The entire culture said: ***build something extraordinary and customers will come.***

But Even Apple Is Shifting...

Apple Services revenue hit ~$96B in FY2024 (~25% of total revenue). Apple One bundles, iCloud storage tiers, and the App Store ecosystem are ***customer-retention plays*** — designed to increase switching costs and lifetime value per user.

CASE STUDY

Hermès

"We don't have a marketing department. We have a creation department."

— Axel Dumas, CEO of Hermès

The Craftsman Model

A single Birkin bag takes 18–25 hours of hand-stitching by one artisan. Hermès deliberately limits production — not because they can't make more, but because scarcity IS the product. ***Customers don't choose Hermès; Hermès chooses its customers.*** Waitlists of 2–6 years are a feature, not a bug.

Why It Works

€13.4B revenue in FY2023
~40% operating margin
(highest in luxury, double LVMH average)

No discounts. No outlet stores. No influencer campaigns. The product's heritage, quality, and scarcity create a self-reinforcing moat that no amount of customer analytics could replicate.

So When Does Product-Centricity Break Down?

These companies succeed because they have an extraordinary product moat — breakthrough innovation, irreplaceable brand heritage, or deliberate scarcity. But most firms don't have that luxury.

- You have genuine innovation moat (Apple)
- Scarcity IS the product (Hermès)
- Switching costs are built into the product itself
- Customers can't comparison-shop easily
- Brand heritage takes decades to build

- You sell commoditised products (protein, CPG, fashion)
- Competitors are one click away (e-commerce, marketplaces)
- Growth depends on retention, not just acquisition
- You can't out-innovate 50 rivals simultaneously
- Your margin comes from repeat buyers, not new ones

From the Analytics Ladder to the Right Questions

Why we are pivoting away from the 5-stage maturity model

Sophistication is a tool, not a strategy.

Davenport & Harris's maturity ladder describes capability. Bruce & Fader remind us that capability without a customer-as-unit-of-analysis mindset just gets you a more expensive version of product centricity.

Customer-centricity starts with descriptive clarity about who your customers are — and only then earns the right to climb the ladder.

So before we ask 'what model should we build?', what should we be asking?

Enter the Customer-Base Audit

The deliberate, descriptive first step on the journey to customer centricity

Why it comes first

- Sophistication amplifies whatever question you point it at — including the wrong one.
- The audit forces senior leaders to look at the firm through a customer lens, often for the first time.
- It exposes customer heterogeneity — the empirical bedrock of the customer-centric case.
- Only after the audit do CLV, churn, uplift and recommender models earn the right to be built.

Source: Bruce, Fader & Ross, The Customer-Base Audit (2022), Introduction & Conclusion.

Why We Need a New View of the Data

Most reporting hides the customer behind the product

What most firms see

- **Product × Time reports.** Rows = SKUs or product lines. Columns = months/quarters/years.
- **Aggregate revenue & margin.** Total sales by product, growth vs. last period, top sellers.
- **Useful for operations.** Inventory, supply chain, category management.
- **Blind to the customer.** Reveals nothing about who is buying, how often, or whether the customer base is healthy.

What they should see

- Customer × Time view: rows = customers, columns = time periods.
- Reveals heterogeneity — not all customers are equal.
- Surfaces acquisition, retention and development as the real engines of growth.
- Built from data the firm already has — no new collection required.
- The starting point for any customer-centric strategy.

A customer-base audit is…

WHAT IT IS NOT	WHAT IT IS
Demographics Age, income, location profiles	**Actual buying behaviour** What customers do, not what they say
Attitudes & surveys NPS scores, satisfaction ratings	**Transaction-level analysis** Orders, frequency, spend, retention
Market research Focus groups, brand perception studies	**Heterogeneity revealed** How customers differ from each other
"Knowing the customer" Personas built on psychographics	**Longitudinal patterns** How behaviour evolves over time

*"We are interested in understanding their **actual buying behaviour**."*

The Data Cube — Three Dimensions of Every Transaction

Every transaction your business has ever recorded can be described by just three things

WHO

Customer

Which individual, identifiable customer made this transaction?

WHEN

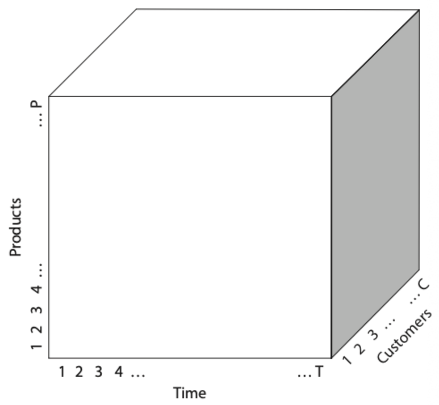
Time

In which calendar period (day, month, quarter, year) did it happen?

WHAT

Product

Which SKU or service was actually purchased?



"Pivot the orientation of the data cube so the primary focus moves to the customer × time face."

This seemingly simple shift in orientation opens up a whole new set of ways to think about your firm's revenues and profits — and is the foundation of every analysis in the customer-base audit.

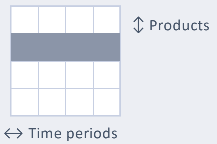
THE DATA CUBE

Three Faces, Three Conversations

Same transaction database — three different ways to look at it. Each one drives a different question.

LEGACY VIEW

Product × Time

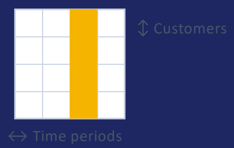


Asks: "What sold this quarter?"

- Sums OVER the customer dimension.
- Standard P&L, dashboard, category reports.
- Built for product managers and operations.

AUDIT VIEW

Customer × Time

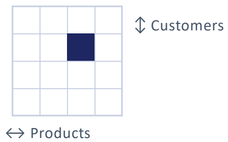


Asks: "How healthy is our customer base?"

› Sums OVER the product dimension.
› Reveals heterogeneity, retention, cohorts.
› The five lenses live here.

NEXT PIVOT

Customer × Product



Asks: "Is the mix of what they buy healthy?"

- Sums OVER the time dimension.
- Cross-buying

Source: Bruce, Fader & Ross, The Customer-Base Audit (Wharton School Press, 2023), Ch. 2 (Faces 1 & 2) and Ch. 8 (cross-buying / Face 3).

Most companies live on Face 1. The audit pivots to Face 2. Face 3 is where you go once you understand who your customers are.

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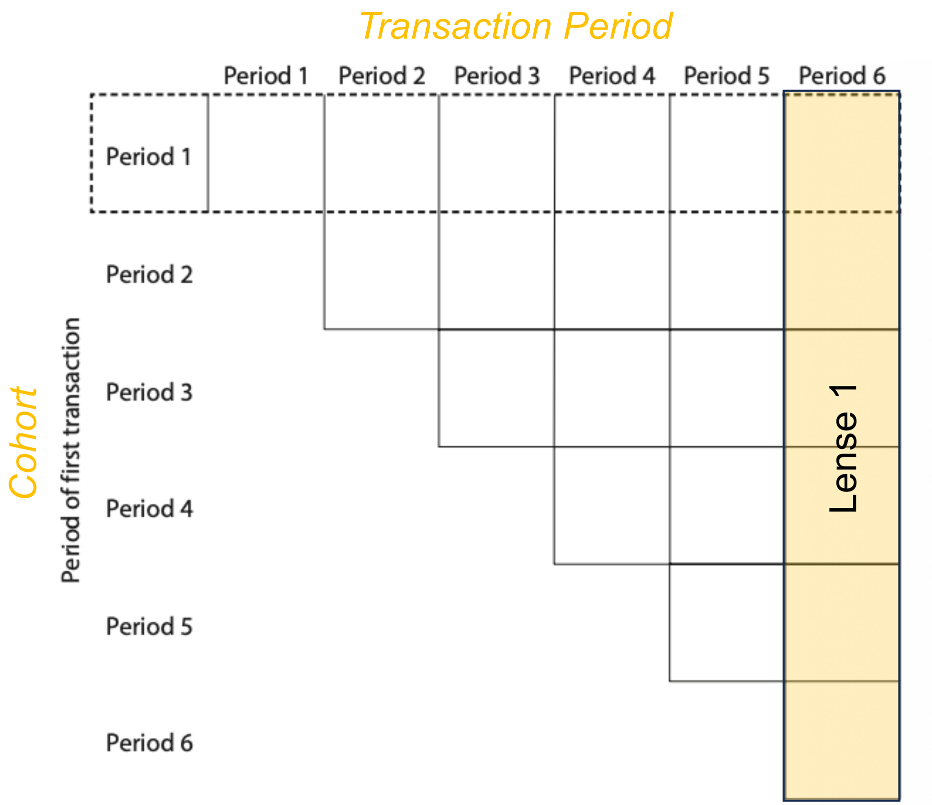
The Five Lenses at a Glance

Each lens answers a different question about your customer base

Lens 1 *One period, all active customers*	How different are our customers from each other right now? Who drives most of the revenue this period? (Heterogeneity)
Lens 2 *Two adjacent periods*	Why did the top-line move? Was it new customers, lost customers, or changed spend among repeats? (Period-on-period decomposition)
Lens 3 *One cohort tracked over time*	How does the behaviour of customers we acquired together evolve as they age? (Retention and decay curves)
Lens 4 *Two cohorts compared at the same age*	Are the customers we are acquiring now better or worse than the ones we acquired before? (Acquisition quality)
Lens 5 *The whole customer base, all cohorts and periods*	Looking at everything together, how healthy is our customer base — and what does that imply for future revenue?



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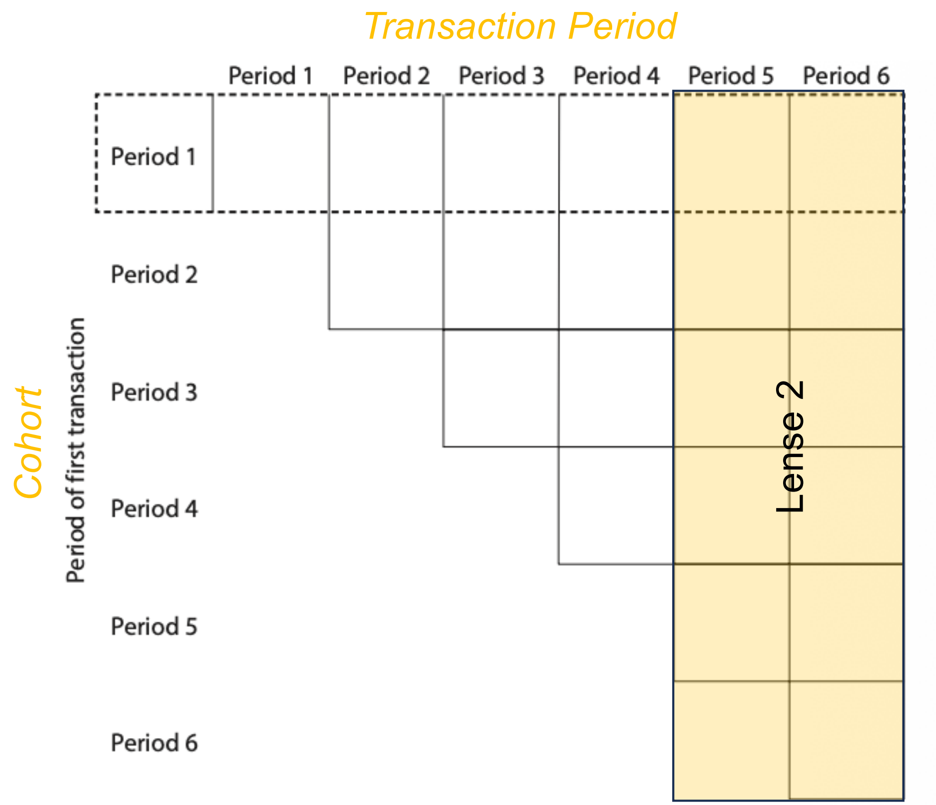

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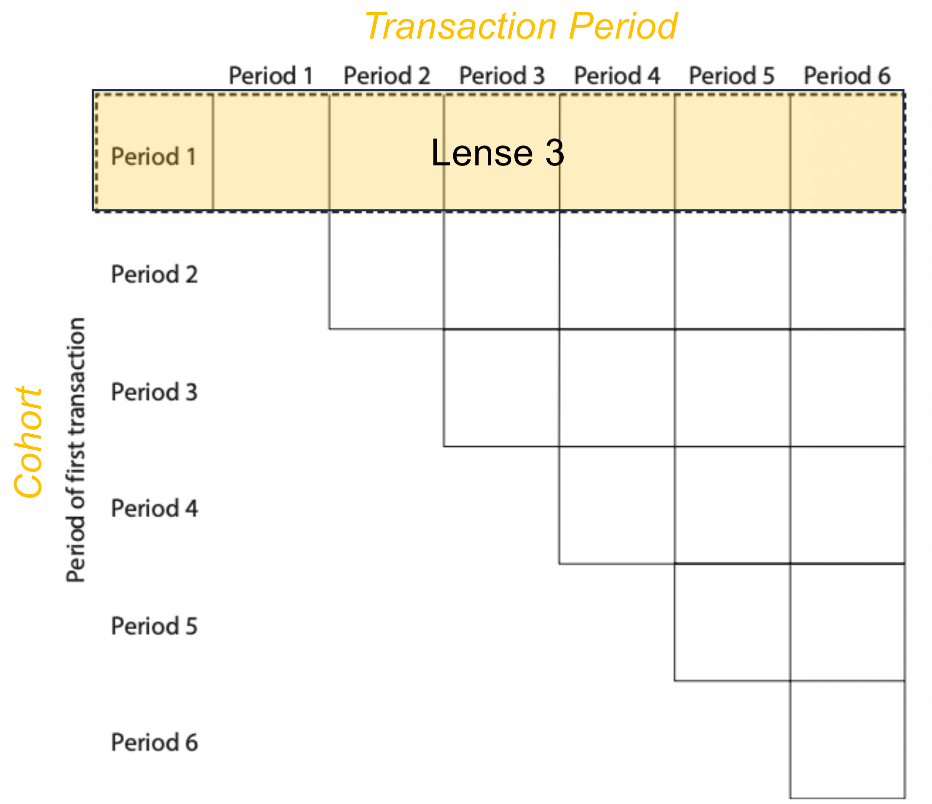

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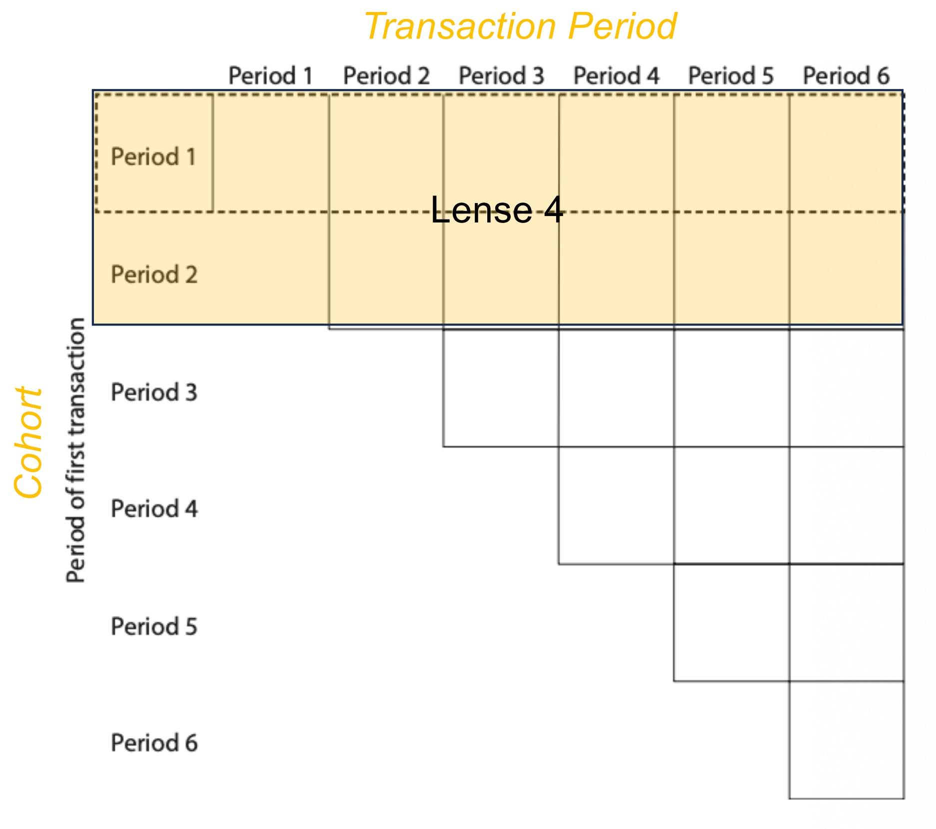

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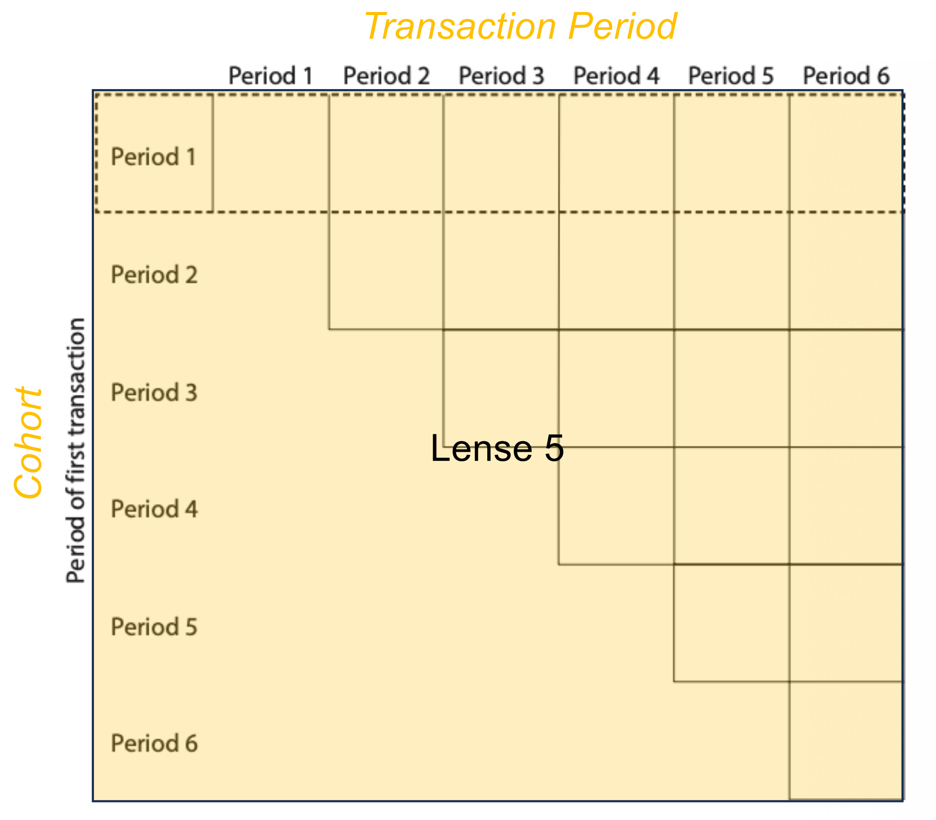


Introducing the Five Lenses

Five different ways to slice the customer × time face

	Period (vertical slice)	Cohort (horizontal slice)
One slice	**Lens 1** Single period — all active customers in it *"Who is shopping with us right now, and how different are they from each other?"*	**Lens 3** Single cohort — tracked over time *"How does the behaviour of customers we acquired in Q1 2022 evolve as they age?"*
Two slices	**Lens 2** Two adjacent periods — what changed? *"What changed between 2023 and 2024, and was it because of new customers, lost customers, or changed spend?"*	**Lens 4** Two cohorts compared at the same age *"Are the customers we acquired in 2023 actually better or worse than the ones we acquired in 2022?"*
Three+	**Lens 5** The whole customer × time face — many cohorts, many periods *"Looking at the whole customer base over time, how healthy is it as a system, and what does that imply about future revenue?"*	

Adapted from Bruce, Fader & Ross, The Customer-Base Audit (2022), Figure 2.9.

Why This Framework Matters

From a vocabulary for analysis to a foundation for customer centricity

A shared vocabulary

Every analysis in the rest of the customer-base audit can be described as "a Lens 1 analysis," "a Lens 4 analysis," and so on. It gives a senior team a common language.

Exposes how partial the standard view is

Most management teams have never seen a Lens 3 or Lens 4 chart of their own firm. Rotating the cube changes the conversation.

Unashamedly descriptive

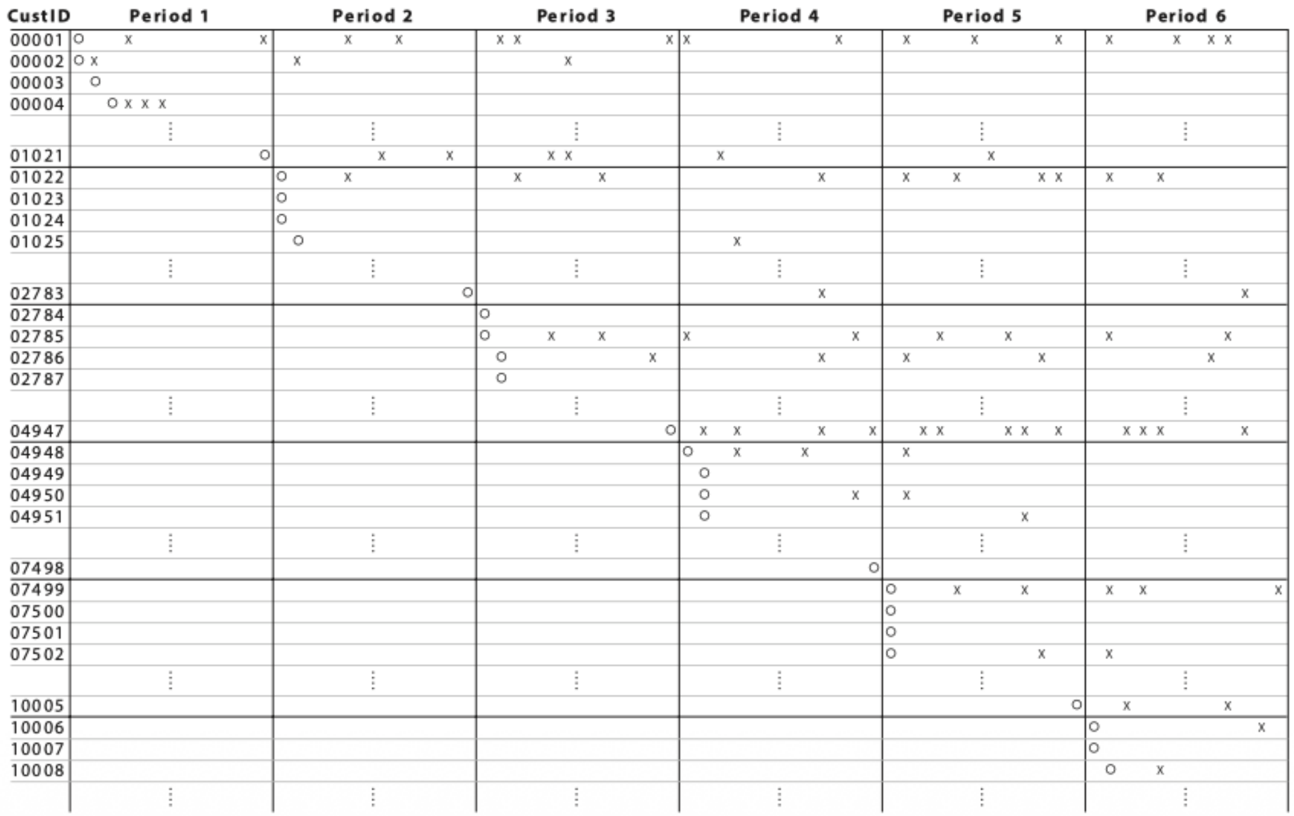
No models. No machine learning. Just a disciplined way of looking at data the firm already has — the prerequisite for any predictive work.

The starting point of customer centricity

Once you can see the customer × time face through all five lenses, the right questions about CLV, retention and acquisition begin to ask themselves.

Source: Bruce, Fader & Ross, The Customer-Base Audit (2022), Chapter 2 — "The Data Cube and the Five Lenses."

Exercise: Given a Table containing CustID and Transaction Period, what are some of the questions we can ask?



Customer Transaction Over Time

Why the chart is a triangle, not a rectangle

A customer cannot transact before they are acquired. If your first ever purchase was in Period 3, you simply do not exist as a customer in Periods 1 or 2. So the grid staircases down from top-left to bottom-right — a lower-right triangle.

Mental model: imagine an attendance register. Rows = students grouped by year of enrolment. Columns = school terms. A tick = "this student showed up this term." That is exactly the customer × time face.

Meet Madrigal

A FICTITIOUS COMPANY — FADER, HARDIE & ROSS (2022)

The Company

Madrigal is a direct-to-consumer (DTC) online retailer that sells consumer packaged goods — think household essentials, personal care, and similar repeat-purchase products shipped directly to customers.

The Data

7 years of transaction records covering ~25,000 customers across 7 annual acquisition cohorts. Each record captures who bought, when, and how much they spent — nothing more.

The Problem

Revenue has been growing, but the CEO isn't sure if this is healthy growth. Are we just adding more customers, or are existing ones spending more? Is the growth coming from a few whales — or the broad base? Are newer cohorts performing like older ones, or worse?

The Five Questions

That lead us to the five-lens audit:

- Lens 1 — How different are your customers?
- Lens 2 — What changed since last period? (two periods)
- Lens 3 — How does customer behaviour evolve? (single cohort over time)
- Lens 4 — Comparing cohorts (is acquisition quality shifting?)
- Lens 5 — How healthy is our customer base? (integrating all of the above)

Madrigal's data is designed so every lens yields a clear, teachable insight.

1. Set up your Python Development Environment

2. By end of Week 2, form your groups - submit your group members via email to

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SMU SINGAPORE MANAGEMENT UNIVERSITY | School of Computing and Information Systems

Lesson 2:
Preparing Data for Customer Analytics

LEARNING OUTCOMES

What You'll Leave Lesson 2 With

Four outcomes to carry into the rest of the course — and into your sponsor project.

1

Map the consumer data landscape

Match data sources to different business use cases — and know which source to reach for when.

2

Build the pipeline the right way

Apply the medallion architecture (bronze → silver → gold) with source-to-target mapping (STTM) and quality checks at each layer.

3

Run Data Quality Check

Profile a real customer dataset and identify data related issues

4

Meet a live industry brief

Connect lesson concepts to practice via the Lush Protein sponsor presentation.

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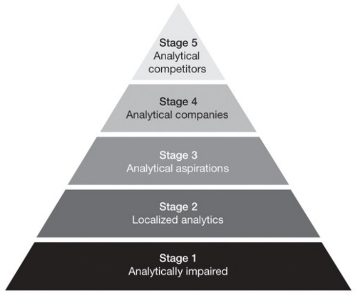
Recap of Lesson 1

The Analytics Maturity Ladder — "Sophisticated Equals Better"

- Four types of analytics capability *(Davenport & Harris, widely cited)*:
  - **Descriptive** — *What is happening / what has happened?*
  - **Diagnostic** — *Why did it happen?*
  - **Predictive** — *What will happen next?*
  - **Prescriptive** — *What should we do about it?*
- The conventional belief: firms must *climb the ladder* from descriptive to prescriptive
- Value is assumed to increase with sophistication — the fancier the model, the bigger the payoff
- Reinforced by the current hype around machine learning and AI: *"sophisticated equals better"*
- Implication for practice: jump straight to predictive models, churn scoring, CLV forecasting, recommender systems

The five stages of analytical capability



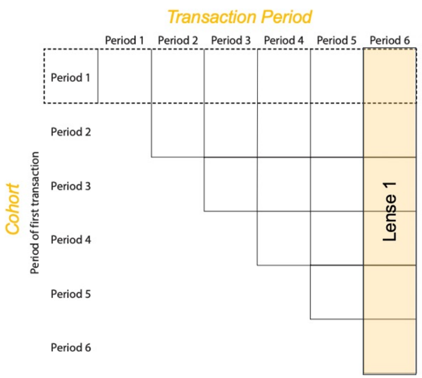
	Distinctive capability/level of insights	Questions asked	Objective	Perceived value
Stage 1 Analytically impaired	Negligible; "flying blind"	What happened in our business?	Get accurate data to improve operations	None
Stage 2 Localized analytics	Local and opportunistic; may not be supporting company's distinctive capabilities	What can we do to improve this activity? How can we understand our business better?	Use analytics to improve one or more functional activities	ROI of individual applications
Stage 3 Analytical aspirations	Begin efforts for more integrated data and analytics	What's happening now? Can we extrapolate existing trends?	Use analytics to improve a distinctive capability	Future performance and market value
Stage 4 Analytical companies	Enterprise-wide perspective; able to use analytics for point advantage; know what to do to get to next level, but not quite there	How can we use analytics to innovate and differentiate?	Build broad analytic capability—analytics for differentiation	Analytics are an important driver of performance and value
Stage 5 Analytical competitors	Enterprise-wide, big results, sustainable advantage	What's next? What's possible? How do we stay ahead?	Analytical master—fully competing on analytics	Analytics are a primary driver of performance and value

Davenport, T. H., & Harris, J. G. (2017). *Competing on Analytics: The New Science of Winning* (Updated ed., Ch. 2). Harvard Business Review Press.

The Five Lenses at a Glance

Each lens answers a different question about your customer base

Lens 1 *One period, all active customers*	How different are our customers from each other right now? Who drives most of the revenue this period? (Heterogeneity)
Lens 2 *Two adjacent periods*	Why did the top-line move? Was it new customers, lost customers, or changed spend among repeats? (Period-on-period decomposition)
Lens 3 *One cohort tracked over time*	How does the behaviour of customers we acquired together evolve as they age? (Retention and decay curves)
Lens 4 *Two cohorts compared at the same age*	Are the customers we are acquiring now better or worse than the ones we acquired before? (Acquisition quality)
Lens 5 *The whole customer base, all cohorts and periods*	Looking at everything together, how healthy is our customer base — and what does that imply for future revenue?



Why This Framework Matters

From a vocabulary for analysis to a foundation for customer centricity

A shared vocabulary

Every analysis in the rest of the customer-base audit can be described as "a Lens 1 analysis," "a Lens 4 analysis," and so on. It gives a senior team a common language.

Exposes how partial the standard view is

Most management teams have never seen a Lens 3 or Lens 4 chart of their own firm. Rotating the cube changes the conversation.

Unashamedly descriptive

No models. No machine learning. Just a disciplined way of looking at data the firm already has — the prerequisite for any predictive work.

The starting point of customer centricity

Once you can see the customer × time face through all five lenses, the right questions about CLV, retention and acquisition begin to ask themselves.

GROUP DISCUSSION

Where Would You Get the Data?

Each group receives a different business scenario. You have 15 minutes.

1 What questions do you need to answer?
5 min

2 What data do you need — and where would you get it?
7 min

3 What can't you measure — and what would you do?
3 min

Your Scenario:

- A — FMCG Multinational
- B — Small Retailer
- C — DTC Brand

SCENARIO A

FMCG Multinational — Hair Care Division

THE BRIEF

Your shampoo brand has dropped from #2 to #4 in market share in Southeast Asia over the past 12 months. The regional VP wants to know why — and what to do about it. You sell through supermarkets, pharmacies, and convenience stores. You don't sell online directly.

Discussion Prompts

1. What's happening in the overall category — is it growing or shrinking? Where would you find out?
2. Are you losing share to specific competitors, or to everyone? How would you know?
3. Is the problem pricing, distribution (fewer shelves), or advertising that isn't working?
4. Where would you get competitor data if you can't see their internal systems?
5. How would you measure whether your TV and digital advertising is actually driving sales?

Hint: Think about what data you own vs. what data you'd need to buy.

SCENARIO B

Small Retailer — A 12-Outlet Bakery Chain

THE BRIEF

You run analytics for a local bakery chain with 12 stores across Singapore. Sales have been flat for 6 months despite opening 2 new outlets. The founder suspects customers are visiting less often, but isn't sure. She wants to know: are we losing customers, or are existing customers spending less?

Discussion Prompts

1. What data do you already have from your own systems (POS, loyalty app, delivery platforms)?
2. How would you identify individual customers if many pay by cash or GrabPay without a loyalty card?
3. Can you tell the difference between "customer stopped coming" vs. "customer switched to a different outlet"?
4. What external data might help you understand foot traffic or neighbourhood demographics?
5. How would you measure the impact of a new menu item or a promotion?

Hint: Think about the gap between "transactions you can see" and "customers you can identify."

SCENARIO C

DTC Brand — Online Protein Supplements

THE BRIEF

You're the data analyst at a DTC protein supplement brand. You sell through your own website (Shopify), Lazada, and Shopee. Growth has slowed — customer acquisition cost is rising, and the founder suspects repeat purchase rates are dropping. She asks: "Are we acquiring the wrong customers, or failing to retain the right ones?"

Discussion Prompts

1. What data do you get from Shopify vs. Lazada vs. Shopee — and can you link a customer across platforms?
2. How would you identify your most valuable customers vs. one-time buyers?
3. What data would you need to understand why customers aren't coming back?
4. Can you see what competitors are doing on the same marketplaces?
5. How would you measure whether your Google/Meta ads are bringing in high-value or low-value customers?

Hint: You're data-rich on your own customers but data-blind on the market. What's missing?

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Your Scenario:

A — FMCG Multinational | B — Small Retailer | C — DTC Brand

DEBRIEF

What the Industry Actually Uses

The data sources and analytics studies behind the questions you just asked.

Placeholder for discussion slides

From Source to Analytics

How raw data becomes analytics-ready:

Source-to-Target Mapping, Medallion Architecture, and Data Quality

Source-to-Target Mapping

Medallion Architecture

Data Quality Checks

GOOD PRACTICES

Source-to-Target Mapping (STTM)

An STTM is a document that defines how each field in your source systems maps to fields in your target analytics tables. It is the contract between the data engineer who builds the pipeline and the analyst who consumes the output.

Example: Shopify → Customer Signature

Source System	Source Table	Source Field	Target Table	Target Field	Transformation
Shopify	`orders`	`created_at`	`customer_signature`	`first_purchase_date`	MIN(created_at) per customer_id
Shopify	`orders`	`total_price`	`customer_signature`	`total_revenue`	SUM(total_price) per customer_id
Shopify	`orders`	`id (count)`	`customer_signature`	`order_count`	COUNT(id) per customer_id
Shopify	`order_line_items`	`product_type`	`customer_signature`	`top_category`	MODE(product_type) per customer_id
Google Ads	`ad_clicks`	`utm_source`	`customer_signature`	`acquisition_channel`	First-touch attribution from earliest click

Without an STTM, two analysts building the same metric from the same source data can get different answers — and neither knows who is right.

BryteFlow (2024), "Source to Target Mapping Guide

STTM Good Practices

Version Your STTM

Treat the mapping document like code. Store it in Git or a shared drive with change history. Every pipeline change should update the STTM first.

One Source of Truth

Every target field should have exactly one mapping. If two sources can populate the same field, document the priority rule (e.g., CRM overrides Shopify).

Include Transformations

Don't just map field names — specify the logic. "SUM(amount) WHERE status = 'completed' GROUP BY customer_id" leaves no ambiguity.

Document Assumptions

Record grain (one row per customer? per order?), filters applied, how NULLs are handled, and timezone conventions.

Flag Data Gaps

If a target field has no source (e.g., "NPS score" when no survey exists), mark it explicitly. Missing data is not the same as zero.

Validate With Stakeholders

Walk through the STTM with the business user before building the pipeline. Their definition of "active customer" might differ from yours.

DATA ARCHITECTURE

The Medallion Architecture

A three-layer pattern for progressively refining raw data into analytics-ready tables. Used by Databricks, Snowflake, and modern data teams worldwide.

BRONZE
Raw / Landing

Exact copy of source data. No transformations applied. Append-only ingestion with load timestamps. Schema matches source system 1:1.

Examples: Shopify orders JSON, Google Ads CSV exports,

Data Quality

Duplicates, nulls, inconsistent formats — all preserved as-is

▼

SILVER
Cleaned / Conformed

Deduplication, type casting, NULL handling, and schema enforcement applied. Common identifiers resolved (e.g., customer_id matched across sources).

Examples: Deduplicated orders table, unified customer_id, standardised timestamps

Data Quality

Validated schemas, referential integrity checks, business rule filters

▼

GOLD
Business-Ready / Curated

Aggregated, enriched, and modelled for specific use cases. This is the customer signature — the table analysts query directly.

Examples: customer_signature, product_performance, cohort_metrics

Data Quality

Business-validated, documented, tested, ready for dashboards and models

The medallion pattern maps directly to STTM: bronze preserves source fields, silver applies transformations, gold produces target fields.

https://www.databricks.com/blog/what-is-medallion-architecture

Medallion in Action: E-Commerce Pipeline

orders.json	*Raw JSON*
order_id: "#1234"	
created_at: "2024-03-15T..."	
total_price: "89.90"	*← string!*
customer.email:	
"John@Gmail.COM"	*← messy*
line_items: [...]	
Duplicates exist	⚠
NULLs preserved	⚠

→

orders_clean	*Table*
order_id: INT 1234	*✓typed*
created_at: TIMESTAMP	*✓parsed*
total_price: DECIMAL 89.90	*✓cast*
customer_email:	
john@gmail.com	*✓normalised*
customer_id: FK	*✓resolved*
Deduplicated	✓
Schema enforced	✓

→

customer_signature	*View*
customer_id: 42	
first_purchase: 2024-01-12	
total_revenue: $347.60	
order_count: 4	
avg_order_value: $86.90	
top_category: protein	
acq_channel: google_ads	
days_since_last: 18	
segment: "active"	

Each layer adds value: Bronze preserves the raw record. Silver fixes quality issues. Gold answers business questions.

DATA QUALITY

Data Quality (DQ) Checks

"Garbage in, garbage out" is not just a cliché — it's the #1 reason analytics projects fail. DQ checks are the gates between medallion layers.

DQ Dimension	What It Checks	Example Check	When to Apply
Completeness	Are required fields populated?	`customer_id IS NOT NULL`	Bronze → Silver
Uniqueness	Are there duplicate records?	`COUNT(order_id) = COUNT(DISTINCT order_id)`	Bronze → Silver
Validity	Do values fall within acceptable ranges?	`total_price >= 0 AND order_date <= CURRENT_DATE`	Silver
Consistency	Do related fields agree?	`SUM(line_items.price) ≈ order.total_price`	Silver → Gold
Timeliness	Is data arriving on schedule?	`MAX(load_ts) >= CURRENT_DATE - INTERVAL 1 DAY`	Bronze (ingestion)
Accuracy	Does data reflect reality?	`Revenue in Gold = Revenue in source system report`	Gold (reconciliation)

These six dimensions come from the DAMA Data Management Body of Knowledge (DMBOK). They apply regardless of infrastructure — SQL tables, lakehouses, or event streams.

DQ Checks: Good Practices

DO

Automate checks in your pipeline

Use tools like Great Expectations, dbt tests, or Soda Core to run DQ checks as part of every data refresh — not manually.

Set thresholds, not just pass/fail

"< 1% NULLs in email" is more useful than "no NULLs" — real data is messy, and zero-tolerance rules generate noise.

Alert, don't silently drop

When a check fails, log it and notify the team. Never silently remove rows — that hides problems downstream.

Test at each medallion layer

Bronze: row counts match source. Silver: no duplicates, types correct. Gold: business metrics reconcile with source-of-truth reports.

Document your DQ rules in the STTM

Every transformation in the STTM should have a corresponding DQ check. If you map total_price, check it's >= 0.

DON'T

Check only at the end

If bad data enters at Bronze and you only check at Gold, you've wasted all transformation compute on garbage data.

Rely on "the source is always right"

Source systems have bugs too. An order with negative quantity or a future-dated transaction is a source error you must catch.

Use DQ checks as a one-time exercise

Data quality degrades continuously. Sources change schemas, new edge cases appear. DQ must run on every pipeline execution.

Ignore row-count reconciliation

If Bronze has 10,000 rows and Gold has 9,500 — where did 500 rows go? Track row counts at each layer.

Build a 100-check suite on day one

Start with 5-10 critical checks (NULLs in key fields, duplicate PKs, row counts). Expand as you learn the data's failure modes.

Great Expectations (2017), greatexpectations.io

Putting It All Together

STTM Define the mapping → **BRONZE** Ingest raw data → **DQ** → **SILVER** Clean & conform → **DQ** → **GOLD** Aggregate

▼

CUSTOMER SIGNATURE — The analytics-ready, one-row-per-customer table

STTM is your blueprint

Before writing any code, map every source field to its target. The STTM is the single document that prevents "my numbers don't match" conversations.

Medallion is your assembly line

Bronze → Silver → Gold gives structure to your pipeline. Each layer has a clear purpose and clear DQ gates. Don't skip layers.

DQ is your quality control

Automated checks at each medallion boundary catch problems before they propagate. Start simple (5-10 rules), expand as the data teaches you where it breaks.

Session 2

Preparing Data for Customer Analytics

Agenda

Section 1: Data Quality Challenges

Why data preprocessing matters – common data quality problems

Section 2: Cleaning & Fixing Data

Missing values, duplicates, outliers and transformations

Section 3: Data Preparation Techniques

Scaling, binning, feature engineering, reduction and sampling

Section 4: Customer Data Structure

Customer signature – what it is and how to build it

Section 5: Analytics Foundations

Visualization, Supervised vs unsupervised, parametric/non-parametric techniques

Learning Outcomes

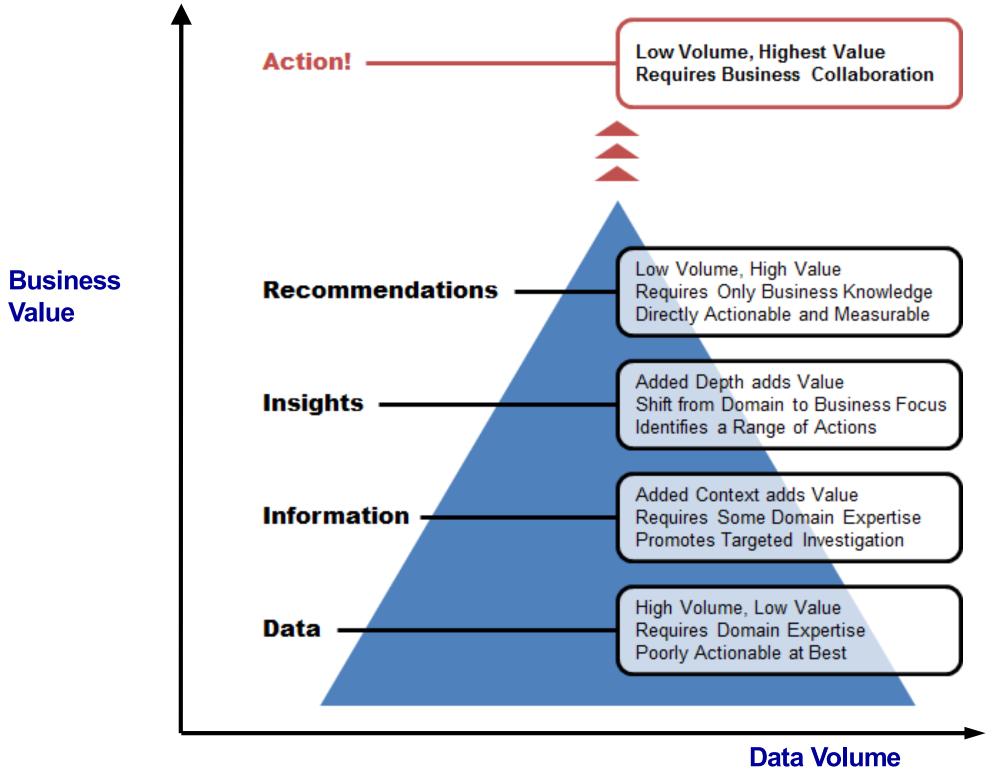
Upon completion of this session, students will:

GET AN INTRODUCTION TO REAL WORLD CASE STUDY ON LUSH PROTEIN

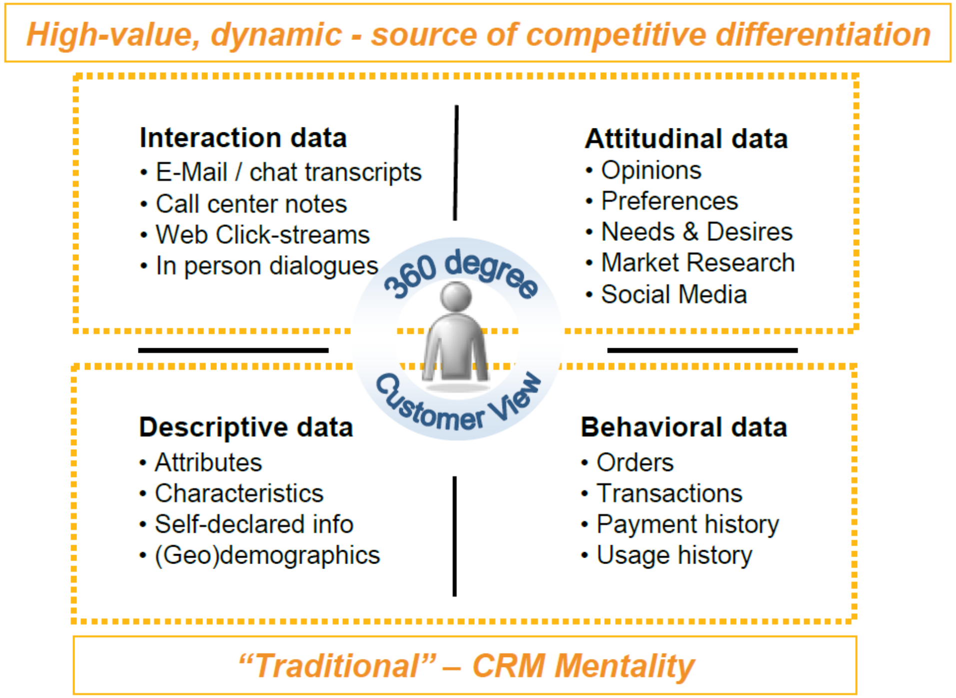
UNDERSTAND COMMON DATA ISSUES AND PROBLEMS

LEARN DATA PREPARATION TECHNIQUES FOR CUSTOMER ANALYTICS

From Customer Data to Value

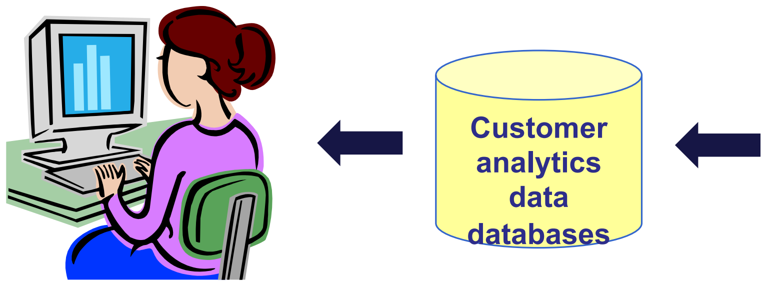


Data at the heart of customer analytics

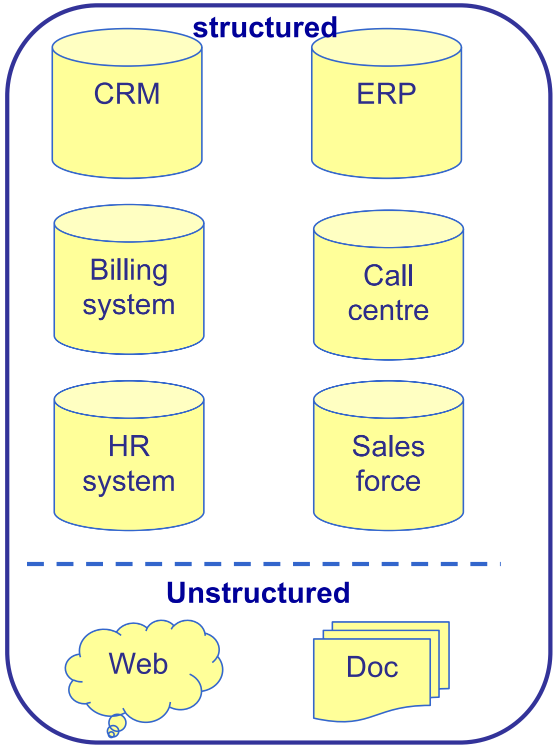


A Central Database for All Needs

- The key is data integration.



One version of the truth



Data sources for customer analytics

Section 1

Data Quality Challenges

Why preprocessing matters – common data quality problems

Why Data Preprocessing?

Data in the real world is dirty

- incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
- noisy: containing errors or outliers
- inconsistent: containing discrepancies in codes or names

No quality data, no quality mining results!

- Quality decisions must be based on quality data
- Data warehouse needs consistent integration of quality data

Types of Raw Data Quality Problems

Data imperfectly describes the features of the real world e.g.

- Data might be missing or empty
- Samples might not be representative
- Categorical variables might have too many values
- Numerical may have unusual distributions and outliers
- Data might be coded inconsistently

Types of Raw Data Quality Problems

Errors, Outliers, and Missing

cking	#cking	ADB	NSF	dirdep	SVG	bal
Y	1	468.11	1	1876	Y	1208
Y	1	68.75	0	0	Y	0
Y	1	212.04	0	6		0
	.	.	0	0	Y	4301
y	2	585.05	0	7218	Y	234
Y	1	47.69	2	1256		238
Y	1	4687.7	0	0		0
	.	.	1	0	Y	1208
Y	.	.	.	1598		0
	1	0.00	0	0		0
Y	3	89981.12	0	0	Y	45662
Y	2	585.05	0	7218	Y	234

Common Data Quality Issues

Issue Type	Impact	Solution
Missing Values	Reduces sample size, introduces bias	Impute or Drop
Duplicates	Skews statistics	Tag and Remove
Outliers	Distorts means / regressions	Cap, or transform or drop
Inconsistent	Cases errors	standardize

Data Cleaning - Benefits

- Data scientists often spend **60-80% of their time** on data preprocessing
- Data cleaning is critical first step that directly impacts the quality of insights

Source: Lecture notes at Department of Statistics, Northwestern University**.**
https://lizhen0909.github.io/nu-stat303-1-sec20-coursebook

Section 2

Cleaning & Fixing Data

Missing values, duplicates, outliers and transformations

Types of Missing Data

- Understanding the type of missing data helps in selecting the right imputation method and mitigating potential biases in the analysis.

Missing Completely at Random (MCAR)

Missing at Random (MAR)

Missing Not at Random (MNAR)

Options for missing value imputation

Options	Actions
Throw out the records with missing values	*No. This creates a bias for the sample.*
Replace missing values with a “special” value (-99)?	Yes. This action is possible when data is missing for a reason, such as insufficient history
Replace with a suitable value	Maybe. Model the imputed value using domain knowledge.
Replace with a “typical” value?	*Maybe. Replacement with the mean, median, or mode changes the distribution, but predictions*
Use data mining techniques that can handle missing values?	Yes. One of these is decision trees
Partition records and build multiple models?	Yes. This action is possible when data is missing for a reason, such as insufficient history

Duplicates

Check	Display	Drop
Check how many duplicate rows exist • The duplicated() method returns a boolean Series indicating whether each row is a duplicate	Display duplicates • filter the DataFrame using the boolean mask returned by duplicated(). To see all occurrences of duplicate rows (including the first occurrence), use keep=False:	Drop duplicates

Outliers – how to identify

An outlier is an observation that is significantly different from the rest of the data.

- Descriptive stats
  (min/max/median)
- Visualization

  e.g. Boxplots, Histograms, Scatterplot
- Statistical

Outliers – How to Handle

- Removing extreme values
- Capping or Flooring
- Replacing with the median
- Transforming Data – e.g log, square root

Section 3

Data Preparation Techniques

Scaling, binning, feature engineering, reduction and sampling

Data – Attribute Types

Nominal

- Examples: ID numbers, eye color, zip codes

Ordinal

- Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}

Interval

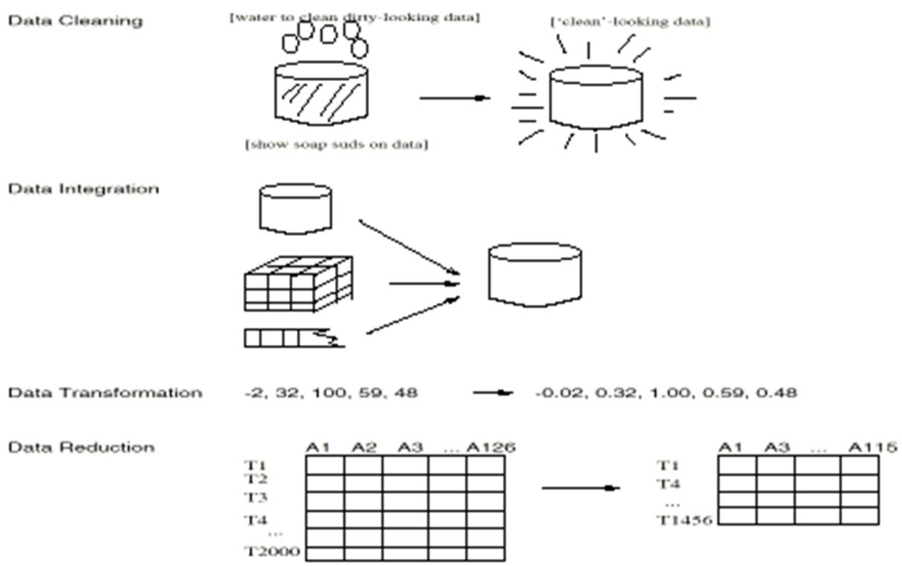
- Examples: calendar dates

Ratio

- Examples: length, time, counts

Source: https://www.iitr.ac.in/media/facspace/patelfec/16Bit/slides/Lecture-2-Data-Preprocessing-Part-1.pdf

Forms of data preprocessing



Source: Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann Publishers. (Slides for textbook, Chapter 3)

Why Data Preparation Matters?

Challenge	Issue	Solution
Information Loss	Continuous data hides patterns	Bin
Categorical Data	Algorithms need #	Dummy Variables
Scale Difference	e.g. Income vs Age have diff magnitude	Standardize
Model Performance	Bias or Inaccurate	Transformations

Example of Binning

$$0 \leq \text{Age} < \infty$$

Bin	Age Range
1	(0, 20]
2	(20, 50]
3	(50, 70]
4	(greater than 70)

Feature Scaling

- **Common scaling methods:**
- **Min-Max Scaling:** Rescales features to a fixed range, usually [0, 1].
- **Standardization (Z-score):** Centers features by removing the mean and scaling to unit variance

Scaling ensures that numerical features contribute equally to model training and prevents features with larger ranges from dominating those with smaller ranges.

Many algorithms (e.g., k-nearest neighbors, k-means, principal component analysis, gradient descent-based models) compute distances or rely on feature magnitude.

Features with larger scales can unduly influence the model if not scaled.

When and What to Scale

Always scale continuous numerical features before using algorithms sensitive to feature magnitude (e.g., k-means, k-NN, PCA, neural networks).

Never scale categorical variables or their encoded versions (label-encoded or one-hot/dummy variables). Scaling these destroys their categorical meaning and can confuse models.

For tree-based models (e.g., decision trees, random forests), scaling is not required, as these models are insensitive to feature scale.

Feature Transformation

Transformations stabilize variances, remove nonlinearity, and correct nonnormality in inputs to improve the fit of the model.

Interval Inputs

Mathematical Functions
- Centering
- Exponential
- Inverse
- Inverse Square
- Inverse Square Root
- Log_e / Log_{10}
- Range Standardization
- Square
- Square Root
- Standardize

Binning
- Bucket
- Quantile
- Tree-Based Binning

Best

Class Inputs

- Bin Rare Nominal Levels
- Level Encoding
- Level Frequency Encoding
- Level Proportion Encoding
- Target Encoding
- WOE Encoding

Feature Transformation – make the data mean more

- Single-variable transformations
  - Standardizing numeric variables
    - Centering and rescaling
  - Turning numeric values into percentiles
  - Turning counts into rates
  - Relative measures
  - Replacing categorical variables with numeric ones

Derived features: making the data mean more

Rolling up information to Customer level

donor_id	pay_mode	pay_freq	pay_mth	pay_year	date_deducted	donate_amt
00451600A	CPF Payroll	1	2	2009	2009-03-05	8
00469400K	CPF Payroll	1	11	2009	2009-12-08	19
00499100A	CPF Payroll	1	11	2009	2009-12-11	2
01302400X	CPF Payroll	1	8	2009	2009-08-17	11
02076000X	CPF Payroll	1	10	2009	2009-12-07	5
02906800B	CPF Payroll	1	4	2009	2009-05-05	5
02906800B	CPF Payroll	1	10	2009	2009-11-04	5
02906800B	CPF Payroll	1	5	2009	2009-06-09	5
02906800B	CPF Payroll	1	6	2009	2009-07-10	5
02906800B	CPF Payroll	1	8	2009	2009-09-03	5
02906800B	CPF Payroll	1	7	2009	2009-08-11	5
02906800B	CPF Payroll	1	2	2009	2009-03-10	5
02906800B	CPF Payroll	1	9	2009	2009-10-05	5
02906800B	CPF Payroll	1	3	2009	2009-04-13	5
02906800B	CPF Payroll	1	1	2009	2009-02-05	5
02906800B	CPF Payroll	1	12	2009	2010-01-14	5
02906800B	CPF Payroll	1	11	2009	2009-12-14	5
03891200K	CPF Payroll	1	6	2009	2009-07-10	5
03908400X	CPF Payroll	1	9	2009	2009-10-14	13
04374700X	CPF Payroll	1	3	2009	2009-04-17	80

donor_id	N(pay_freq)	N(pay_mth)	Mean(donate_amt...	Sum(donate_amt)
00451600A	1	1	8	8
00469400K	1	1	19	19
00499100A	1	1	2	2
01302400X	1	1	11	11
02076000X	1	1	5	5
02906800B	12	12	5	60
03891200K	1	1	5	5
03908400X	1	1	13	13
04374700X	12	12	83.333333	1000
05179100E	1	1	4	4

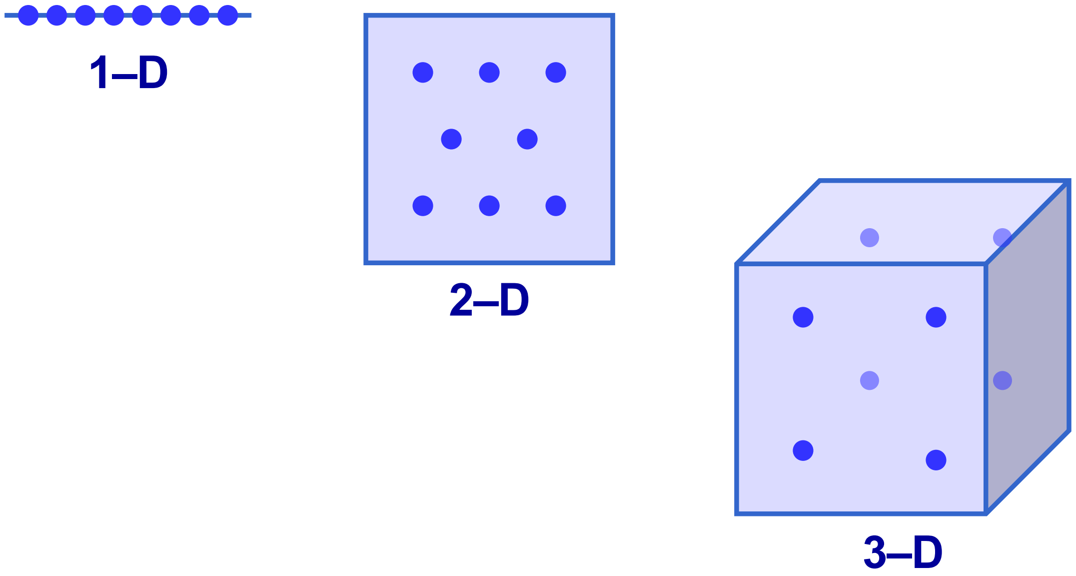
Rolling Up Longitudinal Data

Decide the optimal level of time series roll up – weekly, monthly , annual etc. .

Frequent Flier	Month	Flying Mileage	VIP Member
10621	Jan	650	No
10621	Feb	0	No
10621	Mar	0	No
10621	Apr	250	No
33855	Jan	350	No
33855	Feb	300	No
33855	Mar	1200	Yes
33855	Apr	850	Yes

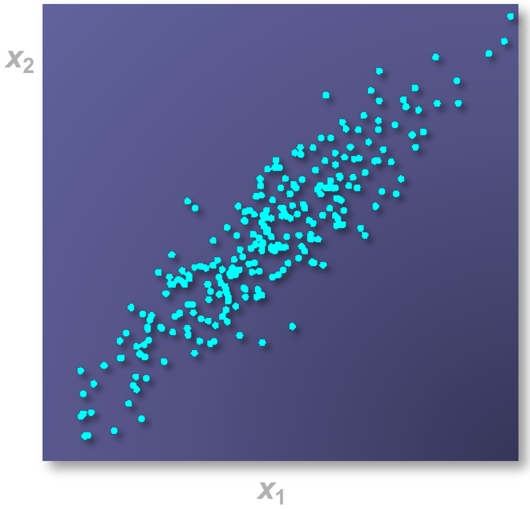
Massive Data - Curse of Dimensionality

Can I (more importantly, *should* I) analyse all the data that I have?

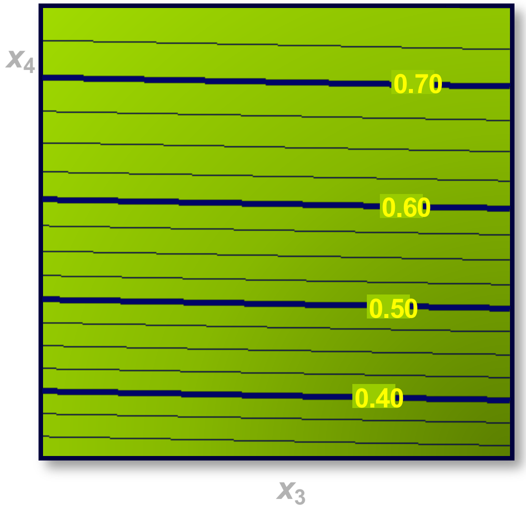


Reasons for Data Reduction

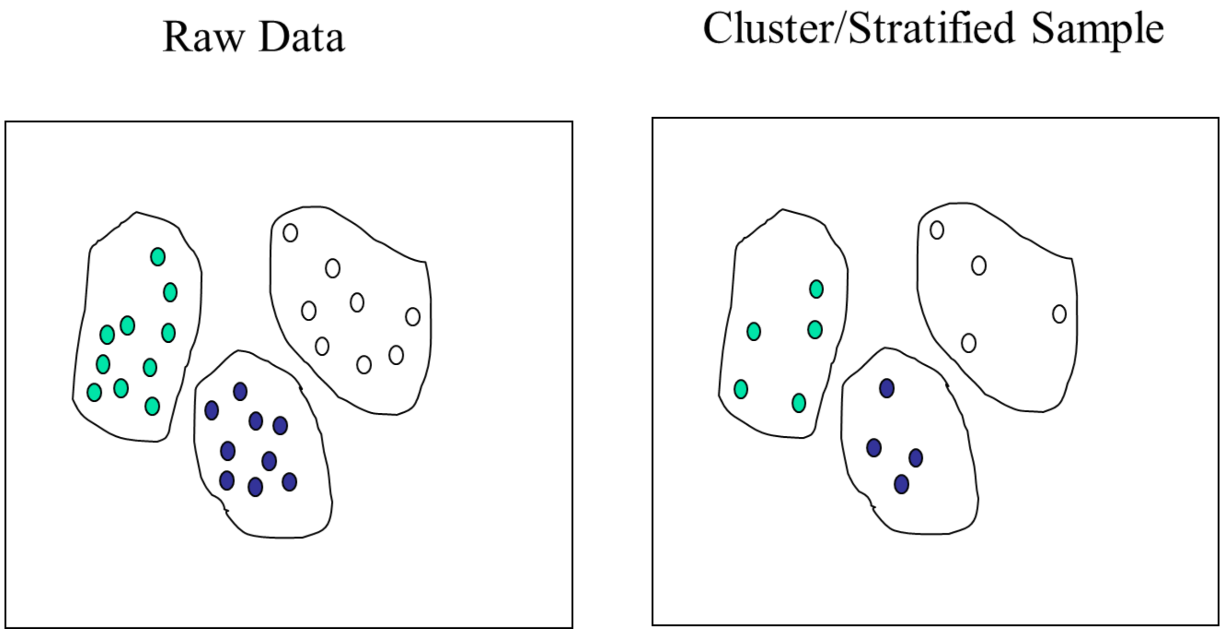
Redundancy



Irrelevancy

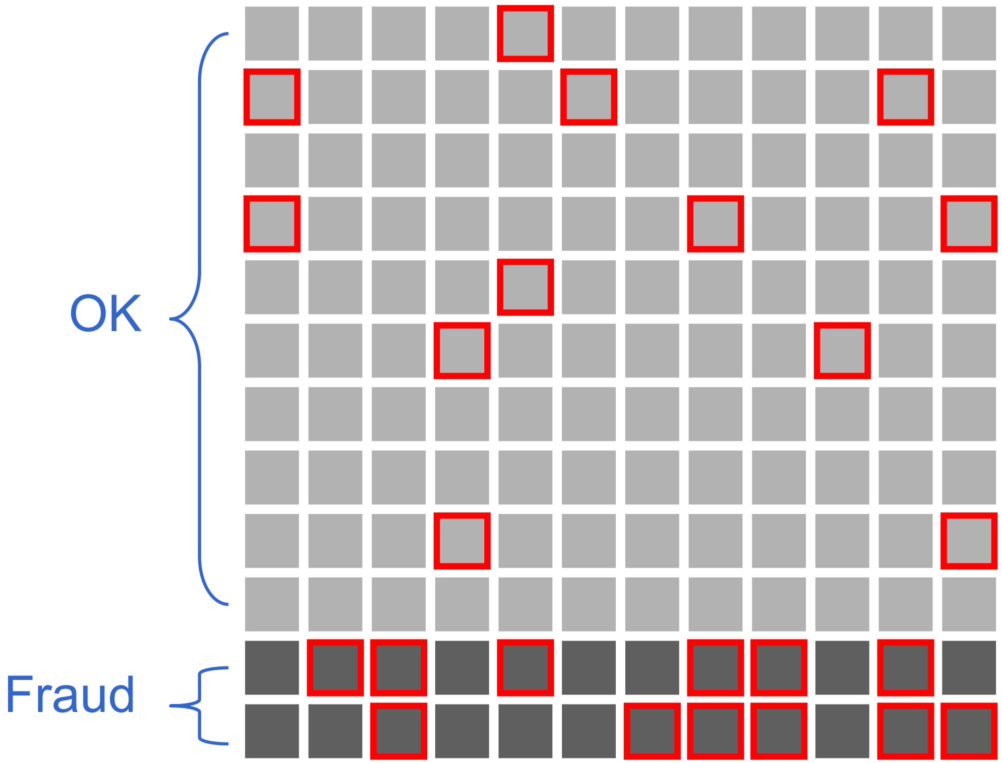


Sampling



Source: Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann Publishers. (Slides for textbook, Chapter 3)

Oversampling (Example from Fraud Analytics)



Domain Expertise – is important!

Data quality gurus:

“We found these peculiar records in your database after running sophisticated algorithms!”

Domain Experts:

“Oh, those apples - we put them in the same baskets as oranges because there are too few apples to bother. Not a big deal. We knew that already.”

Source: Lecture notes on “DataQuality and Data Cleaning: An Overview” by Theodore Johnson, ATT Research Labs

Section 4

Customer Data Structure

Customer signature – what it is and how to build it

What the customer data should look like

- Data format expectations for customer analytics
  - All data in a single table (rows/columns)
  - Each row corresponds to an entity (customer)
  - Columns with single value should be ignored
  - Character Columns with many unique values to be ignored
  - Target column identified for customer analytics

What is a customer signature

- Provides a continuous "snapshot" of customer behavior

Each row represents the customer and whatever might be useful for data mining

This column is an id field where the value is different in every column. It gets ignored for data mining purposes.

This column is from the customer information file.

This column is the target, what we want to predict.

2610000101	010377	14		A	19.1		14 Spring . . .	TRUE
2610000102	103188	7		A	19.1		NULL	TRUE
2610000105	041598	1		B	21.2		71 W. 19 St.	FALSE
2610000171	040296	1		S	38.3		3562 Oak. . .	FALSE
2610000182	051990	22		C	56.1		9672 W. 142	FALSE
2610000183	111192	45		C	56.1		NULL	TRUE
2620000107	080891	6		A	19.1		P.O. Box 11	FALSE
2620000108	120398	3		D	10.0		560 Robson	TRUE
2620000220	022797	2		S	38.3		222 E. 11th	FALSE
2620000221	021797	3		A	19.1		10122 SW 9	FALSE
2620000230	060899	1		S	38.3		NULL	TRUE
2620000231	062099	10		S	38.3		RR 1729	TRUE
2620000300	032894	7		B	21.2		1920 S. 14th	FALSE

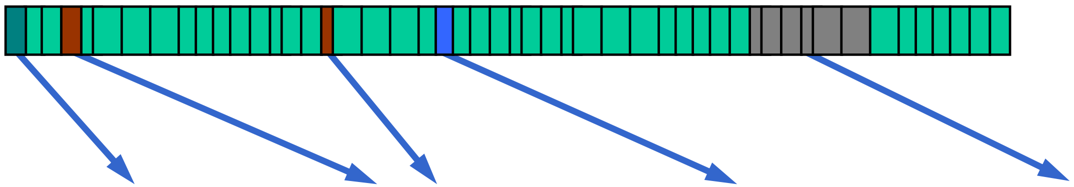
These rows have invalid customer ids, so they are ignored.

This column is summarized from transaction data.

This column is a text field with unique values. It gets ignored (although it may be used for some derived variables).

How a signature looks like

The fields of a customer signature have various roles



The **primary key** uniquely identifies each row, often corresponding to customer ID.

The **target columns** are what you are looking for. Sometimes, the information is in multiple columns, such as a churn flag and churn date.

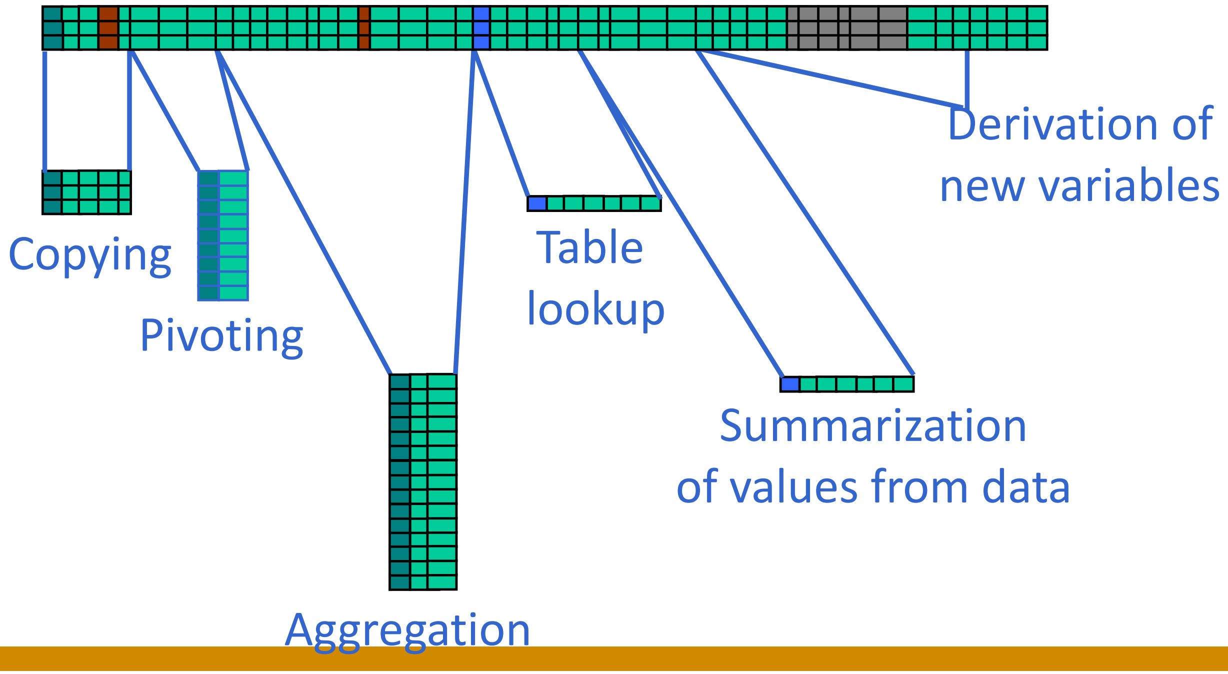
A **foreign key** gives access to data in another table, such as ZIP code demographics.

Some columns are ignored because the values are not predictive or they contain future information, or for other reasons.

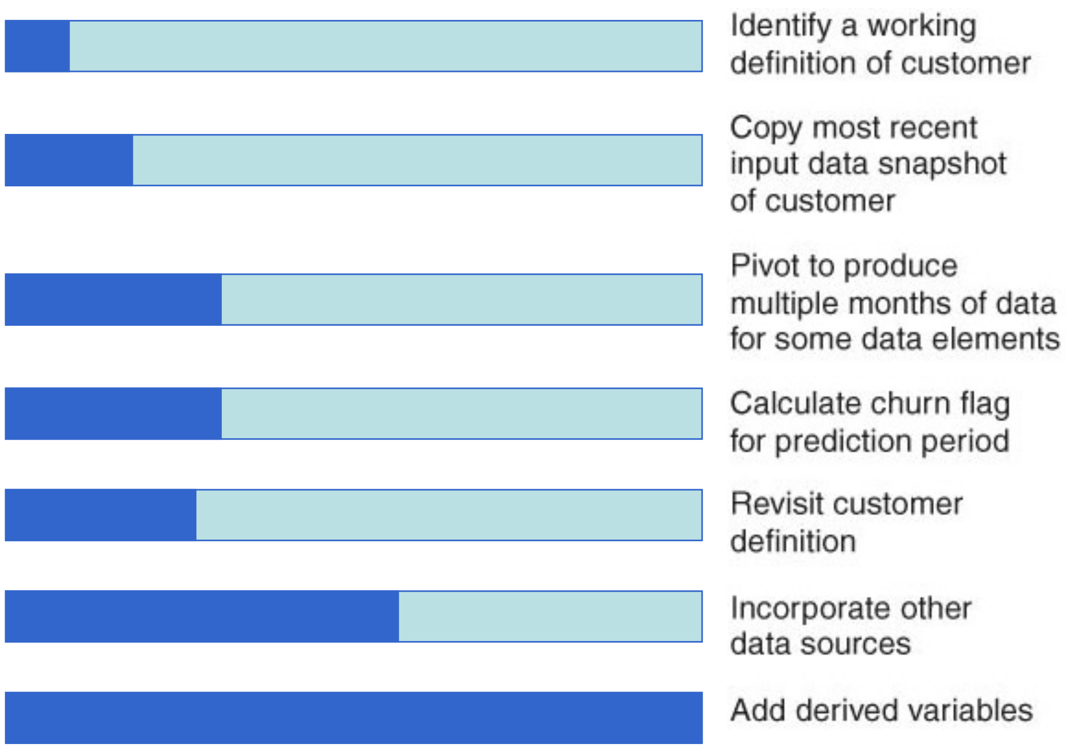
Each row generally corresponds to a customer.

Customer signature assembly

- Data from most sources must be transformed in various ways before it can be incorporated into the signature.



Constructing the Customer Signature



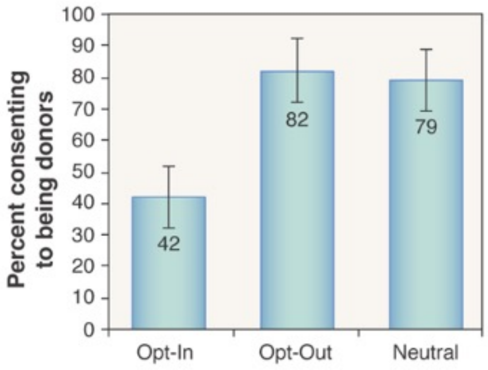
Section 5

Analytics Foundations

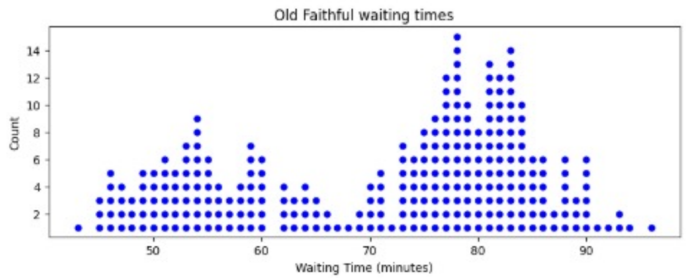
Visualization, Supervised vs unsupervised, parametric/nonparametric

Distributions and Visualizations

- The distribution of a variable describes the pattern of the variable across the different observational units.
- The best way to represent this pattern is often with a visualization.



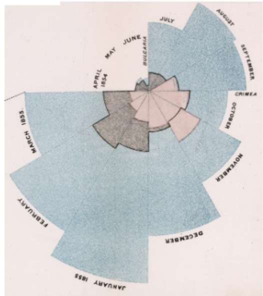
Organ donation results



Data Visualization in history

Florence Nightingale (1820-1910)

- Nightingale is best known as the founder of modern nursing.
- But she was also a statistician, the first female member of the Royal Statistical Society in 1879.
- She drew public attention to the importance of nursing by making visualizations, such as the one on the next slide, which depicts deaths during the Crimean War.



Rose Diagram –
Diagram of the causes of mortality in the army in the East

"The blue wedges measured from the centre of the circle represent area for area the deaths from Preventable...diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes."

Source: Lecture notes at Department of Statistics, Stanford University
https://web.stanford.edu/class/stats60/lectures/01-lecture-welcome.html

Data Visualization

- Purpose of data visualization is to *communicate* data to others.
- It is as much psychology and art, as it is math.

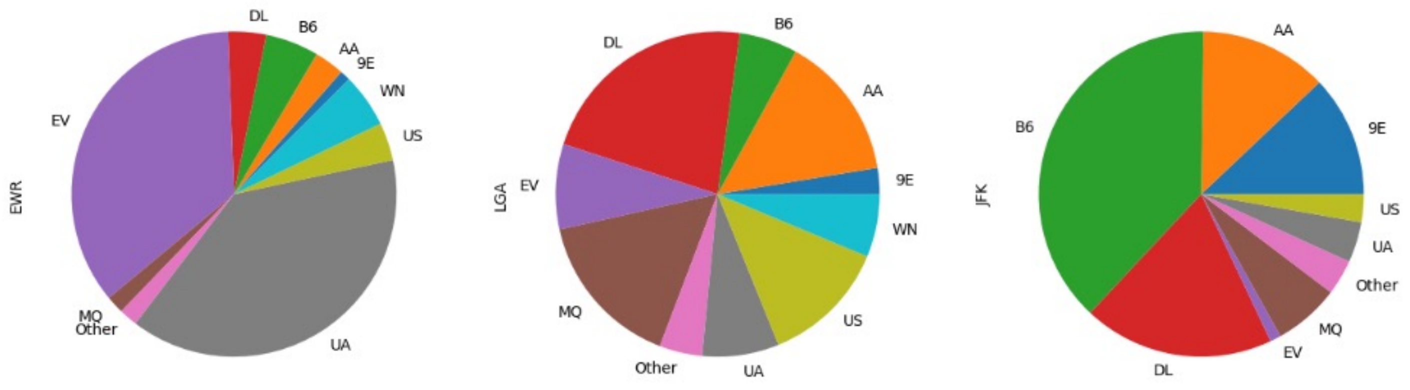
The following data set consists of flights departing from NYC airports in 2013.

	year	month	day	dep_time	dep_delay	arr_time	arr_delay	carrier	tailnum	flight	origin	dest	air_time	distance	hour	minute
0	2013	6	30	940	15	1216	-4	Other	N626VA	407	JFK	LAX	313	2475	9	40
1	2013	5	7	1657	-3	2104	10	DL	N3760C	329	JFK	SJU	216	1598	16	57
2	2013	12	8	859	-1	1238	11	DL	N712TW	422	JFK	LAX	376	2475	8	59
3	2013	5	14	1841	-4	2122	-34	DL	N914DL	2391	JFK	TPA	135	1005	18	41
4	2013	7	21	1102	-3	1230	-8	9E	N823AY	3652	LGA	ORF	50	296	11	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
32730	2013	10	8	752	-8	921	-28	9E	N8505Q	3611	JFK	PIT	63	340	7	52
32731	2013	7	7	812	-3	1043	8	DL	N6713Y	1429	JFK	LAS	286	2248	8	12
32732	2013	9	3	1057	-1	1319	-19	UA	N77871	1545	EWR	IAH	180	1400	10	57
32733	2013	10	15	844	56	1045	60	B6	N258JB	1273	JFK	CHS	93	636	8	44
32734	2013	3	28	1813	-3	1942	-23	UA	N36272	1053	EWR	CLE	59	404	18	13

32735 rows × 16 columns

There is too much information in a data set. We have to decide what to show and how to show it.

Data Visualization - Categorial

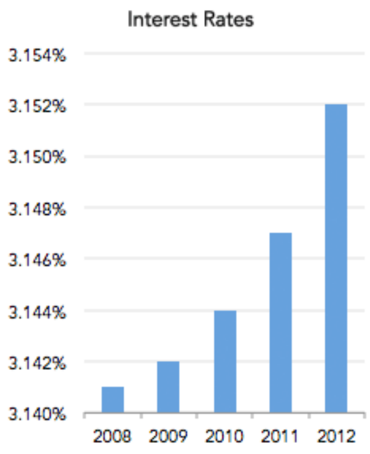


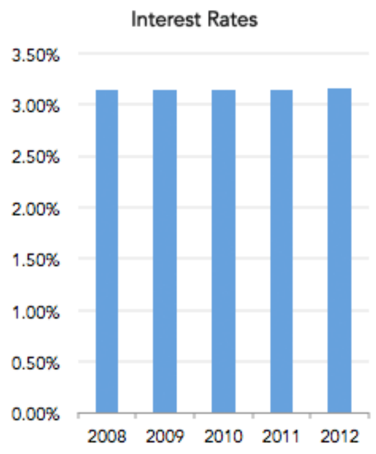
Can you draw some interesting insights?
How about bar chart? Line Charts? Combined?

Source: Lecture notes at Department of Statistics, Stanford University
https://web.stanford.edu/class/stats60/lectures/01-lecture-welcome.html

Time Series Data

- Data collected over time is called time-series data





The change is completely obscured by the "correct" bar chart on the right!

Source: Lecture notes at Department of Statistics, Stanford University
https://web.stanford.edu/class/stats60/lectures/01-lecture-welcome.html

Data Visualization - Numeric

In a histogram, values are sorted into *bins*, and the number of values in each bin is plotted as a bar.

It is one of the best ways to quickly learn a lot about data including central tendency, spread, shape, outliers, mode etc.

Differences between histogram and bar chart?

When is a scatterplot useful?

Data Visualization – In summary

- When making a visualization, think about the **number of variables** and the **type of variable** (quantitative or categorical).

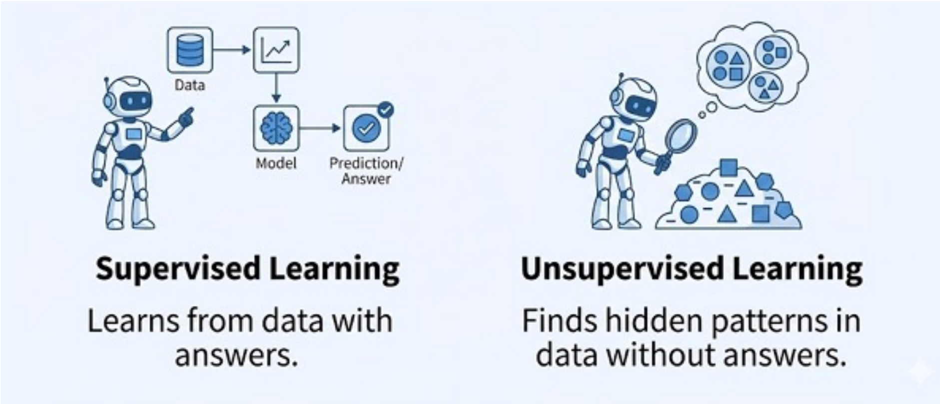
Single
- Categorical
- Numeric

Multiple
- Two Categorical
- Two Numeric
- One each

Over Time

Over Location

Two ways to learn from data - Supervised vs Unsupervised

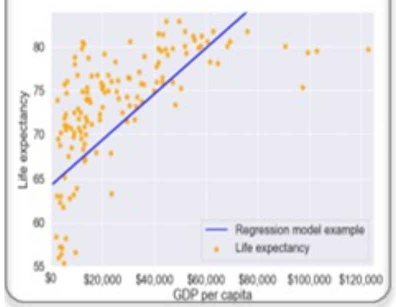


Source: Chapter 2, Statistical Learning by Trevor Hastie and Robert Tibshirani

Supervised Learning

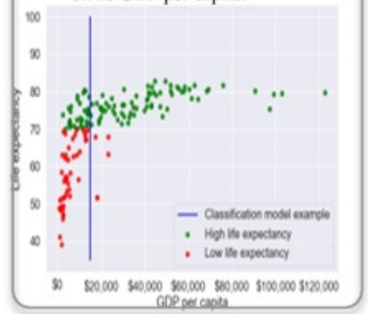
Regression:

- The response is a continuous variable. For example, predicting life expectancy of a country based on its GDP per capita.



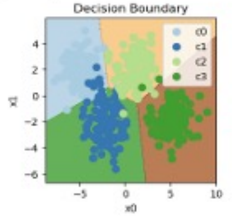
Binary Classification:

- The response is a categorical variable. For example, classifying a country as having low or high life expectancy based on its GDP per capita.



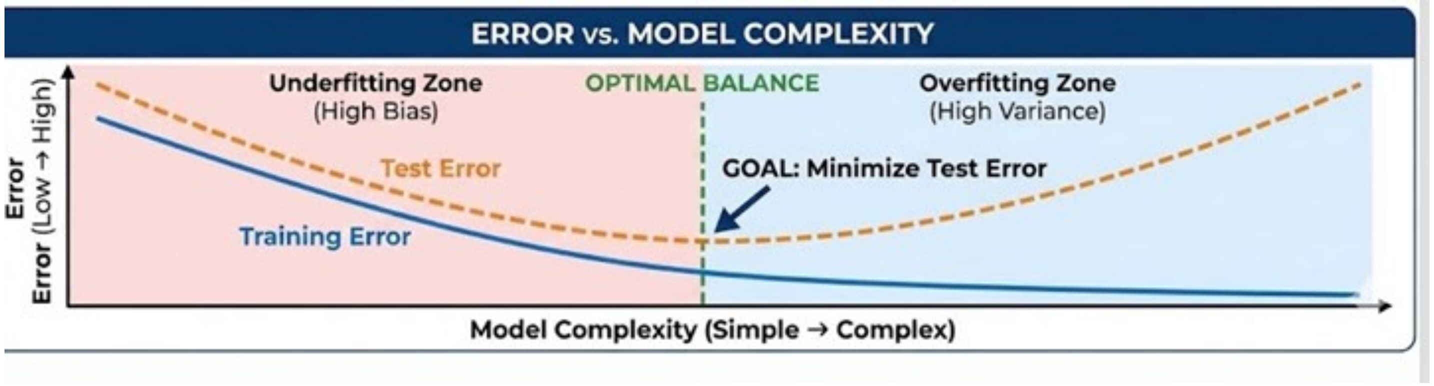
Multiclass Classification:

- The response is a categorical variable. For example, classifying emails into spam, promotions, social, and primary.



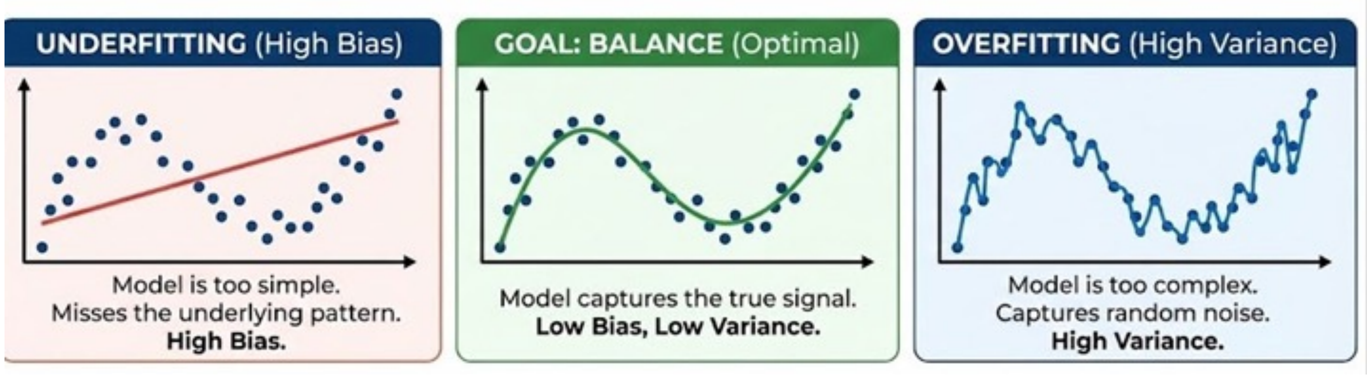
Source: extracted from Statistical Leaning by Trevor Hastie and Robert Tibshirani

Bias-Variance Trade-off



Source: extracted from Statistical Leaning by Trevor Hastie and Robert Tibshirani

Undefitting vs Overfitting



Parametric vs Non-parametric

Category	Parametric	Non-parametric
Model Form	Fixed functional form	Not fixed
Assumptions	Strong	Minimal
Flexibility	Low	High
Interpretability	High	Low
Data required	Less data	More Data
Bias	Higher	Lower
Variance	Lower	Higher

Source: extracted from Statistical Leaning by Trevor Hastie and Robert Tibshirani

Consumer Data Protection

- From the corporation's standpoint there are questions of
  - ✓what can be collected,
  - ✓how it can be stored and
  - ✓how it can be used.

- From the customer's standpoint there are questions of
  - ✓what *is* being collected and
  - ✓exactly how and when –
  - ✓and by whom - it is being applied.

Summary – Best Practices for analytical data prep

- Understand the business process
- Establish analytics friendly protocols for extracting data from data warehouses
- Prepare data to meet the needs of the analytic method that you've chosen
- Preserve analytical data's rich details, because they enable discovery
- Improve analytical data *after* working with it, not before

“It is better to understand the problem and use what ‘tools’ you have, than to know every advanced procedure and not quite understand the problem.”
– Fred Ruland

Source: Junker, B. (2007). *Statistical practice (Course notes for 36-726).* Carnegie Mellon University.
http://www.stat.cmu.edu/~brian/726 (Chapter 1 – Intro)

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Reference

- Davenport, T.H., Harris, J.G. and Morison, R. (2010) ***Analytics at Work***. Harvard Business Press, Boston, Massachusetts. Chapter 2.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009) ***The Elements of Statistical Learning***. Springer (https://hastie.su.domains/ElemStatLearn/).
- Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann Publishers.
- Johnson, T. “Data Quality and Data Cleaning: An Overview.” Lecture notes, AT&T Research Labs.
- Junker, B. (2007). *Statistical practice (Course notes for 36-726).* Carnegie Mellon University. http://www.stat.cmu.edu/~brian/726
- Stanford University, Department of Statistics. ***STATS 60: Introduction to Statistical Methods***. Course lecture notes (https://web.stanford.edu/class/stats60/lectures/01-lecture-welcome.html).
- Northwestern University, Department of Statistics. ***STAT 303-1: Data Science with Python***. Course materials (https://lizhen0909.github.io/nu-stat303-1-sec20-coursebook).
- https://www.iitr.ac.in/media/facspace/patelfec/16Bit/slides/Lecture-2-Data-Preprocessing-Part-1.pdf
- Georges, J. (2003). Data preparation for data mining [Course notes]. SAS Institute
- IDA Singapore – Policies & Regulation: Consumer Protection (http://www.ida.gov.sg/policies%20and%20regulation/20060627155443.aspx).
- Report on a Model Data Protection Code for the Private Sector (http://www.agc.gov.sg/publications/docs/Model_Data_Protection_Code_Feb_2002.pdf).
- Put consumer data privacy first – analytics value will follow (http://searchbusinessanalytics.techtarget.com/feature/Put-consumer-data-privacy-first-analytics-value-will-follow).
- Analytics Community Must Address Privacy Issues (http://www.allanalytics.com/author.asp?section_id=1412&doc_id=240547).

Note: Some slides are taken from the prior 2025 version of the course

Lens 1: How Different Are Your Customers?

The Customer-Base Audit — with Meridian & Co.

Meet Meridian & Co.

A FICTITIOUS US LIFESTYLE RETAILER — BUILT FOR ISSS603

The Company

Meridian is a US-based lifestyle retailer selling Home goods, Apparel, Accessories, Footwear and Beauty through three channels — website, mobile app, and a legacy mail-order catalog.

The Data

Four years of line-item transactions (2016–2019): 173,079 rows across 140,297 orders placed by 59,510 unique customers. Every row tells us who bought, what, when, through which channel — and at what margin.

The Problem

Revenue grew every year from 2016 to 2019, but the CMO suspects the growth is fragile. Are we adding new customers or re-engaging old ones? Is the growth coming from a few whales — or the broad base? Catalog or app?

The Five Questions

That lead us to the five-lens audit:

- Lens 1 — How different are your customers?
- Lens 2 — What changed since last period? (two periods)
- Lens 3 — How does customer behaviour evolve? (single cohort over time)
- Lens 4 — Comparing cohorts (is acquisition quality shifting?)
- Lens 5 — How healthy is our customer base? (integrating all of the above)

Meridian's dataset is designed so every lens yields a clear, teachable insight — and a surprise or two.

The Three Ds of Customer Analysis

A practical framework for understanding customer heterogeneity

Distribution

- Plot the full distribution, not just the average
- Look for right-skewness (Mean > Median)
- Ask: what % of customers are below the mean?
- Apply to: spend, transactions, AOV, profit, margin

Decomposition

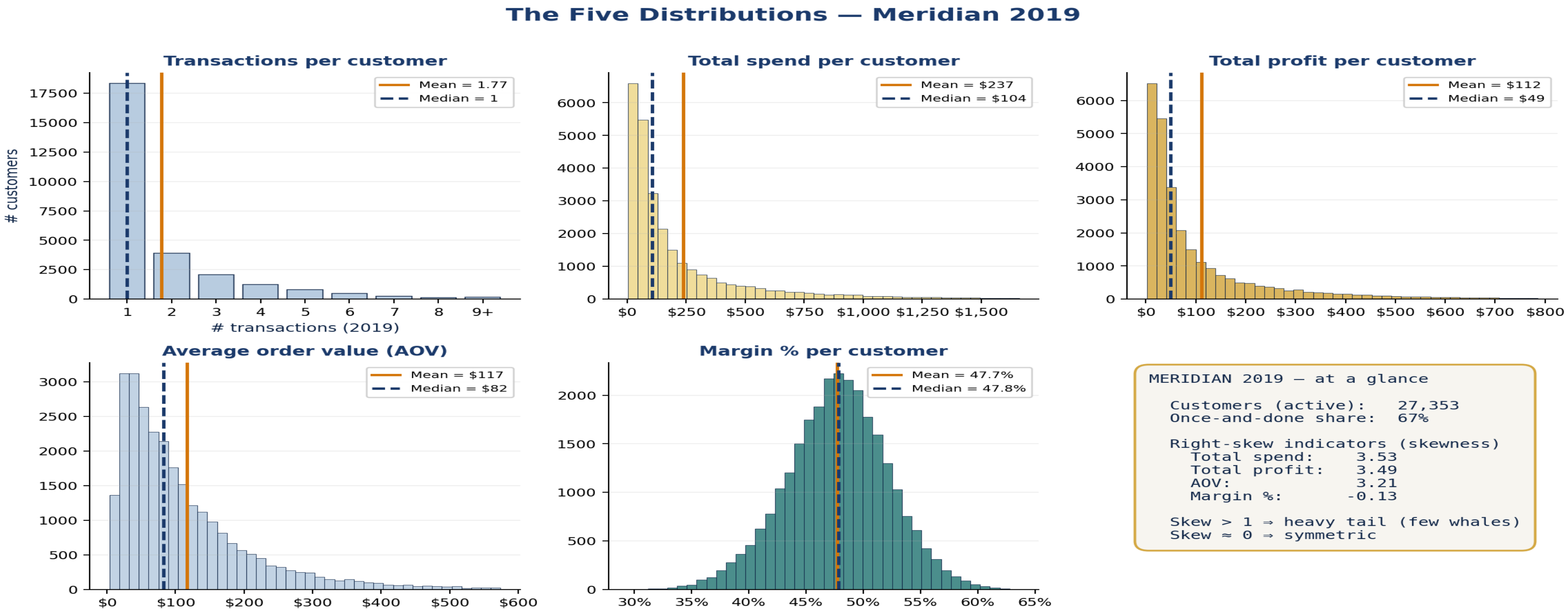
- Break totals into multiplicative components
- Spend = Frequency × Basket Size (AOF × AOV)
- Profit = Spend × Margin
- Identify which lever drives the variation

Decile

- Sort customers into 10 equal groups
- Compare value concentration across groups
- Decompose each decile's behavior
- Understand what makes top customers different

The Five Distributions — Meridian 2019

Every metric you care about is heavy-tailed — except margin.



Take-away: Mean ≫ Median for spend, profit, and AOV — the 'average customer' is a myth. Margin % is nearly symmetric because Meridian's pricing architecture is stable across categories.

“The Vital Few and the Useful Many”

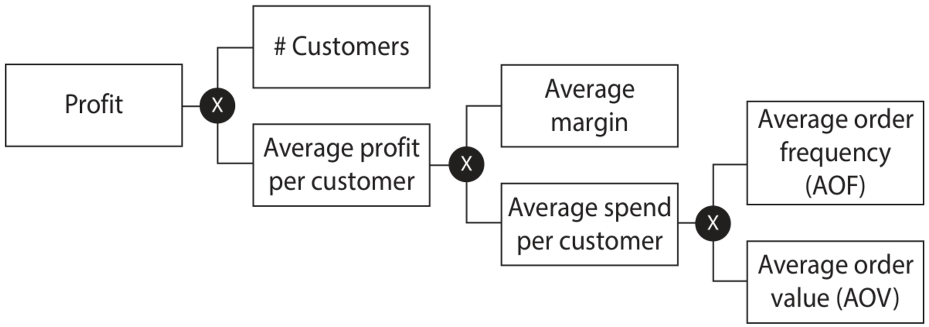
The distributions show us customers are wildly different.
Now: how concentrated is the value?

Not all customers are equal in their value to the firm.
A lot of the value is concentrated in a small subset of them.

Multiplicative Decomposition of Profit

Every driver becomes inspectable individually.

Figure 3.6. A Multiplicative Decomposition of Profit



Why decompose?

A single 'avg profit per customer' number hides four different levers:

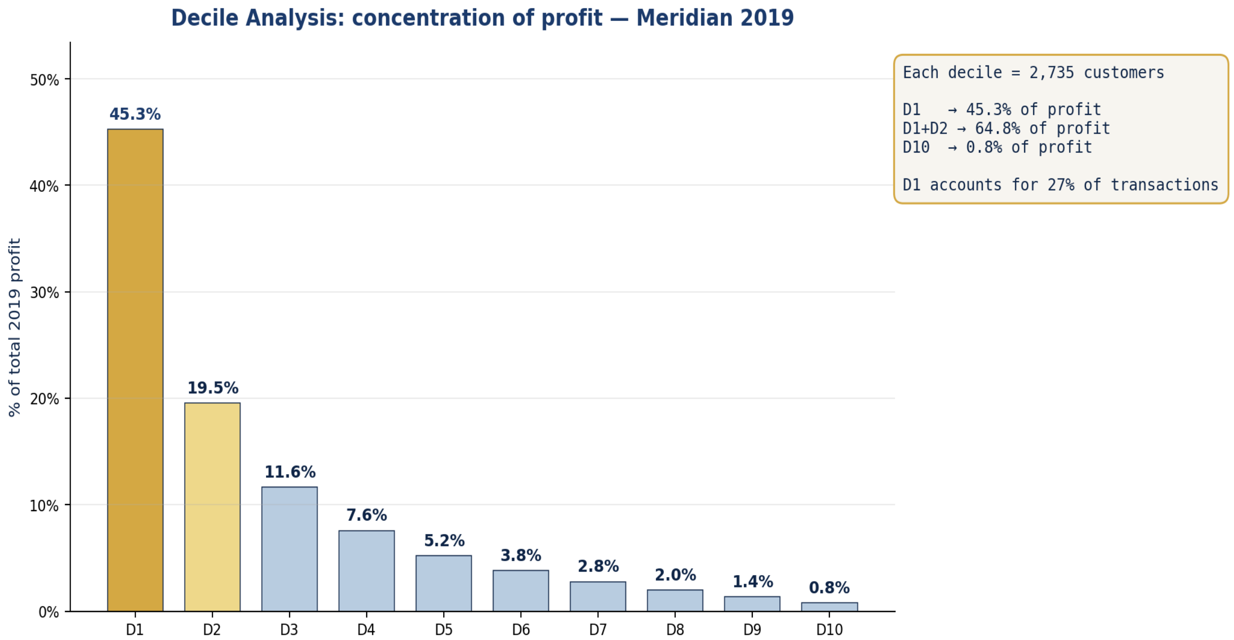
- Are we acquiring enough customers?
- Are they coming back (AOF)?
- Is each trip bigger (AOV)?
- Are we selling the right mix (margin)?

Meridian 2019: 27,353 active customers × avg profit $112 = $3.06M total profit.

AOF = 1.77 · AOV (unweighted) = $117 · Avg margin = 47.7%. The heavy tail lives almost entirely in AOF and AOV — not in margin.

Decile Analysis: Concentration of Value

Sorting 27,353 Meridian 2019 customers into 10 equal groups by profitability (most → least profitable).



Key Findings

45%	of profit from top 10% (D1)
65%	of profit from top 20% (D1+D2)
0.8%	of profit from bottom 10% (D10)
27%	of transactions from D1 (vs 6% from D10)

Each decile has exactly 2,735 customers — same # of people, vastly different value.

Decile Decomposition — three views of the same 27,353 customers

Table 3.2 — Summary of 2019 Buyer Behavior by Customer Decile

Decile	Customers	Transactions	Revenue	Profit	% Cust	% Txns	% Rev	% Profit
D1	2,736	12,882	$2,939,544	$1,384,577	10.0%	26.6%	45.3%	45.3%
D2	2,735	7,764	$1,267,510	$597,287	10.0%	16.0%	19.5%	19.5%
D3	2,735	5,382	$755,747	$355,983	10.0%	11.1%	11.7%	11.6%
D4	2,739	4,206	$490,947	$231,695	10.0%	8.7%	7.6%	7.6%
D5	2,733	3,558	$339,704	$160,087	10.0%	7.4%	5.2%	5.2%
D6	2,734	3,201	$245,486	$116,099	10.0%	6.6%	3.8%	3.8%
D7	2,735	3,016	$179,837	$84,959	10.0%	6.2%	2.8%	2.8%
D8	2,735	2,874	$128,093	$60,754	10.0%	5.9%	2.0%	2.0%
D9	2,737	2,778	$88,071	$41,767	10.0%	5.7%	1.4%	1.4%
D10	2,734	2,744	$48,580	$23,192	10.0%	5.7%	0.7%	0.8%
Totals	27,353	48,405	$6,483,519	$3,056,400	100.0%	100.0%	100.0%	100.0%

Table 3.2 — Totals per decile

What each 10% slice CONTRIBUTES. D1 produces $1.38M profit — 45% of the total.

Table 3.3 — Decomposition of Customer-Decile Profit (AOF × AOV × Margin)

Decile	% Cust	% Profit	Avg Spend/Cust	Avg Profit/Cust	AOF	AOV	Avg Margin
D1	10.0%	45.3%	$1,074	$506	4.71	$279	47.3%
D2	10.0%	19.5%	$463	$218	2.84	$220	47.4%
D3	10.0%	11.6%	$276	$130	1.97	$180	47.4%
D4	10.0%	7.6%	$179	$85	1.54	$140	47.6%
D5	10.0%	5.2%	$124	$59	1.30	$107	47.6%
D6	10.0%	3.8%	$90	$42	1.17	$82	47.8%
D7	10.0%	2.8%	$66	$31	1.10	$62	47.8%
D8	10.0%	2.0%	$47	$22	1.05	$46	48.0%
D9	10.0%	1.4%	$32	$15	1.01	$32	48.0%
D10	10.0%	0.8%	$18	$8	1.00	$18	48.3%

Table 3.3 — Decomposition

Why D1 is different: AOF 4.71 (vs D10's 1.00) and AOV $279 (vs $18). Margin barely moves (47–48% all rows).

Table 3.4 — Equal-Profit Deciles (each row = 10% of total profit)

Profit-Decile	% Customers	% Profit	Avg Spend/Cust	Avg Profit/Cust	AOF	AOV	Avg Margin
PD1	1.1%	10.0%	$2,168	$1,020	6.79	$365	47.2%
PD2	1.7%	10.0%	$1,375	$648	5.54	$307	47.2%
PD3	2.3%	10.0%	$1,023	$483	4.69	$281	47.4%
PD4	3.0%	10.0%	$801	$377	4.13	$258	47.2%
PD5	3.8%	10.0%	$621	$292	3.47	$240	47.3%
PD6	5.0%	10.0%	$473	$223	2.92	$219	47.3%
PD7	6.9%	10.0%	$342	$161	2.27	$200	47.5%
PD8	10.3%	10.0%	$231	$109	1.78	$162	47.5%
PD9	17.4%	10.0%	$136	$64	1.34	$115	47.6%
PD10	48.4%	10.0%	$49	$23	1.06	$47	48.0%

Table 3.4 — Equal-profit slicing

Flip the question: to make 10% of profit we need just 1.1% of customers (PD1). To make the LAST 10%, we need 48% of the base.

Hands-On — Your Turn with the Meridian Data

Open Lab1_Student_Notebook_20260423.ipynb in Jupyter. Each task maps to a slide you just saw.

Q1

Recreate the five distributions

Filter meridian_transactions.csv to 2019 activity. Roll up to customer level. Plot #transactions, total spend, total profit, AOV and margin% with mean/median annotated. Which is symmetric — and why?

Q2

Build the decile concentration chart

Sort customers by total profit, descending. Assign deciles D1…D10 (equal N). Plot % of profit per decile. What share does D1 capture? D1+D2? D10?

Q3

Assemble Decile Tables

Three tables, one data frame each.

Lesson 3:

Customer-Base Audit: Lens 2

Period-on-Period Decomposition

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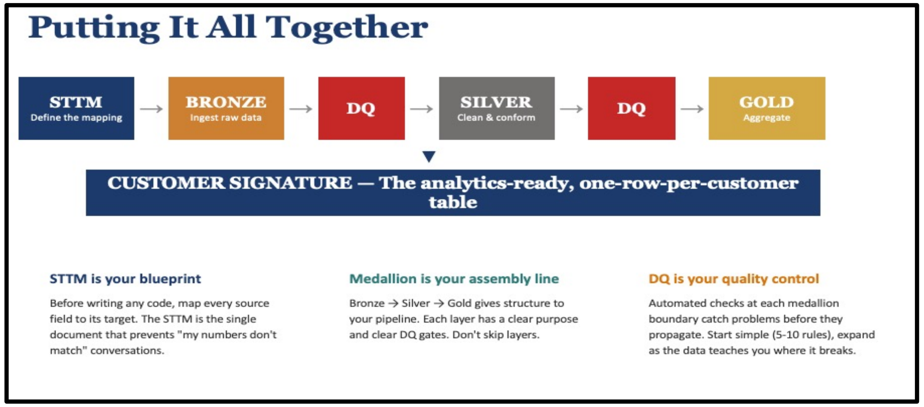
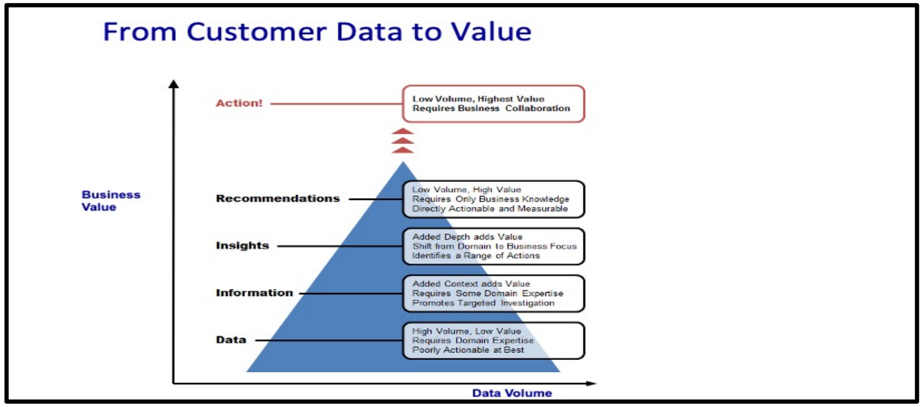
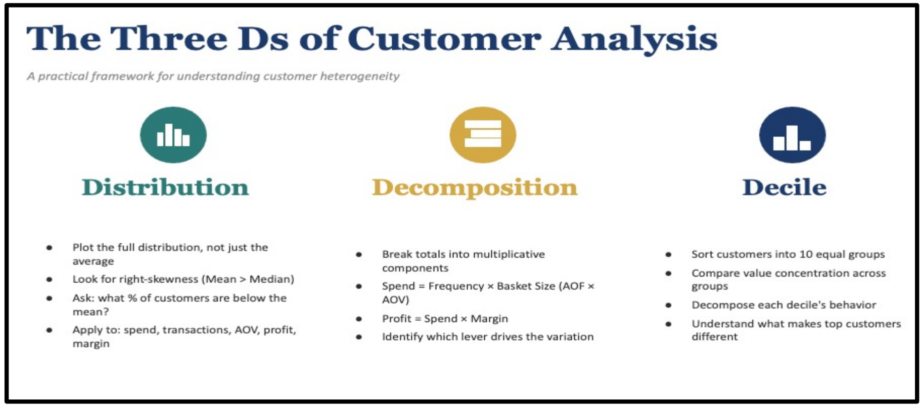
Recap of Lesson 2







FRAMEWORK

The Five Lenses at a Glance

Lens 1

How Different?

Single-period snapshot
Distributions & deciles

Lens 2

What Changed?

Period-on-period
Retained / Lost / New

TODAY

Lens 3

How Do They Evolve?

Cohort over time
Retention & lifecycle

Lens 4

Cohorts Degrading?

Vintage comparison
Quality drift

Lens 5

Is It Sustainable?

Overall health
Integrated diagnostic

Each lens answers a question the previous one structurally cannot.

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LEARNING OUTCOMES

What You'll Be Able to Do After Lens 2

Period-on-Period Decomposition — techniques for comparing customer behaviour across time periods

1

Decompose period-on-period profit changes

Break down profit changes into customer groups (retained, lost, new) and behavioural drivers (AOF, AOV, margin) — and read a migration matrix.

2

Read a profit decile migration matrix

Track how customers move between profit ranks across years. Apply the ±1 decile stability threshold to distinguish real movement from measurement noise.

3

Perform up-down decomposition

Split both-years customers into profit improvers vs decliners. Identify which components (frequency, spend, margin) drive changes in each direction.

4

Identify the selection effect

Recognise when observed differences between customer groups reflect who self-selects into the group — not what being in the group causes.

LENS 2

The Headlines: Meridian 2018 vs 2019

$6.48M	$3.06M	27,353
Revenue 2019	**Profit 2019**	**Customers 2019**
up from $4.92M in 2018	up from $2.32M in 2018	up from 21,415 in 2018
+31.8%	+31.9%	+27.7%

What changes in customer behaviour lie behind these increases?

At first glance, both revenue and profit grew by about 32%. **But with customers growing by 28%, is this just more people spending the same way — or are customers actually changing?**

LENS 2

Per-Customer Metrics: Nearly Identical

Despite 32% revenue growth, per-customer metrics barely moved.

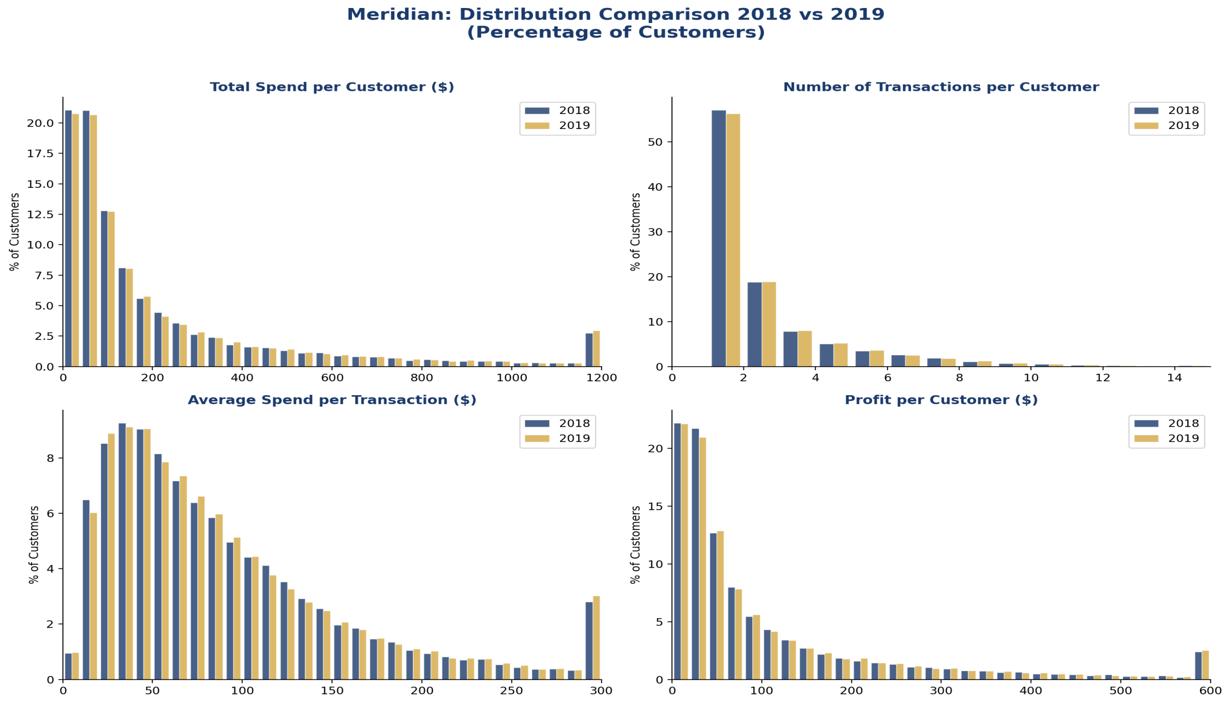
Metric	2018	2019	Change
Avg Total Spend / Customer	$229.70	$237.03	↑ +3.2%
Avg Transactions / Customer	2.16	2.20	↑ +1.9%
Avg Spend per Transaction	$92.70	$93.65	↑ +1.0%
Avg Profit / Customer	$108.18	$111.74	↑ +3.3%

Key Insight: Per-customer behaviour is essentially flat. The ~32% revenue growth is driven almost entirely by a 28% increase in the number of customers.

LENS 2

Distribution Comparison: 2018 vs 2019

The four key distributions are virtually unchanged year-on-year, confirming that growth is customer-count driven.



Source: Meridian transaction data (meridian_transactions.csv) | Navy = 2018, Gold = 2019

LENS 2

So What Does This Mean?

What Lens 1 Told Us

- The distributions look the same across both years
- Per-customer averages barely changed
- Growth appears to come entirely from more customers
- This seems like a healthy, stable business

But Lens 2 Asks...

- Are these the same customers, or did we swap old for new?
- Are retained customers spending more or less?
- Are new customers behaving the same as existing ones?
- Is the "stable average" hiding offsetting changes?

The key question: Lens 1 snapshots can't distinguish healthy growth from a leaky bucket being refilled with new customers.

The First Questions to Ask

How many people were customers in both years? And how many only purchased in one year?

Meridian had **21,415 customers in 2018** and **27,353 customers in 2019**. But these are not the same people.

We need to split these customers into three groups to understand what is really happening:

2018-Only

12,496

Purchased in 2018 but not 2019

58.4% of 2018 base

Both Years

8,919

Purchased in both 2018 and 2019

22.4% of all unique

2019-Only

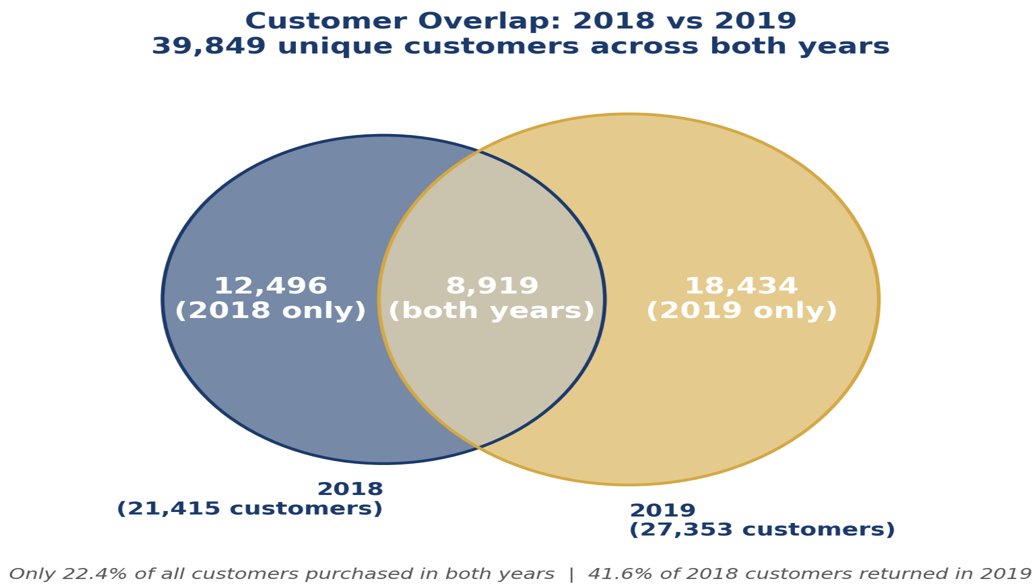
18,434

Purchased in 2019 but not 2018

67.4% of 2019 base

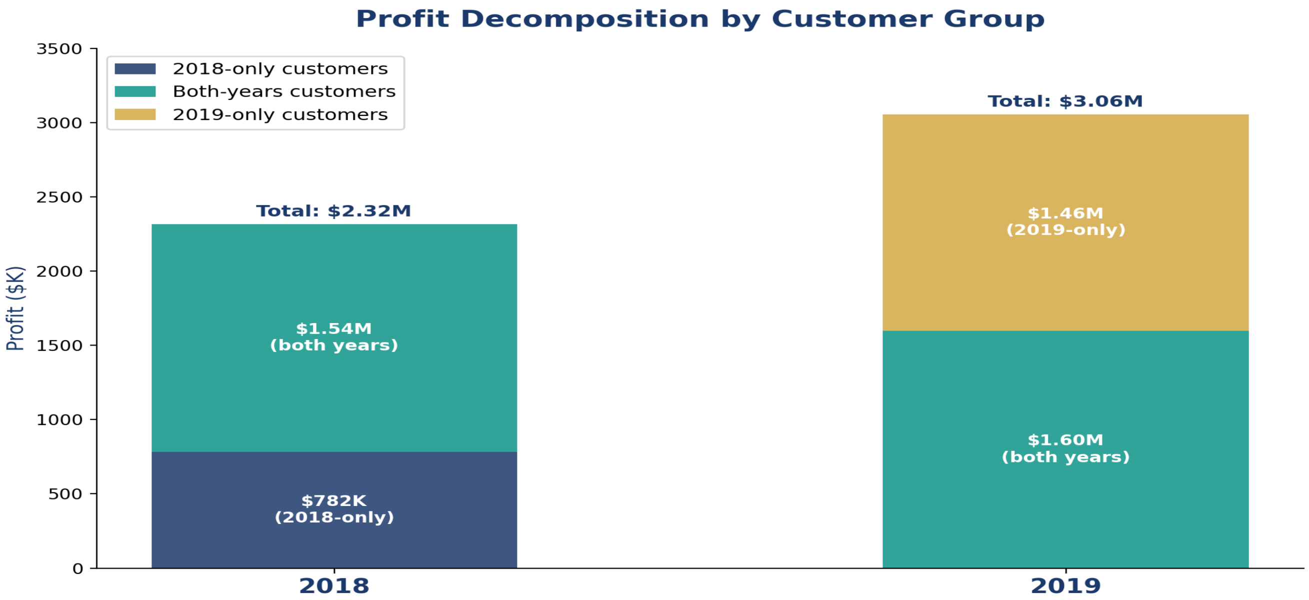
Total unique customers across both years: **39,849**

Customer Overlap: 2018 vs 2019



Only 41.6% of 2018 customers returned in 2019. It may be tempting to think those customers who purchased in 2018 and not in 2019 are "lost." But it is too premature to draw that conclusion — some are one-time buyers, others are simply light buyers who skip years.

Profit Decomposition by Customer Group



Key finding: The 2019-only group's profit ($1.46M) is 87% higher than the 2018-only group ($782K). The both-years group grew modestly from $1.54M to $1.60M (+$61K).

Multiplicative Decomposition of Performance

Profit = Customers × AOF × AOV × Margin

TEMPORAL DECOMPOSITION

Multiplicative Decomposition of Performance

Profit = # Customers × AOF × AOV × Average Margin

Customer Group	Metric	2018	2019	
2018-only (12,496 customers)	AOF	1.33	—	
	AOV	$100	—	
	Avg Margin	47%	—	
2019-only (18,434 customers)	AOF	—	1.41	
	AOV	—	**$120**	
	Avg Margin	—	47%	
Both years (8,919 customers)	AOF	**2.32**	**2.52**	Selection Effect
	AOV	**$157**	**$151**	
	Avg Margin	47%	47%	

Profit Decile Migration

How do individual customers move between profitability tiers?

DECILE MIGRATION

How Decile Migration Works

1 **Pool & Threshold**
Combine all customer-year profits. Calculate 9 thresholds that create 10 bins of equal total profit.

2 **Assign Deciles**
Each customer gets a decile in 2018 and a decile in 2019 using the same common thresholds.

3 **Cross-Tabulate**
Create a 10×10 matrix showing how many customers moved from each 2018 decile to each 2019 decile.

4 **Read the Diagonal**
Customers on the diagonal stayed in the same decile. Off-diagonal = movement (improvement or decline).

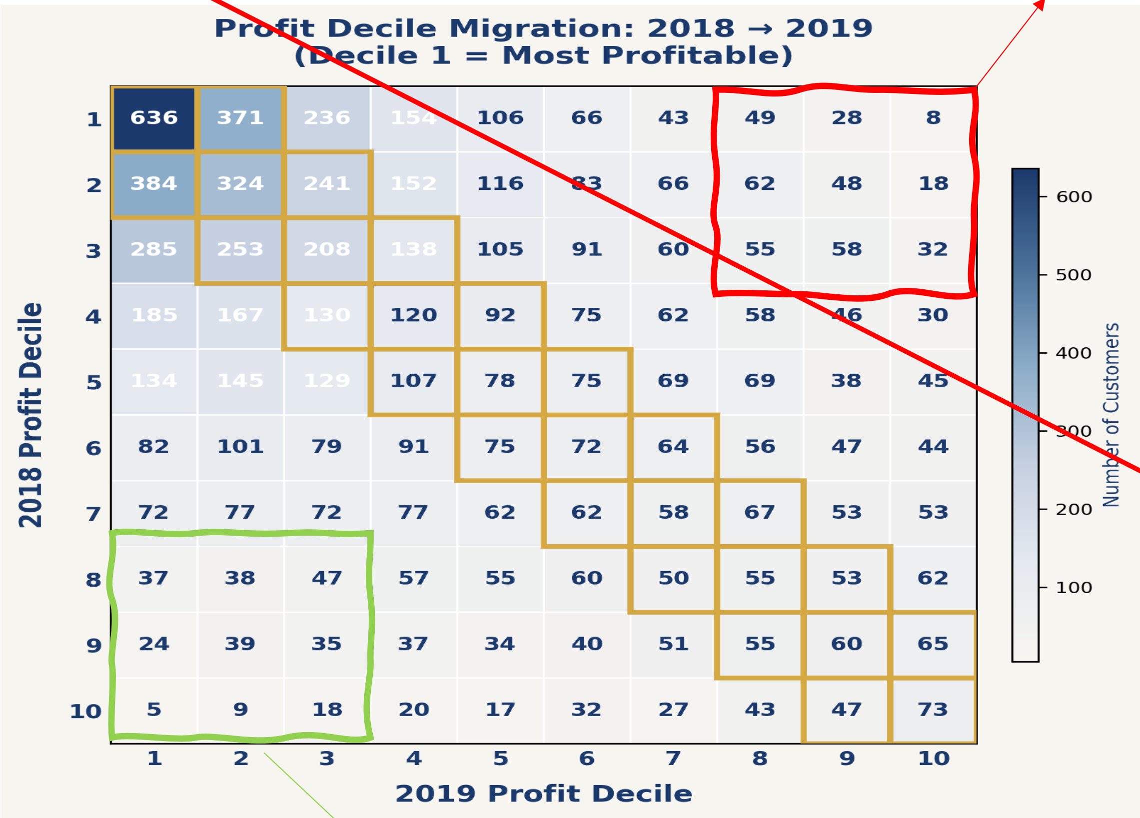
~~**Meridian's profit thresholds:** $287 | $160 | $101 | $69 | $49 | $36 | $26 | $19 | $12 *(Decile 1 = most profitable)*~~

Meridian's profit thresholds: 427 | 284 | 202 | 144 | 103 | 72 | 49 | 33 | 19 *(Decile 1 = most profitable)*

DECILE MIGRATION

Profit Decile Migration Heatmap

On the flip side: 369 customers dropped from top 3 → bottom 3 deciles. Can we identify what drove that behaviour?



On the bright side: 252 customers rose from bottom 3 → top 3 deciles. Can we identify what drove their improvement?

HOW TO READ THIS HEATMAP

Y-axis = 2018 profit decile. X-axis = 2019. Each cell = number of customers who migrated. Gold outline marks the ±1 stability band. Darker shading = more customers.

TOP-DECILE STICKINESS

Cell (1,1) = 636 customers — by far the largest cell. Top-decile profitability is sticky and self-reinforcing. These are committed high-value repeat buyers.

±1 DECILE = PRACTICAL STABILITY

Exact diagonal = only 19% of customers. But ±1 decile captures 45%. Adjacent thresholds can be just $5–$7 apart — minor noise causes apparent “movement.”

MIDDLE ROWS SPREAD EVENLY

Rows 5–7 spread across all columns. A decile 6 customer was equally likely to end up in decile 4 as decile 8. Middle-tier profitability is inherently unpredictable.

BOTTOM-RIGHT IS SURPRISINGLY SPARSE

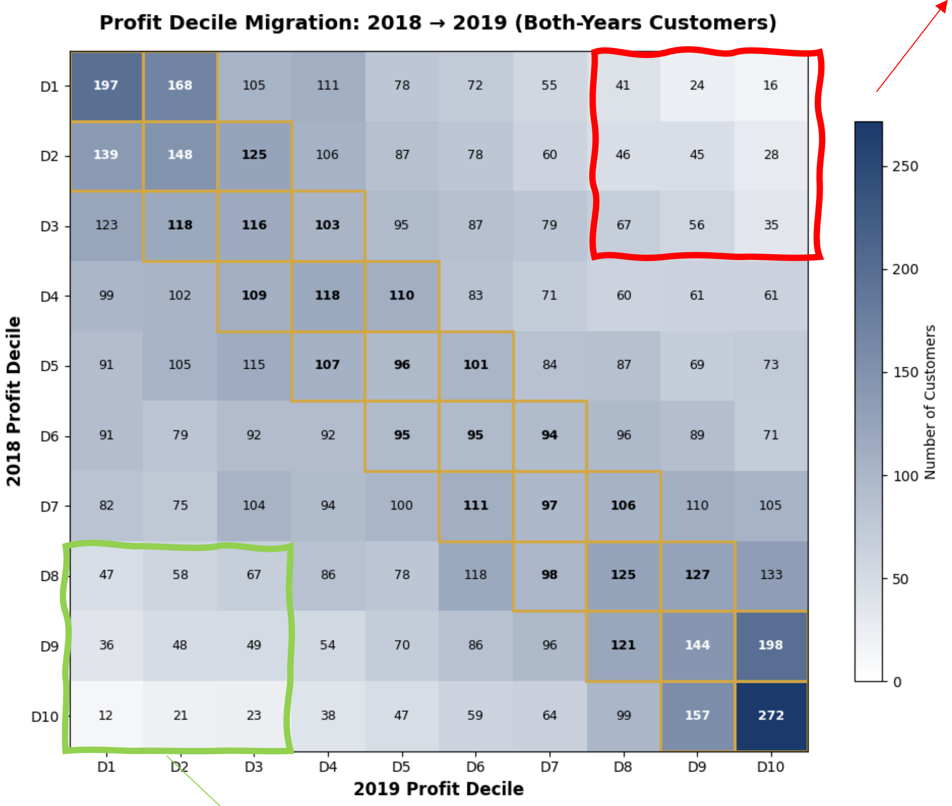
Rows 8–10 show small numbers everywhere. Most bottom-decile customers churned out entirely — they became 2018-only and don’t appear in this matrix. Selection effect.

Revised Copy – Threshold is determined by profit of the 8919 both_year group, instead of all customers in 2018 and 2019

DECILE MIGRATION

Profit Decile Migration Heatmap

On the flip side: 358 customers dropped from top 3 → bottom 3 deciles. Can we identify what drove that behaviour?



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On the bright side: 361 customers rose from bottom 3 → top 3 deciles. Can we identify what drove their improvement?

Up-Down Decomposition

Drilling down into same-customer profit changes

The Multiplicative Decomposition Per Customer

Customer Profit = # Orders × Avg Spend per Order × Avg Margin

For each both-years customer, we ask four questions:

- ↑↓ Is their 2019 profit ≥ 2018 profit?
- ↑↓ Did they make the same or more orders in 2019?
- ↑↓ Is their avg spend per order the same or higher?
- ↑↓ Is their avg margin the same or higher?

4 binary questions → **16 possible groups** → Each group's size and total profit change reveals what's driving individual-level dynamics.

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Up-Down Analysis: Key Groups

Profit = # Orders × Avg Spend/Order × Avg Margin

Profit	#Orders	Avg Spend	Margin	# Cust	2018 ($K)	2019 ($K)	Change ($K)
↑	↑	↑	↓	1,764	133.8	464.6	**+330.8**
↑	↑	↑	↑	1,184	84.6	288.3	**+203.7**
↑	↑	↓	↑	631	89.4	161.6	**+72.2**
↑	↑	↓	↓	474	73.2	135.3	**+62.1**
				— Profit Decreased —			
↓	↓	↓	↑	962	332.5	81.8	**-250.7**
↓	↓	↓	↓	588	213.2	51.8	**-161.4**
↓	↑	↓	↑	1,215	188.8	102.1	**-86.7**
↓	↑	↓	↓	941	147.5	82.1	**-65.4**
			Both years total	**8,919**	**1,535.0**	**1,596.3**	**+61.2**

Up-Down Analysis: Key Groups

Profit = # Orders × Avg Spend/Order × Avg Margin

Profit	#Orders	Avg Spend	Margin	# Cust	2018 ($K)	2019 ($K)	Change ($K)
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Both years total				**8,919**	**1,535.0**	**1,596.3**	**+61.2**

UP-DOWN ANALYSIS

The Big Picture: Profit Up vs Down

4,471

Profit Increased/Same

+$706K total gain

50%

4,448

Profit Decreased

-$645K total loss

50%

Net effect: +$706K – $645K = **+$61K** modest gain from both-years customers

Among profit-up customers:

66% had ALL 3 components improve

Clean, broad-based improvement

Among profit-down customers:

Only 13% had all three components decline

Losses are fragmented across many patterns

Revised Copy – 84% had 2 or more components improving, instead of claiming all 3 components improving

UP-DOWN ANALYSIS

The Big Picture: Profit Up vs Down

4,471	4,448
Profit Increased/Same	Profit Decreased
+$706K total gain	-$645K total loss
50%	50%

Net effect: +$706K – $645K = **+$61K** modest gain from both-years customers

Among profit-up customers:	**84% had two or more components improve** *Clean, broad-based improvement*
Among profit-down customers:	**83% had declining average spend** *Spend erosion is the common thread*

Among profit-up customers, 84% had two or more factors improve. Improvement is broad-based.

Among profit-down customers, 65% had two or more factors decline, which at first glance also looks broad-based. But look deeper: 83% of all profit-down customers had declining average spend, regardless of what happened to orders or margin. Spend erosion is the dominant thread, and the single biggest lever to address.

LENS 2 CONCLUSION

Period-on-Period Decomposition: Key Takeaways

Multiplicative Decomposition

Profit = Customers × AOF × AOV × Margin

KEY INSIGHTS

87% profit gap between one-year groups — volume, frequency, AND spend all compound

Both-years: AOF ↑9%, AOV ↓4% — offsetting changes, modest net +$61K

Selection effect: both-years AOF (2.3) is 2× one-year (1.3) — frequent buyers self-select

Margins flat at 47% across all groups — frequency drives the entire profit story

ACTIONS

Frequency levers: replenishment reminders, subscriptions, loyalty rewards

Skip product mix changes — margins are already consistent

Profit Decile Migration

10×10 heatmap tracking profit-rank movement

KEY INSIGHTS

Top deciles sticky: 636 stayed in D1 — largest single cell by far

±1 decile = 45% stable vs 19% exact diagonal — gaps as low as $7

Middle tiers (D5–7) spread across all columns — inherently unpredictable

Bottom deciles sparse — most churned out entirely (selection effect)

ACTIONS

Protect top with loyalty & outreach — they don't need acquisition

Target middle with predictive models — highest intervention ROI

Accept bottom churn — redirect budget to movable segments

Up-Down Decomposition

Split both-years into profit ↑ vs ↓, decompose

KEY INSIGHTS

~50/50 split between improvers and decliners in both-years group

Large gross movements in both directions cancel to modest net change

Frequency is the driver — AOF change dominates profit shifts both ways

Individual volatility hidden by aggregate stability — averages mislead

ACTIONS

Monitor direction of change, not just segment averages

Flag early decliners — frequency drop is the leading warning signal

The Big Picture: Customer profitability is dynamic, not fixed. Effective strategy decomposes changes to find real drivers, distinguishes signal from noise in profitability movements, and invests differently across segments based on their migration patterns.

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Lens 3: How Does Customer Behaviour Evolve Over Time?

Cohort Analysis and Customer Lifecycle

School of Computing and Information Systems

Recap of Lesson 3

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Profit Decile Migration Heatmap

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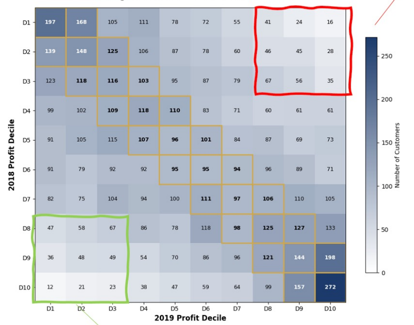
Profit Decile Migration: 2018 → 2019 (Both-Years Customers)



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FRAMEWORK

The Five Lenses at a Glance

Lens 1	Lens 2	Lens 3	Lens 4	Lens 5
How Different?	**What Changed?**	**How Do They Evolve?**	**Cohorts Degrading?**	**Is It Sustainable?**
Single-period snapshot Distributions & deciles	Period-on-period Retained / Lost / New	Cohort over time Retention & lifecycle	Vintage comparison Quality drift	Overall health Integrated diagnostic
		TODAY		

Each lens answers a question the previous one structurally cannot.

LEARNING OUTCOMES

What You'll Be Able to Do After Lens 3

Cohort Analysis — understanding who your best and worst customers are

1

Diagnose customer value heterogeneity

Use volume-to-date (VTD) distributions and decile analysis to quantify how concentrated value is among your customer base — and why the top 20% matter disproportionately.

2

Analyse purchase timing and RFM patterns

Examine inter-purchase intervals, recency-frequency-monetary patterns, and natural repurchase cycles to identify opportunities for timely marketing interventions.

3

Compute Cohort Retention Curves

Construct cohort retention curves to track how a fixed group of customers evolves from acquisition onward.

LENS 3

What Is a Cohort Analysis?

Lens 1

Who are our customers?

All customers active in a period. Cross-sectional snapshot.

Lens 2

How do customers change?

All customers compared across two consecutive periods.

Lens 3

How does behavior evolve?

One acquisition cohort tracked from first purchase onward.

Meridian Q1/2016 Cohort

7,830
Cohort Size
customers acquired

4 years
Observation Window
Q1/2016 – Q4/2019

$3.05M
Total Revenue
over 4 years

$1.44M
Total Profit
over 4 years

LENS 3

Cohort Analysis vs Customer Segmentation

	Cohort Analysis	Customer Segmentation
Defined by	When the customer was acquired (first purchase date)	Current behavioral or demographic characteristics (e.g., spend, frequency)
Membership	Fixed forever — you never leave your cohort	Fluid — customers move between segments over time
Analogy	Your birth year — it never changes	Your job title — it can change anytime
Purpose	Track how a fixed group evolves from acquisition onward	Classify customers at a point in time for targeting or analysis
Example	"Q1/2016 cohort" — all 7,830 customers acquired that quarter	"High-value" — top 20% by spend this year (refreshed annually)

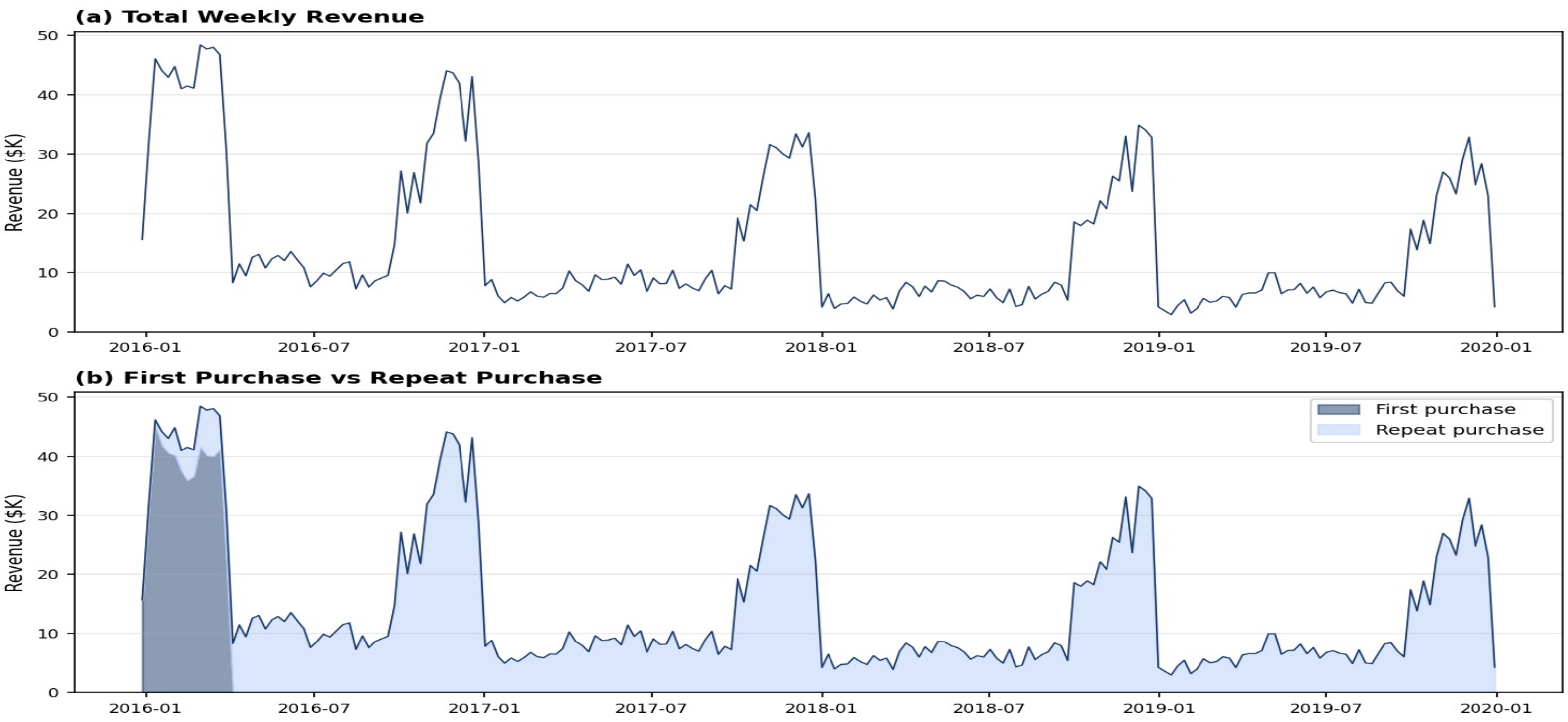
Why it matters: Tracking a "high-value segment" over time can create an illusion of stability — the group is constantly refreshed with new members replacing those who churned. Only cohort analysis reveals the true retention and evolution story.

Initial Analysis

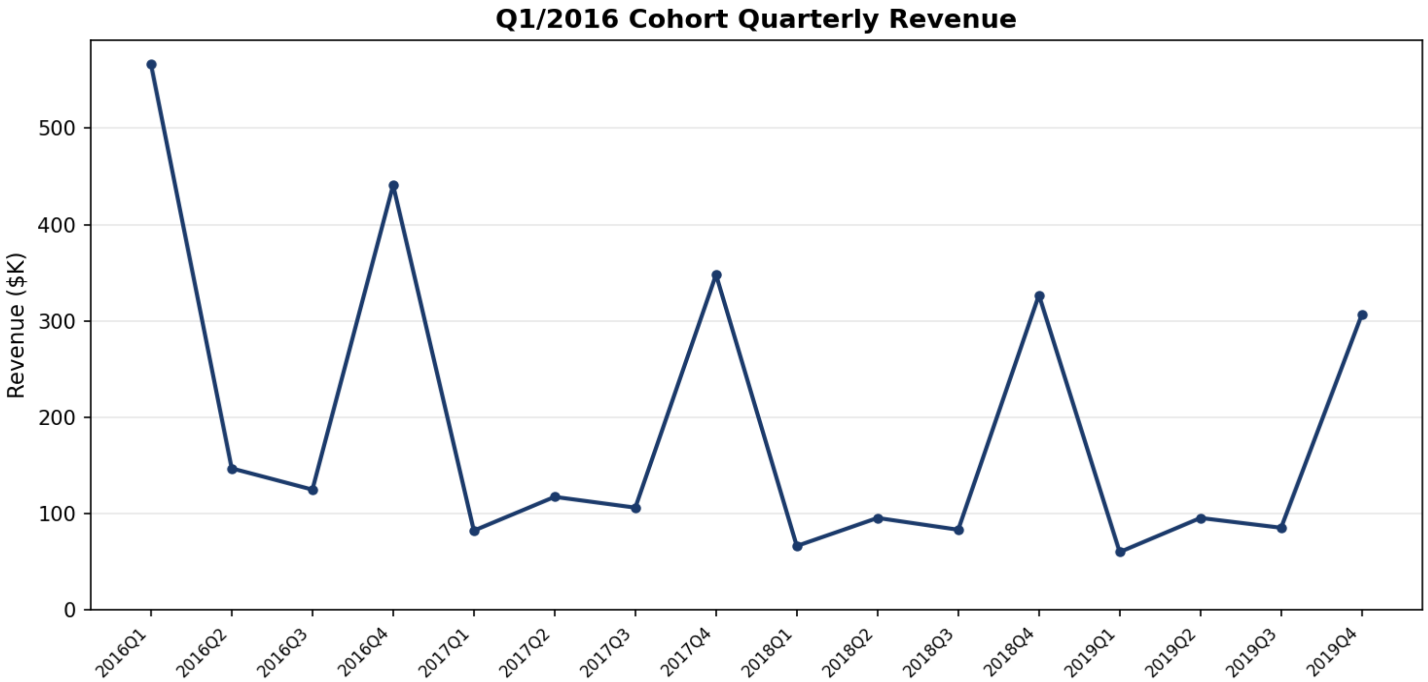
Tracking cohort revenue from acquisition through four years

INITIAL ANALYSIS

Q1/2016 Cohort Weekly Revenue



Q1/2016 Cohort Quarterly Revenue



Q1/2016: $567K (acquisition) **Q4/2016:** $441K **Q4/2017:** $348K **Q4/2018:** $327K **Q4/2019:** $307K

Multiplicative Decomposition

Unpacking what drives cohort revenue changes over time

DECOMPOSITION

Multiplicative Decomposition of Period Revenue for a Cohort

Cohort Revenue

=

Active Cohort Members × Avg Spend per Active Member

Cohort Size × % Cohort Active

Average Order Frequency (AOF) × Average Order Value (AOV)

The Formula

Cohort Revenue = Cohort Size × % Active × AOF × AOV

Cohort size is fixed (7,830). All changes in revenue come from three moving parts:

- % Active — are fewer customers buying?
- AOF — are they ordering less often?
- AOV — are they spending less per order?

LENS 3 DEFINITIONS

What Does "% Cohort Active" Mean?

Definition

% Cohort Active = # cohort members who made ≥ 1 purchase in the period ÷ total cohort size × 100

The denominator is always the original cohort size — it never changes. Only the numerator moves.

Meridian Q1/2016 Cohort — Worked Example

Quarter	# Active	Cohort Size	% Active	What's Happening
Q1/2016	7,830	7,830	**100%**	Acquisition quarter — all members active by definition
Q2/2016	912	7,830	**11.6%**	Sharp drop once acquisition ends; only repeat buyers remain
Q4/2016	1,146	7,830	**14.6%**	Holiday season lifts activity temporarily
Q4/2019	885	7,830	**11.3%**	Four years later, ~11% still purchasing each quarter

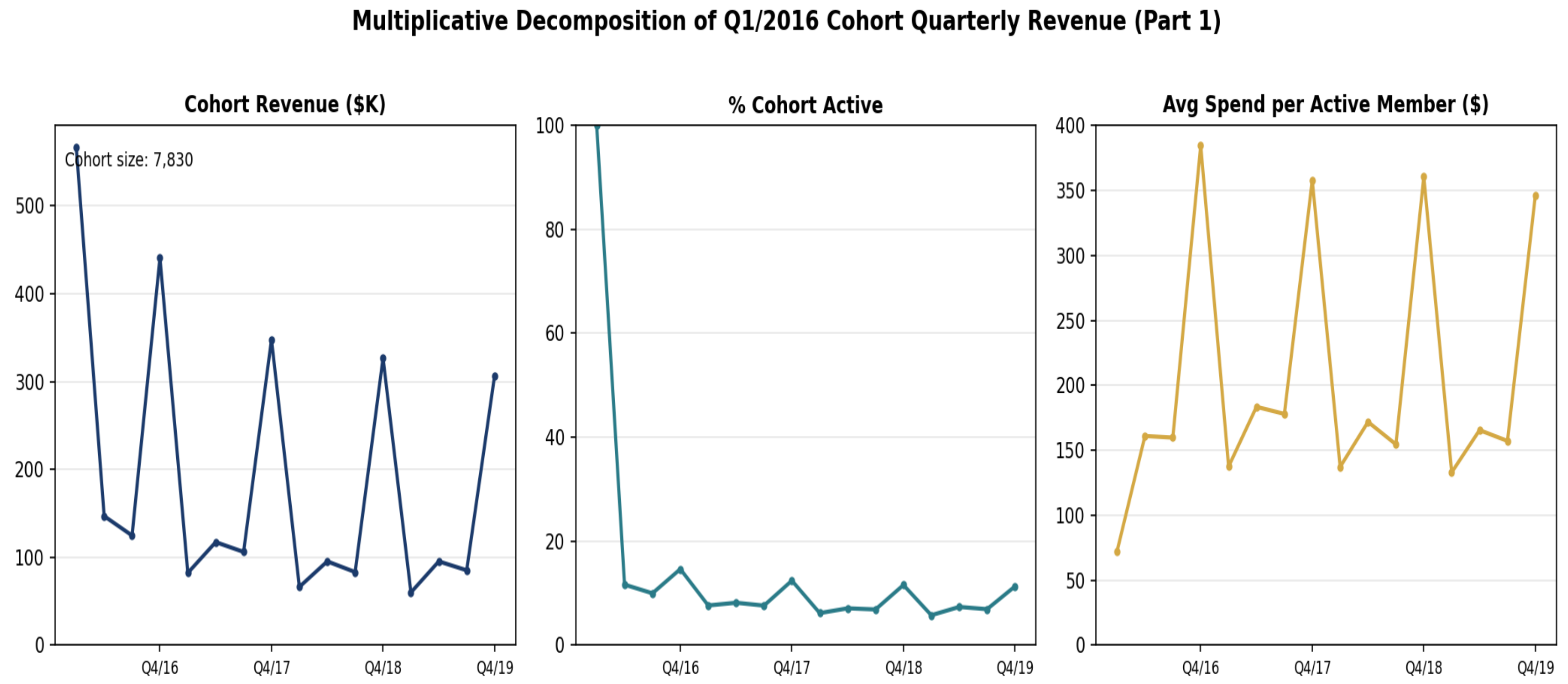
Why This Metric Matters

= **Fixed denominator** — Unlike Lens 1 (where total active customers changes each year), cohort size is constant. This isolates changes in customer engagement from changes in acquisition volume.

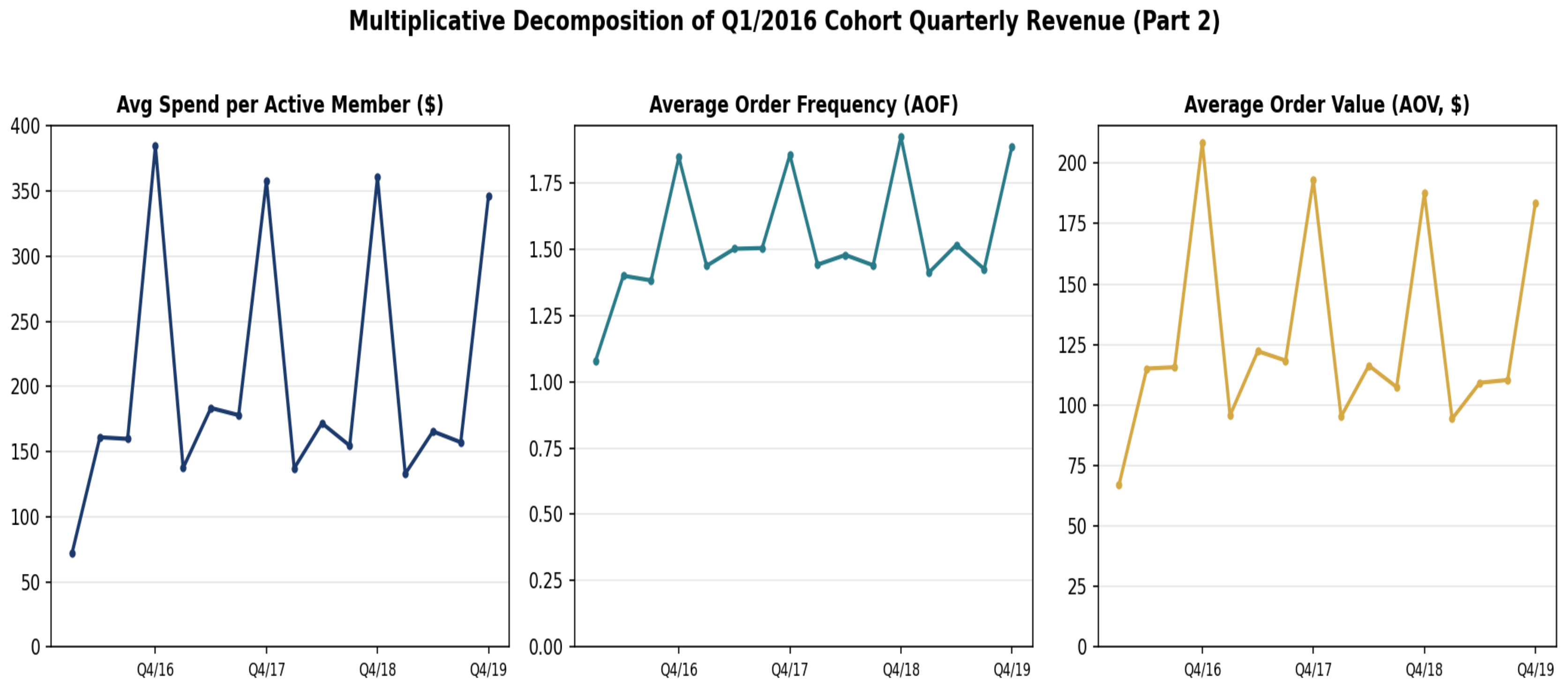
↓ **Primary driver of revenue decay** — Meridian's cohort revenue decline is almost entirely explained by this metric falling from 100% to ~7-12%. Per-customer spending stays roughly flat.

≠ **Active ≠ retained** — A customer inactive in one quarter may return later. 16% of Meridian's repeat buyers had at least one gap year. "% active" is a snapshot, not a verdict.

Decomposition Part 1: Revenue = Cohort Size × % Active × Avg Spend



Decomposition Part 2: Avg Spend = AOF × AOV



Evolution of Purchase Incidence

Understanding annual buying patterns and repeat purchase behavior

SMU Singapore Management University | School of Computing and Information Systems

PURCHASE INCIDENCE

Patterns of Annual Buying for the Q1/2016 Cohort

2016*	2017	2018	2019	Count	% Cohort
Y	Y	Y	Y	**799**	**10.2%**
Y	Y	Y	N	341	4.4%
Y	Y	N	Y	341	4.4%
Y	Y	N	N	369	4.7%
Y	N	Y	Y	274	3.5%
Y	N	Y	N	305	3.9%
Y	N	N	Y	297	3.8%
Y	N	N	N	473	6.0%
N	N	N	N	**4,631**	**59.1%**

** 2016 = purchases after Q1/2016 (post-acquisition)*

59% — **One-and-Done**
Never bought again after Q1/2016. The single biggest group.

10% — **Loyal Core**
Active every year. Small but disproportionately valuable.

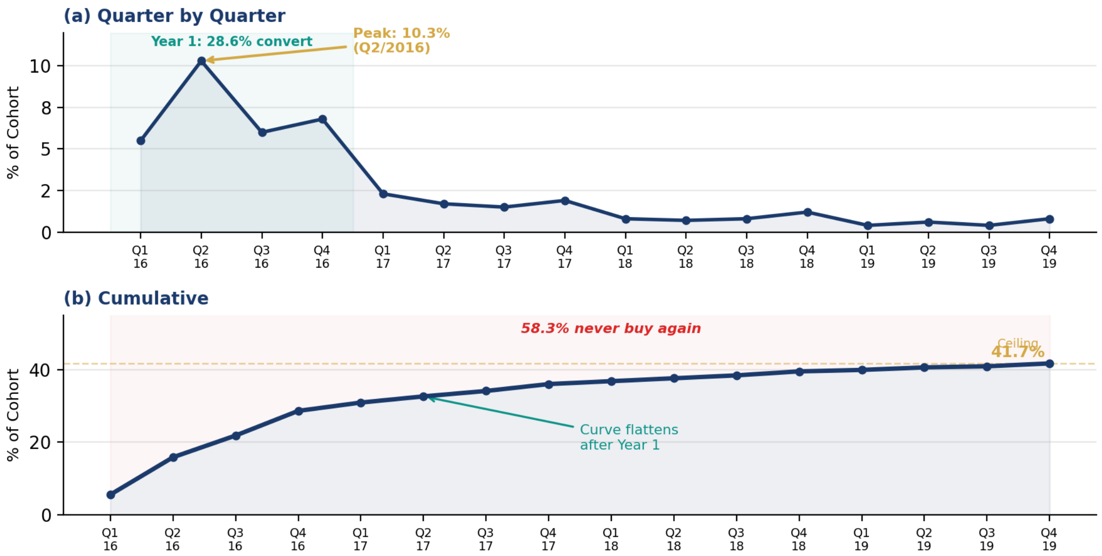
41% — **Ever Repeated**
Made at least one more purchase. Generated nearly all repeat revenue.

Year-to-year repeat buying rates: 2016→2017: 58% 2017→2018: 62% 2018→2019: 62% *Once customers come back, they tend to stay.*

PURCHASE INCIDENCE

Time to Second Purchase

% of Q1/2016 Cohort Making Their Second-Ever Purchase



1 How It's Built

Take the Q1/2016 cohort (**7,830 first-time buyers**). For each quarter, count how many made their ***second-ever*** purchase. (a) shows the quarterly rate; (b) is the running total.

2 The Peak & Drop-off

Peak is Q2/2016 (**10.3%**) — one quarter after acquisition. By Year 2, the rate drops below 2%. If a customer hasn't returned by Q4/2016, they probably never will.

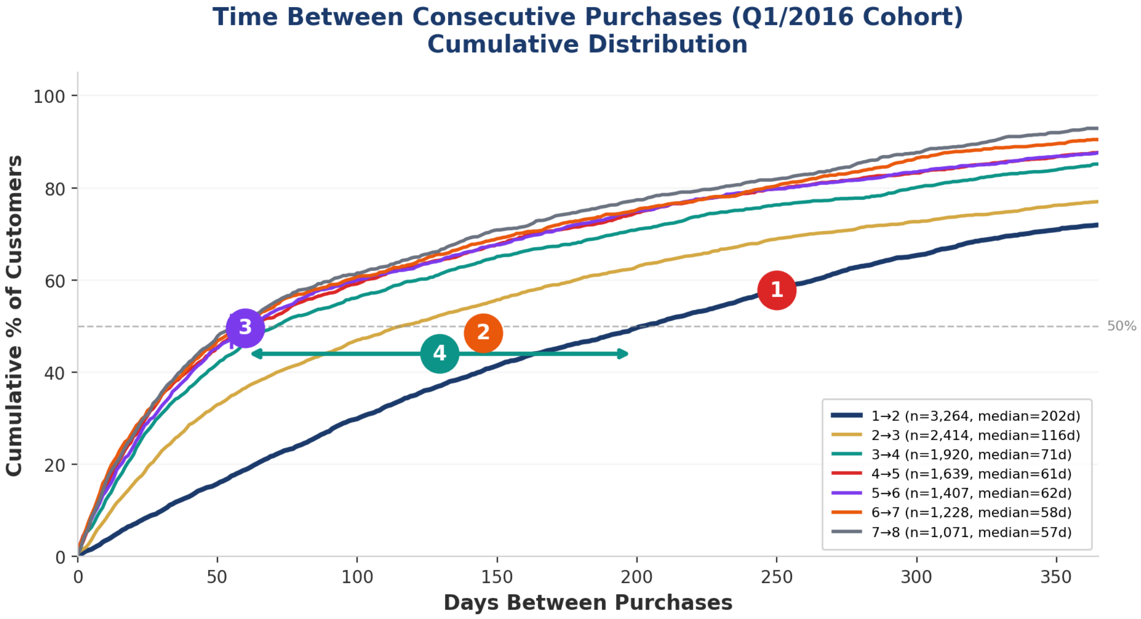
3 The One-and-Done Ceiling

The cumulative curve plateaus at **41.7%**. That means **58.3% of the cohort never made a second purchase** — even after 4 years. This is the one-and-done rate.

Key takeaway: **Year 1 is the window.** If you're going to convert a one-time buyer into a repeat customer, it must happen in the first 3–4 quarters. After that, the probability drops to near zero.

LENS 3 — PURCHASE INCIDENCE

Reading the Inter-Purchase Time Chart



1 **The Slowest Curve — 1st → 2nd Purchase**

This is the lowest line because the starting population includes everyone — heavy buyers, light buyers, and the 59% who never buy again. It plateaus at ~72%, meaning 28% of those with a 2nd purchase take over a year.

2 **The Big Gap — Selection Effect**

The large gap between 1→2 and 2→3 reflects the most dramatic filtering. Once one-and-done buyers drop out, the remaining pool is fundamentally different — they buy again faster (median drops from 202 to 116 days).

3 **Upper Curves Cluster — Behaviour Stabilises**

Transitions 3→4 through 7→8 are tightly clustered (medians 57–71 days). By purchase 3, customer heterogeneity has been largely resolved — these are committed repeat buyers with a predictable rhythm.

4 **Median Shift — Accelerating Repurchase**

The 50% line is crossed at 202 days for 1→2 but only 57 days for 7→8. This isn't the same customers speeding up — it's the base population becoming increasingly composed of frequent buyers.

Key takeaway: The hardest conversion is 1st → 2nd purchase. After that, each subsequent transition becomes faster and more predictable. If you can move a customer past purchase 2, the odds of continued buying rise sharply — not because of magic, but because reaching purchase 2 signals they were always a higher-propensity buyer.

When Should You Send the First Reactivation Email?

Think About This

You're LushProtein's marketing manager.
A customer just made their **first purchase** of bubble tea protein on Shopify.

You want to set up an automated email sequence to bring them back for a second order.

Using the concept of the inter-purchase time chart you just saw, when would you triggerthe first reactivation email?

Questions for Your Group

1. Would you send the email at the same time for all customers, or would you vary it? What data would inform this?

2. A protein tub lasts ~30 days. Does product consumption cycle matter more than the inter-purchase chart here?

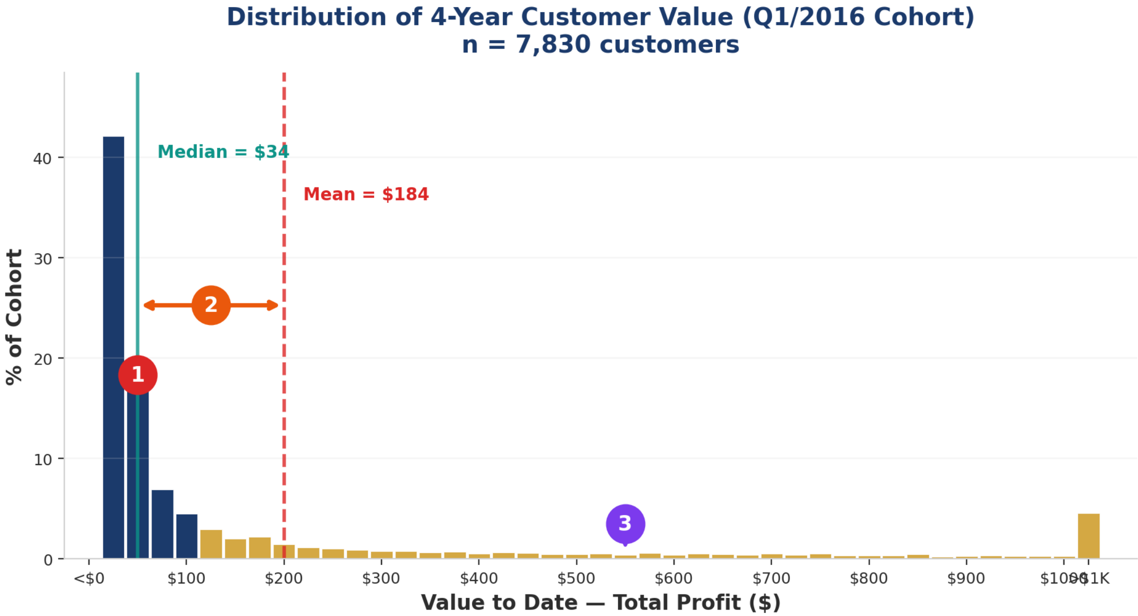
3. After purchase 2→3, if the median drops, should the email cadence change for repeat buyers vs first-timers?

Exploring Differences in Customer Value

Long-term profitability and the VTD (Value to Date) decile analysis

LENS 3 — CUSTOMER VALUE

Reading the VTD Distribution Chart



1

How It's Constructed

Each customer's total profit over all 4 years (2016–2019) is calculated, then binned into $25 buckets. The y-axis shows what % of the 7,830 cohort members fall into each bin. The spike on the left is the 59% one-and-done buyers with very low VTD.

2

What It Reveals — Mean vs Median

The median ($34) is far below the mean ($184) — a 5.4× ratio. This means 78% of customers are worth less than the average. The "average customer" is a fiction — most generate very little profit, while a few drive enormous value.

3

Why the Long Tail Matters

The top 10% of customers account for 60% of total profit ($861K of $1.44M). These high-value customers in the right tail are what make the cohort profitable. This Pareto-like concentration is why decile analysis (next) is essential — averages hide the real story.

7,830
Cohort Size
Q1/2016 customers

5.4×
Mean / Median
$184 vs $34

60%
Top 10% Share
of total profit

Summary of Cohort Behavior by VTD Decile

Decile	%VTD	%Cohort	%Trans	Avg VTD	AOF	AOV	Margin
1	10%	1%	7%	$2,441	36.3	$136	50%
2	10%	1%	8%	$1,247	20.5	$123	49%
3	10%	2%	8%	$842	14.6	$117	49%
4	10%	3%	9%	$609	11.1	$111	49%
5	10%	4%	9%	$451	8.6	$107	49%
6	10%	5%	9%	$334	6.6	$103	49%
7	10%	7%	10%	$243	5	$99	49%
8	10%	10%	10%	$168	3.7	$94	48%
9	10%	16%	11%	$105	2.5	$89	48%
10	10%	51%	20%	$34	1.4	$59	40%
Total	100%	100%	100%	$171	3.7	$97	48%

60%

Extreme Concentration

Top 10% of customers generate 60% of total 4-year profit. Top 20% generate 82%.

26×

Frequency is King

Decile 1 AOF is 36 vs 1.4 for decile 10. Buying frequency drives most of the value difference.

47%

Consistent Margins

Margins are virtually identical across all deciles. Product mix doesn't vary by customer value.

Annual % Active by VTD Decile

Decile	% Cohort	2016	2017	2018	2019
1	1%	100%	90%	86%	84%
2	1%	100%	72%	67%	68%
3	2%	100%	43%	41%	38%
4	3%	100%	19%	16%	17%
5	4%	100%	7%	7%	8%
6	5%	100%	3%	2%	2%
7	7%	100%	2%	0%	1%
8	10%	100%	0%	0%	0%
9	16%	100%	0%	0%	0%
10	51%	100%	0%	0%	0%

Heterogeneity Amplifies Over Time

The initial differences in customer value observed in Lens 1 are not temporary. Over four years, they become even more pronounced:

- Decile 1: 84% still active after 4 years
- Decile 5: only 8% still active
- Deciles 8-10: effectively gone after year 1

The more valuable the customer, the more likely they are to stay active — and this gap widens with each passing year.

But Don't Write Off Low-Value Customers

Decile 10 is 51% of the cohort. Even if only 5% were active in 2019, that's more people in absolute terms than 84% of decile 1 (which is only 1% of the cohort).

Rates vs. absolute numbers — a crucial distinction.

LENS 3 — STRATEGIC IMPLICATIONS

How both Tables Work Together

1 Frequency Drives Everything

If you want to improve cohort value, the lever is getting customers to buy more often, not getting them to spend more per visit.

→ This should shape where marketing dollars go — loyalty programmes, replenishment reminders, and reactivation campaigns rather than upselling.

Table 5.1: AOF varies 26× across deciles (36.3 vs 1.4) while AOV only varies 2× ($136 vs $59)

2 Don't Write Off Low-Value Customers

Decile 10 has 51% of the cohort and only 5% are active in 2019. But 5% of 51% is still a meaningful number of customers in absolute terms — potentially more people than 90% of decile 1.

→ You can't simply dismiss low-value customers. Some of them will emerge as valuable over time.

Table 5.2: D10 = 51% of cohort × 5% active = more bodies than D1's 1% × 90% active

3 The Retention Gradient Is a Diagnostic Tool

If you're running a retention campaign, the retention gradient lets you check which decile it's actually moving.

→ Retaining a D1 customer (avg VTD $2,441) is worth vastly more than retaining a D10 customer (avg VTD $34). The tables help you prioritise where to invest.

Table 5.1 + 5.2: D1 retains at 90% with $2,441 VTD vs D10 at 5% with $34 VTD

4 It's Not Product Mix — It's Frequency

The stable margins across all deciles (48–50%) tell you that profitability concentration isn't driven by product mix. Heavy buyers aren't buying higher-margin products.

→ They're just buying more often. This simplifies the strategic picture considerably — you only need to solve one problem.

Table 5.1: Margin = 48–50% across all 10 deciles; only AOF varies meaningfully

The unifying theme: Tables 5.1 and 5.2 decompose customer heterogeneity from two angles — value composition and retention over time. Together, they tell you not just who your best customers are, but how reliably they stay and where the biggest leverage points lie.

Taking Stock: RFM Analysis

Segmenting the cohort by Recency, Frequency, and Monetary value

RFM ANALYSIS

RFM Segmentation: Three Dimensions of Customer Value

R

Recency

Quarter of last purchase

Q1 (acquisition only)
Q2-Q8 (through 2017)
Q9-Q15 (2018-Q3/2019)
Q16 (Q4/2019 — most recent)

59% are Q1 only

F

Frequency

Total orders over 4 years

1 order
2-4 orders
5-10 orders
11+ orders

58% bought once

M

Monetary

Avg profit per transaction

< $25
$25 - $50
$50 - $75
$75+

Profit per transaction

Key insight: The order of the letters R-F-M is not a coincidence. Recency and frequency are much more important and diagnostic than monetary value. Spend levels are largely independent of transaction flow. A careful customer-base audit objectively reflects their true importance.

RFM SCORING

How RFM Scoring Works

1 **Sort independently**

Sort customers by each dimension: most recent purchase, purchase frequency, average spend

2 **Assign quintile codes**

Split each sorted list into 5 equal groups, coded 1 (best) to 5 (worst)

3 **Combine into RFM cells**

Each customer gets a 3-digit code (e.g., 111, 234, 555) — 125 possible cells

4 **Run test campaign**

Send a promotion to a test group and measure actual response rate per RFM cell

Source: Kumar & Reinartz (2018), Fig 6.5

Example: 5 × 5 × 5 = 125 RFM Cells

Code	R	F	M	Meaning
111	Best	Best	Best	**Champions**
113	Best	Best	Mid	Loyal mid-spend
311	Mid	Best	Best	At-risk high value
555	Worst	Worst	Worst	Lost / inactive

RFM Cell Sorting

An alternative approach sorts sequentially: first by R into 5 groups, then each group by F into 5, then each by M into 5 — ensuring exactly equal cell sizes (320 customers each in the textbook example).

A Typical RFM Scenario

This slide walks through the mechanics behind RFM-based targeting using the Breakeven Index. The textbook example (Kumar & Reinartz, 2018) imagines a direct-mail campaign where each piece costs $1 to send and a successful response generates $45 in net profit. From those two numbers we derive a breakeven response rate of 2.22 %, then compare every RFM cell's actual response rate against that threshold. Cells with a positive BEI are worth targeting; cells below zero would lose money. The table on the right shows the first 15 of 125 RFM cells, ranked from highest to lowest response rate.

Breakeven Value (BE)

BE = unit cost price / unit net profit

Breakeven Index (BEI)

BEI = **(Response rate − BE) / BE** × 100

Worked Example

Cost to mail $150 coupon = **$1**   Net profit per sale = **$45**

BE = $1 / $45 =

0.0222 = 2.22%

If a cell has 3.5% actual response:

BEI = (3.5% − 2.22%) / 2.22% × 100 =

57.66 ← profitable!

Table 6.2 RFM Codes, Breakeven & BEI (excerpt)

Cell #	RFM codes	Cost per mail ($)	Net profit per sale ($)	Breakeven (%)	Actual response (%)	Breakeven index
1	111	1	45.00	2.22	17.55	690
2	112	1	45.00	2.22	17.45	685
3	113	1	45.00	2.22	17.35	681
4	114	1	45.00	2.22	17.25	676
5	115	1	45.00	2.22	17.15	672
6	121	1	45.00	2.22	17.05	667
7	122	1	45.00	2.22	16.95	663
8	123	1	45.00	2.22	16.85	658
9	124	1	45.00	2.22	16.75	654
10	125	1	45.00	2.22	16.65	649
11	131	1	45.00	2.22	16.55	645
12	132	1	45.00	2.22	16.45	640
13	133	1	45.00	2.22	16.35	636
14	134	1	45.00	2.22	16.25	631
15	135	1	45.00	2.22	16.15	627

Interpretation

BEI > 0 → Profitable — target these customers
BEI = 0 → Just broke even
BEI < 0 → Unprofitable — do not target

Source: Kumar & Reinartz (2018), Ch. 6, Eq. 6.1–6.2

RFM SCORING

RFM Sorting: Independent vs Sequential (Cell) Sorting

Method 1: Independent Sorting

(The typical approach)

1. Rank ALL customers by Recency → split into 5 equal quintiles
2. Rank ALL customers by Frequency → split into 5 equal quintiles
3. Rank ALL customers by Monetary → split into 5 equal quintiles
4. Combine: each customer gets a 3-digit code (e.g., 1-1-1, 3-2-5)

✓ Simple & intuitive ✗ Unequal cell sizes (R, F, M are correlated)

Method 2: Sequential (Cell) Sorting

(Kumar & Reinartz, Fig 6.5)

1. Sort ALL customers by Recency → split into 5 groups
2. Within EACH R-group, sort by Frequency → split into 5
3. Within EACH R×F group, sort by Monetary → split into 5
4. Result: 5 × 5 × 5 = 125 cells, each with EQUAL customers

✓ Equal cell sizes ✗ Order of R→F→M matters

RFM Sorting: Independent vs Sequential (Cell) Sorting

Method 2: Sequential (Cell) Sorting

(Kumar & Reinartz, Fig 6.5)

1. Sort ALL customers by Recency → split into 5 groups
2. Within EACH R-group, sort by Frequency → split into 5
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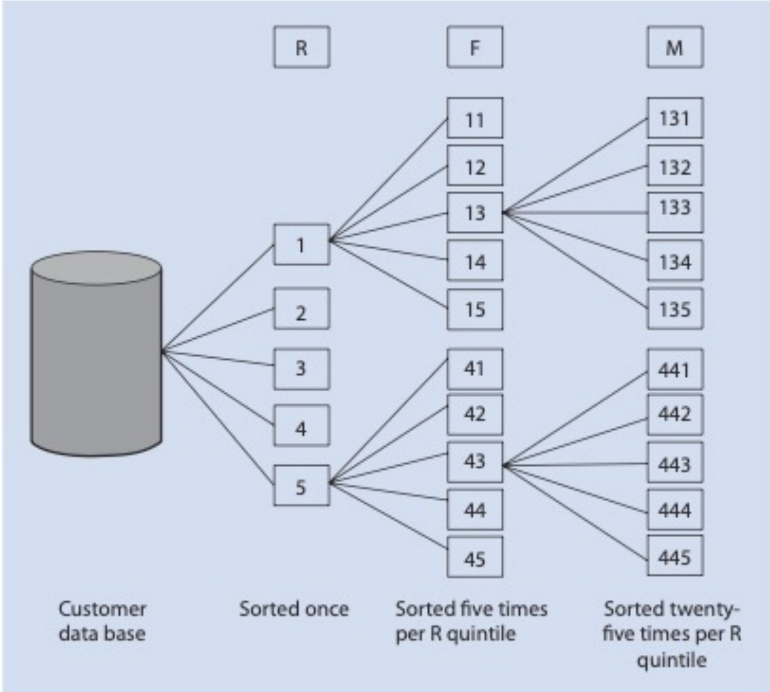
4. Result: 5 × 5 × 5 = 125 cells, each with EQUAL customers

✓ Equal cell sizes ✗ Order of R→F→M matters



RFM SCORING

RFM Variations & When to Use RFM

Variations of the RFM Model

RFD (Recency, Frequency, Duration)

Duration = time spent on content. Best for viewership, readership, and surfing-oriented products (e.g., streaming platforms, news sites, online courses).

RFE (Recency, Frequency, Engagement)

Engagement = composite value (time on page, pages per visit, bounce rate, social media activity). Best for online businesses and digital platforms.

Who Should Use RFM?

Best Suited For

- Rewards & loyalty programs
- Airlines & frequent flyer programs
- Credit card companies
- Ranking suppliers & salespeople
- Subscription / repeat-purchase models

Not Suitable For

- Unique or one-off products
- Large-quantity items (e.g., automobiles)
- Products that are not purchased repeatedly

Should We Analyse RFM at the Cohort Level or the Entire Customer Base?

Think About This

A company has 60,000 customers acquired over 4 years. You run an RFM analysis on the entire base and get 125 cells.

But a colleague says:
"Shouldn't we look at each cohort separately?"

Questions for Your Group

1. What information might you lose when you pool all cohorts into one RFM table?
2. Could two customers land in the same RFM cell but tell very different stories? Give an example.
3. When might the whole-base view actually be more useful than a cohort view?

RFM ANALYSIS

RFM Summary — Entire Customer Base

Frequency →

Recency	Avg Profit per Trans	1	2-4	5-10	11+
Q1	<$25	**3,189**	42	—	—
	$25-50	**974**	19	—	—
	$50-75	244	4	—	—
	$75+	159	—	—	—
Q2-Q8	<$25	**6,001**	**1,097**	12	—
	$25-50	**3,395**	**1,079**	62	—
	$50-75	**1,271**	435	37	4
	$75+	**1,304**	287	43	3
Q9-Q15	<$25	**7,743**	**2,143**	139	5
	$25-50	**3,834**	**2,844**	**799**	124
	$50-75	**1,397**	**1,270**	**794**	274
	$75+	**1,394**	**1,071**	**845**	307
Q16	<$25	**2,128**	459	43	3
	$25-50	**2,183**	**1,060**	**581**	193
	$50-75	**1,095**	**773**	**942**	454
	$75+	**1,843**	**1,126**	**1,347**	**636**

n = 59,510 customers

Upper-Left: Diluted but Still There

Q1 recency drops from 59% to just 8% — but the same 4,631 customers are still there. They look like a small problem in the whole-base view, masking the fact that every cohort likely has a similar one-and-done rate.

Upper-Right: Now Populated!

Unlike the cohort view, the upper-right (old recency + high frequency) now has customers. These are people from later cohorts who were frequent buyers but stopped. The whole-base view treats "once-active-now-lapsed" identically regardless of tenure.

Mixed Tenure Problem

A customer with frequency=2 could be a 2016 customer who bought twice in 4 years (low engagement) or a late-2019 customer who bought twice in 3 months (strong trajectory). The whole-base view cannot distinguish between them — same cell, very different stories.

Practical for Targeting, Not for Diagnosis

Despite these limitations, this is what most companies use day-to-day. "Send the campaign to recent, high-frequency customers" is actionable and doesn't require cohort-level tracking infrastructure.

RFM Summary — Q1/2016 Cohort

Frequency →

Recency	Avg Profit per Trans	1	2-4	5-10	11+
Q1	<$25	**3,189**	42	—	—
	$25-50	**974**	19	—	—
	$50-75	244	4	—	—
	$75+	159	—	—	—
Q2-Q8	<$25	—	323	7	—
	$25-50	—	294	24	—
	$50-75	—	101	19	3
	$75+	—	54	14	3
Q9-Q15	<$25	—	173	46	3
	$25-50	—	273	220	58
	$50-75	—	116	168	114
	$75+	—	54	139	108
Q16	<$25	—	49	13	2
	$25-50	—	66	113	71
	$50-75	—	30	115	131
	$75+	—	27	110	158

n = 7,830 customers

Upper-Left: The One-and-Done Mass

4,566 customers (59%) sit here — Q1 recency, frequency=1. They bought once in their acquisition quarter and never returned. This is the single biggest story in the data. Reactivation ROI here is very low.

Upper-Right: Structurally Empty

All dashes. High frequency + old recency is near-impossible in a cohort — buying many times means being active recently. This diagonal pattern is a structural feature, not a gap to fill.

Lower-Right: The Loyal Core

Q16 recency with 11+ orders — only 362 customers (5%), but they bought recently and frequently. These are your highest-value customers. Protect them with priority service and personalised engagement, not discounting.

Lower-Left: The Growth Opportunity

Q9-Q15 and Q16 recency with frequency 2-4 — recently active but still infrequent. These customers have shown willingness to return but haven't formed a habit. Highest-ROI segment for nurturing campaigns to push them rightward on the frequency axis.

Cohort vs Whole-Base — It Depends on the Question

Whole-Base View

Best for: campaign targeting

"Send the offer to recent, high-frequency customers" — simple, actionable, no cohort tracking infrastructure needed.

Cohort View

Best for: diagnosing behaviour

"Why are 59% of Q1/2016 customers one-and-done?" — reveals structural patterns masked by the aggregate.

The Mixed-Tenure Trap

frequency=2 could be a 2016 customer who bought twice in 4 years (low engagement) **or** a 2024 customer who bought twice in 3 months (strong trajectory). Same cell — very different stories.

What Changes Between the Two Views?

Dimension	Entire Customer Base	Cohort (e.g. Q1/2016)
Upper-left (R=1, F=1)	Looks like 8% of base — small problem	Actually 59% of cohort — dominant pattern
Upper-right (old R, high F)	Populated with customers from later cohorts	Structurally empty — buying often means being active recently
Lower-right (recent, high F)	Mixed tenure — old and new customers blend	The loyal core — 5% but highest-value segment
Actionability	High — directly feeds campaign targeting lists	High — explains why patterns exist and what to fix

Takeaway: Use the whole-base view for **targeting** (who to contact next). Use the cohort view for **diagnosis** (why they behave that way). Best teams use both.

SMU SINGAPORE MANAGEMENT UNIVERSITY | School of Computing and Information Systems

Thank You

Questions?

LushProtein

No Fillers, No Fluff, Building a Premium Nutrition Brand in a Crowded Market

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The Challenge

On a Monday evening in late March 2026, Dzarrin Alidin sat in his office in Singapore reviewing a Shopify dashboard he had come to know all too well. The numbers told a familiar story: a steady stream of new customers arriving each month, but too few of them coming back. For a direct-to-consumer brand that had spent over a decade building a loyal following in Southeast Asia's sports nutrition market, the gap between acquiring a customer and retaining one had become the single most consequential challenge facing the business.

As he scrolled through order histories and ad spend reports scattered across Shopify, Lazada, Google Ads and Shopee, a more fundamental question nagged at him: did he even have the right data to answer the questions that mattered most?

From Track and Field to Tubs of Protein

Dzarrin and Nathaniel had been friends since their days as competitive track-and-field athletes in Malaysia. Both knew the frustration of walking into a supplement store in Southeast Asia and finding shelves dominated by imported American and European brands—products formulated for Western palates, sold at Western prices, with ingredient lists padded with fillers. In 2013, they decided to do something about it. They founded LushProtein with a simple principle: maximum protein per serving, minimal fillers, at a price point accessible to Southeast Asian consumers.[1]

Operating out of Malaysia, they experimented with whey-based and vegetarian protein formulas, selling through local gyms and word of mouth. What set them apart was not just the product quality but their willingness to localise. While global brands offered chocolate and vanilla, Dzarrin and Nathaniel began developing flavours that resonated with regional tastes. Their breakthrough came with a bubble-tea-flavoured protein powder—taro, Thai milk tea—that tapped into Southeast Asia's obsession with boba culture. It was a product that no multinational would think to make and it resonated deeply, particularly with female consumers who had long felt excluded from the hyper-masculine world of sports nutrition.

Over the following years, LushProtein expanded its range to include clear protein drinks, functional supplements, collagen products and even a spin-off brand, Calli, offering low-calorie ice cream—a direct response to their most common customer request.[2]

[1] "Calli offers low-calorie ice cream you can indulge without guilt," Options, The Edge, 2019.
[2] "Eat this now! Guilt-free ice cream packed with fewer calories but full of flavour," Malay Mail, 24 February 2019.

A Regional Footprint, a Digital-First Model

By 2026, LushProtein operated primarily in Singapore and Malaysia, with experimental forays into other markets such as Hong Kong and Indonesia. The company's own Shopify-powered e-commerce store served as the primary channel, supplemented by marketplace listings on Lazada and Shopee—the dominant e-commerce platforms in Southeast Asia. Listings could also be found on smaller marketplaces such as Decathlon Marketplace.

This multi-channel approach reflected the realities of the region's fragmented digital commerce landscape. Each platform had its own customer base, promotional mechanics and data ecosystem. Shopify provided detailed order-level and customer-level data. Lazada and Shopee offered marketplace analytics but with less granularity and, critically, less control over customer relationships. Google Ads and social media campaigns drove traffic across all channels.

LushProtein had also invested in physical touchpoints. Pop-up booths at gyms, partnerships with fitness events like Hyrox—a global fitness competition gaining traction in Asia—and sampling campaigns at wellness expos served as customer acquisition engines. These events attracted serious fitness enthusiasts willing to pay premium prices, but converting a one-time event purchase into a repeat online customer remained an unsolved problem.

The Southeast Asian Opportunity

LushProtein was operating in a market with strong tailwinds. The Asia-Pacific protein market was valued at approximately USD 8.5 billion in 2026, projected to reach USD 12.2 billion by 2031, growing at a compound annual rate of 7.6 per cent.[3] Yet growth brought competition. Global brands with deep pockets were entering the region aggressively. Local competitors were emerging. The ecommerce platforms that enabled LushProtein's direct-to-consumer model also made it easy for consumers to comparison-shop, switch brands and chase the next promotional discount.

The Retention Problem

The core challenge, as Dzarrin saw it, was retention. A tub of protein powder, depending on usage frequency, lasted a customer roughly two to three months. That long repurchase cycle created a dangerous window during which customers could forget the brand, get lured by a competitor's promotion, or simply lose their fitness routine. Acquiring a new customer through digital ads was expensive. If that customer only purchased once, the economics were punishing.

LushProtein had tried several approaches. They ran evergreen advertisements to stay top-of-mind. They experimented with a subscription model, offering auto-replenishment at a discount.

[3]"Asia-Pacific Protein Market Size & Share Analysis — Growth Trends and Forecast (2026–2031)," Mordor Intelligence, 2026.

They hosted pop-ups at gyms and fitness events. The retention rates were not where they should be and further investigation was needed to explain why some customers came back while others did not.

As Dzarrin reflected on the problem, he was increasingly convinced that the highest-leverage intervention was not at the top of the funnel but immediately after the first purchase. "We know a tub lasts about two to three months," he explained. "That gives us a defined window. If we can improve the likelihood that someone makes a second purchase within that window, the economics of everything else—our ad spend, our subscription uptake, our lifetime value—start to shift." He had begun to notice patterns internally. Customers who adopted the subscription option early seemed to behave differently from those who did not. Customers who tried multiple products appeared more likely to return. The timing of the second order seemed to matter more than the size of it. But these were intuitions, not validated findings.

Nathaniel, who managed much of the operational data, added another layer of complexity. *"We have data across Shopify, Lazada, Shopee and Google Ads, but it's not always clean or connected,"* he noted. Different platforms used different customer identifiers. Some of the Shopify data came through various apps that did not always talk to each other. They had relied on basic Shopify reports up to that point, but they knew there was more in the data to be exploited for deeper consumer understanding.

Data: Rich but Fragmented

LushProtein's data ecosystem reflected the typical growing pains of a digital-native SME in the region. The primary data sources included the Shopify e-commerce store, which captured order-level detail—products purchased, order value, timestamps, customer information, traffic sources and discount codes applied. It also tracked customer acquisition channels and repeat purchase behaviour, though not always in a format amenable to sophisticated analysis.

Marketplace platforms such as Lazada and Shopee provided sales volumes, promotional performance and some customer demographics, but these platforms acted as intermediaries, limiting LushProtein's direct visibility into customer identity and behaviour. Google Ads and social media data tracked impressions, clicks and campaign spend, but attributing downstream conversions—especially across platforms—was a persistent challenge. Product and operational data included SKU-level details and product hierarchy.

There was also qualitative knowledge that existed only in the heads of Dzarrin and Nathaniel: which products were "hero SKUs," how pricing decisions were made, which promotional mechanics worked at events versus online and what they believed differentiated their loyal customers from one-time buyers. For a company that aspired to make data-driven decisions about retention, pricing and growth, the first step would be to understand what the data could—and could not—tell them.

Looking Ahead

LushProtein was looking for hypotheses beyond dashboarding. Dzarrin wanted plausible explanations that could be translated into testable ideas that the team could execute within the next 30 to 60 days. *"If someone can identify something worth testing, we have a budget to run it,"* he was fond of saying. *"We're not looking for a report that sits on a shelf . We want to know what to try next—and we're willing to try it within the next 30 to 60 days."*

As LushProtein considered how to make better use of its data, several questions had crystallised.

First, there was the question of the first purchase itself. Did the mix of products a customer chose on their first order predict whether they would return? Were customers acquired through discounts or promotions less likely to repurchase than those who paid full price? If so, did that mean the company was spending money to acquire customers who were structurally unlikely to become profitable?

Second, there was the question of timing and drop-off. At what point after the first purchase did the probability of a second order begin to decline sharply? Was there a critical window—perhaps tied to the two-to-three-month product lifespan—after which a customer was effectively lost? Understanding where customers dropped off could inform when and how to intervene.

Third, among the customers who did return, were there identifiable segments worth prioritising—by product preference, channel, acquisition source or purchasing pattern? Conversely, were there segments the company was over-investing in?

And finally, the broader strategic questions remained. For a small but ambitious Southeast Asian brand competing against global players with far greater resources, how could data analytics serve as a genuine source of competitive advantage rather than just a reporting tool?